CHEMISTRY - FORMULA

NEET/MH-CET



By StudMonk

Chemistry

1. Some Basic Concepts of Chemistry 2
2. Structure of Atom 3
3. State of Matter 5
4. Thermodynamics..... 6
5. Equilibrium 7
6. Solid State 9
7. Solutions..... 10
8. Electrochemistry 13
9. Chemical Kinetics and Nuclear Chemistry..... 16
10. Surface Chemistry 19
11. Hydrogen..... 20
12. S-Block Elements 20
13. P-Block Elements 21
14. Coordination Compound..... 22
15. Organic Chemistry 22
16. Alcohols, Phenols and Ethers 24
17. Aldehydes, Ketones and Carboxylic Acids 36
18. Amines 47
19. Haloalkanes and Haloarenes 58
20. Hydrocarbons 64
21. Polymers 74

Monksays

1. SOME BASIC CONCEPTS OF CHEMISTRY

(1). Number of molecules in W(g) of substance $= \frac{W(g) \times N_A}{GMM}$

(2). Molality (m) = $\frac{\text{No. of moles of solute}}{\text{Mass of solvent in kg}}$

(3). Number of molecules in V litre of gas at S.T.P. $= \frac{VN_A}{22.4}$

(4). Number of gram atoms = $\frac{W(g)}{GAM}$ (GAM $\rightarrow$ gram atomic mass)

(5). Number of gram molecules = $\frac{W(g)}{\text{Gram molecular mass}}$

(6). Dilution formula : $M_1V_1 = M_2V_2$

For mixing two solutions of the same substance

$M_1V_1 + M_2V_2 = M_3(V_1 + V_2)$

Molarity can be directly calculated from % by mass (w/w) if density is known

Molarity $= \frac{\% \times 10 \times d}{GMM}$

(7). Mass of 1 atom of element $= \frac{GAM}{N_A}$

(8). Mass of 1 molecule of substance $= \frac{MM}{N_A}$ (MM $\rightarrow$ Molar mass)

(9). $T(K) = T(^{\circ}C) + 273.15$

(10). Relative atomic mass = $\frac{\text{Mass of an atom of the element}}{\frac{1}{12} \times \text{Mass of an atom of carbon (C-12)}}$

(11). Number of molecules in n moles of substance $= n \times N_A$

(12). Mass % of an element in a compound $= \frac{\text{Mass of that element in 1 mole of the compound}}{\text{Molar mass of the compount}} \times 100$

(13). Mass percent $= \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$

Monksays

(14). $\frac{X_B}{1-X_B} = \frac{\text{molality} \times M_A}{1000}$ where M_A - mass of solvent

(15). Molarity (M) $= \frac{\text{No. of moles of solute}}{\text{Volume of solution in litres}}$ mole / L

(16). Avogadro's No. $N_A = 6.022 \times 10^{23}$

(17). $T(^\circ F) = \frac{9}{5} T(^\circ C) + 32$

(18). Molecular mass $= 2 \times$ vapour density

(19). Mole fraction of A $= \frac{\text{No. of moles of A}}{\text{No. of moles of solution}}$

2. STRUCTURE OF ATOM

(1). Wavelength of matter wave

$$\boxed{\lambda = \frac{h}{mv}}$$

$$\lambda = \frac{h}{p}$$

$$\lambda = \frac{h}{\sqrt{2Em}}$$

Where, E = Kinetic energy

(2). Total number of nodes = n – 1

Radial nodes = n – *l* – 1

Angular nodes = *l*

(3). **Number of neutrons** = A – Z

(4). Number of subshells in n^{th} shell = n

Number of orbitals in n^{th} shell = n^2

Monksays

Number of electrons in n^{th} shell = $2n^2$

Number of orbitals in subshell = $2l + 1$

Number of electrons in subshell = $2(2l + 1)$

(5). Energy of quantum of radiation according to Planck's quantum theory

$$\boxed{E = h\nu}$$

(6). $$\boxed{h\nu = h\nu_0 + \frac{1}{2}m_e v^2}$$

Einstein's photoelectric equation.

(7). **Line spectrum of hydrogen**

$\bar{\nu} = 109677\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) cm^{-1}$ where $\bar{\nu}$ is wave number and $\bar{\nu} = \frac{1}{\lambda}$.

Where $n_1 = 1, 2, ...$

$n_2 = n_1 + 1, n_1 + 2, ...$

(8). **Bohr's model of hydrogen atom**

(a) **Frequency** of radiation absorbed or emitted during transition ; $\nu = \frac{\Delta E}{h}$

$$\nu = \frac{E_2 - E_1}{h}$$

E_1 = Energy of lower energy state

E_2 = Energy of higher energy state.

(b) **Orbit angular momentum** of an electron, $\boxed{m_e vr = n \cdot \frac{h}{2\pi}}$

Where, n = 1, 2, 3,.....

(c) **Energy** of stationary states $\boxed{E_n = -2.18 \times 10^{-18}\left(\frac{Z^2}{n^2}\right) J}$

(d) **Radii** of the stationary states/orbits $\boxed{r_n = 52.9 \times \left(\frac{n^2}{Z}\right) pm}$

Monksays

(9). Energy gap between the two orbits $\boxed{\Delta E = R_H\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)}$

Where $R_H = 2.18 \times 10^{-18}$

Where, n_i = initial orbit

n_f = final orbit

(10). **Atomic number (Z)** = Number of protons in the nucleus of an atom

= Number of electrons in a neutral atom

(11). Heisenberg's uncertainty principle

$\Delta x \times \Delta p \geq \frac{h}{4\pi}$; $\boxed{\Delta x \times m\Delta v \geq \frac{h}{4\pi}}$

(12). Speed of light is equal to the product of frequency and wavelength of light

$\boxed{c = v\lambda}$

(13). **Mass Number (A)** = Number of protons + Number of neutrons

3. STATE OF MATTER

(1). **Van der Waals Equation :** $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

(2). **Combined Gas Equation :** $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$; (n, R constant)

(3). **Dalton's Law of partial pressures :** $P_{Total} = P_1 + P_2 + P_3 + ...$

(4) **Ideal gas equation** : PV = nRT

Monksays

$$M = \frac{dRT}{P}$$

(5) **Charles's Law :** $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ (at constant P and n)

(6). **Avogadro's Law :** V = kn (at constant P and T)

(7). **Partial Pressure in Terms of Mole Fraction :** $P_i = x_i \; P_{total}$

(8). **Boyle's Law :** $P_1 V_1 = P_2 V_2$ (at constant T and n)

(9). Compressibility factor $Z = \frac{PV}{RT}$ (for 1 mole of Gas)

4. THERMODYNAMICS

- **First law of thermodynamics**

  $\Delta U = q + W$

- **Enthalpy of reaction**

  $\Delta H = \sum \Delta_i H^o \text{ (products)} - \sum \Delta_i H^o \text{ (reactants)}$

  By convention heat of formation of every element in its standard state is arbitrarily assumed to be zero.

  $\Delta_{sub} H^o = \Delta_{fus} H^o + \Delta_{vap} H^o$

- **Heat Capacity**

  Specific heat capacity $c_s = \frac{q}{m\Delta T}$

  $C = \frac{q}{\Delta T}$ molar heat capacity $c_m = \frac{q}{n\Delta T}$

- **Energy changes**

  $\Delta U = q_v, \; \Delta H = q_p$

  $q_v \rightarrow$ heat exchange at constant volume

  $q_p \rightarrow$ heat exchange at constant pressure

Monksays

- **Enthalpy**

$$H = U + pV$$

- **Relation between ΔH and ΔU**

$$\Delta H = \Delta U + p\Delta V \quad \text{Or}$$

$$\Delta H = \Delta U + \Delta n_g RT$$

- $C_p - C_v = R$

$$\Delta S = \frac{q_{rev}}{T}$$

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G^o = -RT \ln K$$

Atomicity	γ	Cp	Cv
Monoatomic	5/3	5R/2	3R/2
Diatomic	7/5	7R/2	5R/2
Triatomic non linear	4/3	4R	3R

5. EQUILIBRIUM

(1). $K_w = K_a \times K_b$

(2). $K_i = \frac{[A^+][B^-]}{[AB]}$

(3). $K_{sp} = [A^{y+}]^x \ [B^{-x}]^y$

(4). $\Delta G^o = -2.303\ RT \log\ K$

(5). $K_a = \frac{[A^-][H_3O^+]}{[HA]}$

$$K_b = \frac{[B^-][OH^-]}{[BOH]}$$

(6). $K_w = [H^+][OH^-] = 10^{-16}$ at 25°C

(7). $pH = -\log [H^+]$

(8). $pK_w = pH + pOH = 14$ at 25° C

(9). $K_p = K_c(RT)^{\Delta n}$

(10). Hydrolysis of salts

(i) For a salt of strong acid and weak base

$$pH = \frac{1}{2}[pK_w - pK_b - \log c]$$

(ii) for a salt of weak acid and strong base

$$pH = \frac{1}{2}[pK_w + pK_a + \log c]$$

(iii) For salts of weak acid and weak base

$$pH = \frac{1}{2}[pK_w + pK_a - pK_b]$$

(11). Equilibrium constant, $K_{eq} = \frac{K_a}{K_b}$

(12). K_a or $K_b = c\alpha^2$

(13). Concentration quotient, $Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$

Monksays

6. SOLID STATE

(1). Density of Cubic Crystal System

$$\rho \text{ (or d)} = \frac{Z \times M}{a^3 \times N_A} \text{ g.cm}^{-3}$$

Where, Z = number of atoms per unit cell

N_A = Avogadro's Number

M = Gram atomic weight of element (g. mol^{-1})

a = edge − length

(2). Radius-ratio in ionic solids

Voids	Triangular	Tetrahedral	Octahedral	Cubical
Radius-ratio	$0.155 \le \frac{r^+}{r^-} < 0.225$	$0.255 \le \frac{r^+}{r^-} < 0.414$	$0.414 \le \frac{r^+}{r^-} < 0.732$	$0.732 \le \frac{r^+}{r^-} < 0.999$

(3).

Crystal	Radius of atom (r)	No. of atoms per unit cell	p.f.
Simple cubic	$\frac{a}{2}$	1	52.4 %
F.C.C.	$\frac{a}{2\sqrt{2}}$	2	74 %
B.C.C	$\frac{\sqrt{3}}{4}a$	2	68 %

(4). Packing Fraction (p.f.) = $\frac{\text{Volume occupied by particles (per unit cell)}}{\text{Volume of unit cell}}$

Contribution of each atom present on the corner = $\frac{1}{8}$

Contribution of each atom present on the centre of face = $\frac{1}{2}$

Contribution of each atom present on the body centre = 1

Contribution of each atom present on the edge centre = $\frac{1}{4}$

(5). Seven crystal system with dimensions :

(a) Cubic : $\alpha = \beta = \gamma = 90^{o}$, $a = b = c$

(b) Tetragonal : $\alpha = \beta = \gamma = 90^{o}$, $a = b \neq c$

(c) Orthorhombic : $\alpha = \beta = \gamma = 90^{o}$; $a \neq b \neq c$

(d) Monoclinic : $\alpha = \gamma = 90^{o}$; $\beta \neq 90^{o}$; $a \neq b \neq c$

(e) Hexagonal : $\alpha = \beta = 90^{o}$; $\gamma = 90^{o}$; $a = b \neq c$

(f) Rhombohedral or trigonal : $\alpha = \beta = \gamma \neq 90^{o}$; $a = b = c$

(g) Triclinic : $\alpha \neq \beta \neq \gamma \neq 90^{o}$; $a \neq b \neq c$

7. SOLUTIONS

(1). **Depression in freezing point**

$$\Delta T_f = K_f \times m$$

$$K_f = \frac{RT_0^2}{1000\, l_{fusion}} = \frac{M_A RT_0^2}{1000\, \Delta H_{fusion}}$$

(2) **Dissociation of solute**

$(A)_n \rightarrow nA$

$i = \frac{1 + (n-1)\alpha}{1}$ {α = degree of dissociation}

(3). $P_{sol} = \left.\begin{matrix} P_A^o X_A + P_B^0 X_B \\ (P_B^0 - P_A^0)X_B + P_A^0 \end{matrix}\right\}$ Raoult's law

(4). **Osmotic pressure**

$\pi = CRT$

For isotonic solution, $\pi_1 = \pi_2$

(5) $\frac{1}{P_{sol}} = \frac{Y_A}{P_A^o} + \frac{Y_B}{P_B^o}$

(6). **Relative lowering of vapour pressure**

$\frac{P_A^o - P}{P_A^o} = \chi_B$ $\{\chi_B$ = mole fraction of solute$\}$

(7) $Y_A = \frac{P_A}{P_T}$; $Y_B = \frac{P_B}{P_T}$

Where Y_A and Y_B are mole fractions in vapour phase and $P_A = P_A^o X_A$: $P_B = P_B^o X_B$

(8). **Elevation of boiling point**

$\Delta T_b = K_b \times m$

$K_b = \frac{RT_0^2}{100\ l_{vapour}} = \frac{M_A RT_0^2}{1000\ \Delta H_{vapour}}$

Where, m = molality

M_A = molecular mass of solvent

(9).

Molality (m) = $\frac{\text{Number of moles of solute}}{\text{Weight of solvent (kg)}}$ mole / kg

Molarity (M) = $\frac{\text{Number of moles of solute}}{\text{Volume of solution (L)}}$ mole / L

Monksays

$$\text{Normality (N)} = \frac{\text{Number of gram equivalent of solute}}{\text{Volume of solution (L)}} \text{ g equiv / L}$$

$$\text{Formality (F)} = \frac{\text{Number of gram formula mass}}{\text{Volume of solution (L)}}$$

(10). **Van't Hoff factor**

$$i = \frac{\text{Experimental colligative property (observed)}}{\text{Calculated (normal) colligative property}}$$

$$\frac{P_A^B - P}{P_A^B} = \frac{(i \times n_B)}{(i \times n_B) + n_A}$$

$$\Delta T_b = i \times K_b \times m$$

$$\Delta T_f = i \times K_f \times m$$

$$\pi = iCRT$$

(11). **Association of Solute**

$$nA \rightarrow (A)_n$$

$$i = \frac{(1-\alpha)+\frac{\alpha}{n}}{1}$$ {α = degree of association}

(12). **Henry's Law**

$$\rho = K_H.X \begin{cases} p = \text{partial pressure of gas in vapour phase} \\ K_H = \text{Henry's law constant} \\ X = \text{Mole fraction of gas} \end{cases}$$

Monksays

8. ELECTROCHEMISTRY

(1). **Ohm's law** $\Rightarrow$ $V = RI$

$$R = \rho\frac{\ell}{a}$$

V = Potential difference

R = Resistance

I = Current

ρ = Specific resistance (resistivity)

ℓ = length of conductor

a = cross-section area of conductor

(2). Conductance $G = \frac{1}{R}$

Specific conductance (conductivity) $\kappa = \frac{1}{\rho}$

(3). **Cell constant** $\sigma = \frac{\ell}{a}$

$\kappa = G.\sigma$

(4). **Molar conductance** Λ_M (or ϕ_C) $= \frac{1000 \times \kappa}{C \text{ (or } M)}$ {C = concentration of electrolyte in terms of molarity}

(5). **Equivalent conductance** Λ_M (or A_C) $= \frac{1000 \times \kappa}{C(\text{or } N)}$ {C = concentration (normality)}

$\Lambda_M = \Lambda_N \times (n - \text{factor})$

$\Lambda_0 = \lim_{C \to 0} \Lambda_C$ {Λ_0 = equivalent conductance at infinite dilution (or zero concentration)}

(6) For weak electrolyte, $\Lambda_C \propto \frac{1}{\sqrt{C}}$

Monksays

For strong electrolyte, $\Lambda_C = \Lambda_0 - B(C)^{1/2}$ {B = constant}

(7). At infinite dilution, for an electrolyte A_xB_y

$A_xB_y \rightleftharpoons xA^{y+} + yB^{x-}$

$\Lambda^o_{A_xB_y} = x\lambda^o_A + y\lambda^o_B$ {λ^o_A, λ^o_B = equivalent conductance at infinite dilution of cation and anion}

$= x\lambda^o_+ + \lambda^o_-$

$\lambda^o_C = k\mu^o_c$ {μ = ionic mobility, K = 96500 coulomb}

$$\mu = \frac{\text{Ionic velocity}}{\text{Potential gradient}}$$

(8). For a weak electrolyte, $(CH_3COOH \rightleftharpoons CH_3COO^- + H^+)$

Degree of dissociation, $\alpha = \frac{\Lambda_M}{\Lambda_0}, K_{eq} = \frac{C(\Lambda_M / \Lambda_0)^2}{1-(\Lambda_M / \Lambda_0)}$

(9). For solubility of salt $(AgCl \rightleftharpoons Ag^+ + Cl^-)$

$$K_{sp} = \left(\frac{1000\ K}{\Lambda_0}\right)^2$$

(10). For a cell reaction in an electrochemical cell,

$Zn + CuSO_4 \rightleftharpoons Cu + ZnSO_4$

Cell representation $\Rightarrow$ $Zn \,|\, \underset{\text{Anode}(-)}{Zn^{2+}}(aq.) \,||\, \underset{\text{Cathode}(+)}{Cu^{2+}}(aq.) \,|\, Cu$

(11). For half cell reaction $M \rightarrow M^{n+}(aq.) + ne^-$; $E_{ox} = E_{M/M^-}$

$M^{n+}(aq) + ne^- \rightarrow M$; $E_{red} = E_{M^-/M}$

emf of cell, $E^o_{cell} = E^o_{right} - E^o_{left}$

$= \underset{(R.P.)}{E^o_{cathode}} - \underset{(R.P)}{E^o_{anode}}$ {R.P. = Reduction potential}

Monksays

(12). $\Delta G = -nFE_{cell}$

$= - W$

$= -\Delta G^{o} + 2.303\ RT\ \log\ Q$

$\Delta G^{o} = -nFE^{o}_{cell}$

$E_{cell} = E^{o}_{cell} - \frac{2.303\ RT}{nF}\ \log\ Q$

[ΔG = Change in free energy
W = Useful work done
n = Number of electrons exchanged
F = Faraday constant (96500 coulomb)
Q = Reaction quotient]

At room temperature (25°C)

$\Rightarrow$ Nernst's equation $\Rightarrow E_{cell} = E^{o}_{cell} - \frac{0.0591}{n}\ \log\ Q$

(13). For electrode concentration cell, $(\underset{\text{anode}}{Pt, H_2(P_1) \mid H^{+}} \mid\mid \underset{\text{cathode}}{H_2(P_2), Pt})$

$E_{cell} = \frac{0.0591}{2} \log \frac{P_1}{P_2}$ (P = Pressure)

For electrolyte concentration cell $(\underset{\text{anode}(-)}{Cu \mid Cu^{2+}(C_1)} \mid\mid \underset{\text{cathode}(+)}{Cu^{2+}(C_2) \mid Cu})$

$E_{cell} = \frac{0.0591}{2} \log \frac{C_2}{C_1}$

For concentration cells, $E^{o}_{cell} = 0$

(14). At equilibrium, $E_{cell} = 0$ (as $\Delta G = 0$)

(15). Temperature coefficient (T.C.) = $\left(\frac{\partial}{\partial T} E_{cell}\right)_P$

Change in entropy, $\Delta S = +nF \times (T.C.)$ {ΔH = heat of cell-reaction}

$E_{cell} = \left(\frac{-\Delta H}{nF}\right) + T.(T.C.)$

Monksays

T.C. $> 0 \Rightarrow$ Cell-reaction is endothermic and vice-versa.

(16). Faraday's 1st law of electrolysis

m = Zit

m = mass of substance deposited

Z = electrochemical equivalent

i = current

t = time

$$Z = \frac{\text{Atomic mass}}{n \times F}$$

Faraday's 2nd law of electrolysis

$$\frac{m_1}{m_2} = \frac{E_1}{E_2}$$ (E = equivalent weight)

(17). Oxidation potential for half-cell reaction : $M \rightarrow M^{+n} + ne^-$

$$E_{ox} = E^o_{ox} - \frac{2.303}{nF} \log [M^{n+}]$$

Reduction potential for half-cell reaction : $M^{n+} + ne^- \rightarrow M$

$$E_{red} = E^o_{red} + \frac{2.303\ RT}{nF} \log [M^{n+}]$$

9. CHEMICAL KINETICS AND NUCLEAR CHEMISTRY

(1). For a general chemical transformation

$$nA + mB \rightarrow pC + qD$$

$$\text{Rate} = \frac{-1}{n}\frac{d[A]}{dt} = \frac{-1}{m}\frac{d[B]}{dt} = +\frac{1}{p}\frac{d[C]}{dt} = \frac{1}{q}\frac{d[D]}{dt}$$

(2). For elementary chemical reaction

$mA + nB \longrightarrow products$

Theoretical rate $= \frac{dx}{dt} = k[A]^{rm} \times [B]^{n}$

(3). For a general reaction ; $aA + bB \longrightarrow Pr oducts$

Rate $= \frac{dx}{dt} = k[A]^{m} \times [B]^{n}$

Order of reaction w.r.t. A = m overall order of reaction = (m + n)

Order of reaction w.r.t. B = n

(4). Unit of rate constant = $(mole)^{1-n}$ $(litre)^{n-1}$ $time^{-1}$

Where, n = order of reaction

(5) For a zero order reaction ; $A \longrightarrow B$

Rate $= \frac{-d[A]}{dt} = k[A]^{B} = k$ (cons tan t)

$$\boxed{K = \frac{[A_0] - [A]}{K}}$$

(6). For a first order reaction ; $A \longrightarrow B$

Rate $= \frac{-d[A]}{dt} = k[A]$

$$k = \frac{2.303}{t} \log_{10} \frac{[A]_0}{[A]_1} = \frac{2.303}{t} \log_{10} \left(\frac{a}{a - x} \right)$$

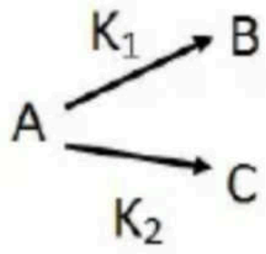
(7). For a zero order reaction, $t_{1/2} = \frac{[A]_0}{2k}$

For a first order reaction, $t_{1/2} = \frac{0.693}{k}$

For an nth order reaction, $t_{1/2} \propto \frac{1}{[A]_0^{n-1}}$ (for $n \geq 2$)

(8). For a parallel reaction

Monksays



$$\frac{-d[A]}{dt} = (k_1 + k_2)[A]$$

(9). For a first order reaction : $A \longrightarrow B + C$, a reagent reacts with all A, B and C

$$k = \frac{2.303}{t} \log_{10}\left(\frac{V_\infty - V_0}{V_\infty - V_t}\right)$$ {V = vol. of reagent}

(10). Temperature coefficient = $\frac{k_{T+10^oC}}{k_T}$

Arrhenius equation, $k = Ae^{-E_a/RT}$

$$\log_{10}\frac{k_2}{k_1} = \frac{E_a}{(2.303\ R)}\left(\frac{T_2 - T_1}{T_2 T_1}\right)$$

$$\log_{10} k = \log_{10} A - \frac{E_0}{2.303RT}$$

$\begin{cases} A = \text{Arrhenius's constant} \\ E_a = \text{Activation energy} \end{cases}$

$$\log_{10}\ k = \log_{10} A - \frac{E_0}{2.303\ RT}$$

(12). Binding energy, B.E. $= \Delta m \times 931.5$ MeV

Δm = mass defect = calculated At. Mass – observed At. Mass

B.E. per nucleon $= \frac{\text{B.E. (total)}}{\text{mass number}}$

1 MeV $= 9.6 \times 10^{10}$ Joule mol^{-1}

(13). Packing fraction. P.F. $= \frac{\text{Isotopic atomic mass-mass no.}}{\text{mass numer}} \times 10^6$

(14). In a radioactive decay, $N_t = N_0 e^{-\lambda t}$

Amount of radioactive substance after 'n' half-life periods

Monksays

$$N = \left(\frac{1}{2}\right)^n N_0$$

(15). Activity, $A_t = A_0 e^{-\lambda t}$, $A = \lambda N$

10. SURFACE CHEMISTRY

(1). **Freundlich Adsorption isotherm**

$$\left(\frac{x}{m}\right) = Kp^{\left(\frac{1}{m}\right)} \; ; \; n \geq 1$$

(2). **Langmuir Adsorption isotherm**

$$\theta = \frac{K'P_A}{1 + KP_A}$$

Or, $\frac{x}{m} = \frac{aP}{1 + bP}$ or $\frac{m}{x} = \frac{1}{aP} + \frac{b}{a}$

(3) $\frac{x}{m} = KC^{\left(\frac{1}{n}\right)}$; C = Concentration of solute in solution.

(4). Zeta potential, $Z = \frac{4\pi\eta\mu}{D}$

11. HYDROGEN

(1) At. Wt. of $H = \frac{\%\ {}^{1}_{1}H \times 1 + \%\ {}^{2}_{1}H \times 2 + \%\ {}^{3}_{1}H \times 3}{100}$

(2) Vapour density of a gas = $\frac{\text{molar mass of gas}}{\text{molar mass of } H_2}$

LHS and RHS both are unitless

(3). Molecular wt. = $2 \times$ (V.D.)

(4). Vol. Strength of H_2O_2 = Molarity $\times$ 11.2

= Normality $\times$ 5.6

12. S-BLOCK ELEMENTS

(1) General Electronic Configuration ns^{1-2}.

(2) Atomic Radii increases down the group.

(3) Hydration enthalpy decreases down the group.

(4). Ionization enthalpy decreases down the group.

(5). On reaction with oxygen give oxide, peroxide and superoxides.

(6) On reaction with water produces hydroxide and hydrogen.

(7). Some important compounds and their general names.

Name	Chemical Formula	Prepared by
Caustic Soda	NaOH	Electrolysis in costner kellner cell
Washing Soda	Na_2CO_3, $10H_2O$	Solvay's process
Baking Soda	$NaHCO_3$	Solvay's process
Glauber's Salt	Na_2SO_4, $10H_2O$	$NaCl + H_2SO_4$
Microcosmic salt	$Na(NH_4)HPO_4$	$NH_4Cl + Na_2HPO_4$

Potash or Pearl Ash	K_2CO_3	Leblanc Process
Caustic potash	KOH	Electrolysis of KCl
Quick lime	CaO	Decomposition of $CaCO_3$
Slaked lime	$Ca(OH)_2$	Hydrolysis of CaO
Gypsum	$CaSO_4$, $2H_2O$	$CaCl_2 + H_2SO_4$
Plaster of Paris	$CaSO_4$, ½ H_2O	By heating gypsum

13. P-BLOCK ELEMENTS

(1). In general metallic character, electropositive character, atomic radii, basic character, reducing character decreases on moving left to right in a period and increases down the group.

(2). Some important compound of p-block elements

i.	Borax (Tincal)	$Na_2B_4O_7, 10H_2O$
ii.	Colomonite	$Ca_2B_6O_{11}, 5H_2O$
iii.	Kemite	$Na_2B_4O_7, 4H_2O$
iv.	Sassolite or boric acid	H_3BO_3
v.	Diborane	B_2H_6
vi.	Borazole	$B_3N_3H_6$
vii.	Alumina	Al_2O_3
viii.	Lithium aluminium hydride	$LiAlH_4$
ix.	Potash Alum	$K_2SO_4, Al_2(SO_4)_3, 24\ H_2O$

14. COORDINATION COMPOUND

(1). Magnetic moment, $\mu = \sqrt{n(n+2)}$ B.M

(2) $\Delta_0(CFSE) = \left[\left(-4 \times n_{\substack{e^- \\ (t_{2g})}}\right) + \left(+6 \times n_{\substack{e^- \\ (e_g)}}\right)\right] Dq$

(3). $\Delta_t(CFSE) = \left[\left(+4 \times n_{\substack{e^- \\ (t_{2g})}}\right) + \left(-6 \times n_{\substack{e^- \\ (e_g)}}\right)\right] Dq$

(4). $\Delta_t = \frac{4}{9}\Delta_0$

(5). For the reaction : $M + 4L \rightleftharpoons ML_4$, overall stability constant of complex (ML_4) is $\beta_4 = \frac{[ML_4]}{[M][L]^4}$

(6). $\underset{\text{(of metal atom/ion)}}{\text{EAN}} = Z - O.N. + 2(C.N.)$

StudMonk

15. ORGANIC CHEMISTRY

Relations for the estimation of elements in organic compounds.

$$\% \text{ of } C = \frac{12}{44} \times \frac{\text{Mass of } CO_2 \text{ formed}}{\text{Mass of the compound}} \times 100$$

$$\% \text{ of } H = \frac{2}{18} \times \frac{\text{Mass of } H_2O \text{ formed}}{\text{Mass of the compound}} \times 100$$

$$\underset{\text{(Dumas's method)}}{\% \text{ of N}} = \frac{28}{22400} \times \frac{\text{Volume of } N_2 \text{ at NTP}}{\text{Mass of compound}} \times 100$$

$$\underset{\text{(kjeldahl's method)}}{\% \text{ of N}} = \frac{1.4 \times \text{Normality of acid used} \times \text{Volume of acid used}}{\text{Mass of the compound}}$$

Monksays

$$\% \text{ of } X(\text{Halogens}) = \frac{\text{At. mass of X}}{(108 + \text{At. mass of X})} \times \frac{\text{Mass of AgX formed}}{\text{Mass of the compound}} \times 100$$

$$\% \text{ of S} = \frac{32}{233} \times \frac{\text{Mass of } BaSO_4 \text{ formed}}{\text{Mass of the compound}} \times 100$$

$$\% \text{ of P} = \frac{62}{222} \times \frac{\text{Mass of } Mg_2P_2O_2 \text{ formed}}{\text{Mass of the compound}} \times 100$$

(1). Inductive effect

(+I effect) $CH_3 < C_2H_5 < C_3H_7 < R_2CH < R_3C < -\dot{O}-R$

(-I effect) $R_4N > NO_2 > -CN > SO_3H > COOH > -F > Cl > 1 > -OR > -OH > -NH_2 > -C_6H_5$

(2). Relative basic strength in aqueous medium.

$R = CH_3$	$2^o > 1^o > 3^o > NH_3$
$R = C_2H_5$	$2^o > 3^o > 1^o > NH_3$
Me_2CH^-	$1^o > NH_3 > 2^o > 3^o$
M_2C	$NH_3 > 1^o > 2^o > 3^o$

(3). Stability of free radical

$$\underset{3^o}{Me_3\dot{C}} > \underset{2^o}{Me_2\dot{C}H} > \underset{1^o}{Me\dot{C}H_2} > \underset{1^o}{Me\dot{}}$$

(4) Heat of hydrogenation of alkenes

1-butene > cis-2-butene > trans-2-butene

(5). Leaving nature in Nucleophilic substitution reaction

$$ArSO_3^- > ROSO_2^- > CH_3COO^- > CN^- > O^-H > MeO^- > H^- > R^-$$

(6). Number of product formed during crossed aldol condensation

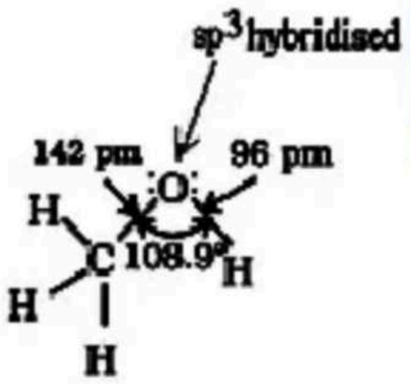
S. No.	Carbonyl compound (I)	Carbonyl compound (II)	Self Condensation Product	Cross Condensation Product	Total Product
1.	One type of similar α H-atom	One type of similar α H-atoms	2	2	4

Monksays

	$CH_3-\overset{\overset{O}{\|}}{C}-H$	CH_3-CH_2-CHO			
2.	Two different types of dissimilar α H-atoms $CH_3-COCH_2CH_3$	One type of similar α H-atoms CH_3-CH_2-CHO	3	3	6
3	Two different types of dissimilar α H-atoms $CH_3CO-CH_2-CH_3$	Two different types of dissimilar α H-atoms $PhCH_2-CO-CH_2-CH_3$	4	4	8

16. ALCOHOLS, PHENOLS AND ETHERS

Structure of Alcohols :



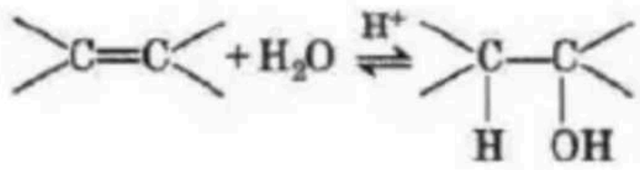
Methanol

(Alcohol)

Preparation of Alcohols

(i) From alkenes

(a) By acid catalysed hydration in accordance with Markownikoff's rule.



(b) By hydroboration-oxidation

Monksays

$$CH_3-CH=CH_2+(H-BH_2)_2 \xrightarrow{B_2H_6,\,H_2O_2/OH^-} CH_3CH_2CH_2OH+B(OH)_3$$

(ii) From carbonyl compounds

(a) By reduction of aldehydes and ketones

$$R-CHO+H_2 \xrightarrow{Pd} RCH_2-OH$$

$$RCOR' \xrightarrow{NaBH_4} R-\underset{\displaystyle OH}{\overset{\displaystyle |}{CH}}-R'$$

(b) By reduction of carboxylic acids and ester

$$RCOOH \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4} RCH_2OH$$

$$RCOOR' \xrightarrow[Catalyst]{H_2} RCH_2OH+R'OH$$

Reduction of aldehyde, ketones and esters with No alcohol is called Bouveault-blanc reduction.

Reaction of carbonyl compound with Grignard Reagent.

$$HCHO+RMgX \longrightarrow RCH_2OMgX \xrightarrow{H_2O} RCH_2OH+Mg\begin{matrix}X\\OH\end{matrix}$$

$$RCHO+R'MgX \xrightarrow{H_2O} R-\underset{\displaystyle OMgX}{\overset{\displaystyle R'}{\overset{|}{CH}}} \xrightarrow[-Mg(OH)X]{H_2O} R-\underset{\displaystyle OH}{\overset{\displaystyle R'}{\overset{|}{CH}}}$$

$$RCOR+R'MgX \longrightarrow R-\underset{\displaystyle OMgX}{\overset{\displaystyle R'}{\overset{|}{\underset{\displaystyle R}{\overset{|}{C}}}}} \xrightarrow[-Mg(OH)X]{H_2O} R-\underset{\displaystyle OH}{\overset{\displaystyle R'}{\overset{|}{\underset{\displaystyle R}{\overset{|}{C}}}}}$$

(iv) Hydrolysis of alkyl halides

$$R-X+KOH_{(aq)} \rightarrow ROH+KX$$

Ease of hydrolysis of alkyl halides RI > RBr > RCl > and 3° > 2° > 1° alkyl halides.

(v) Hydrolysis of ethers

$$R-\ddot{O}-R+H_2O \xrightarrow{H_2SO_4} 2ROH$$

(vi) From primary amines by treatment with nitrous acid

$$RNH_2+HONO \xrightarrow{(NaNO_2+HCl)} ROH+N_2+H_2O$$

Monksays

Methylamine does not give methyl alcohol when treated with HNO_2. It gives CH_3OCH_3 and CH_3ONO.

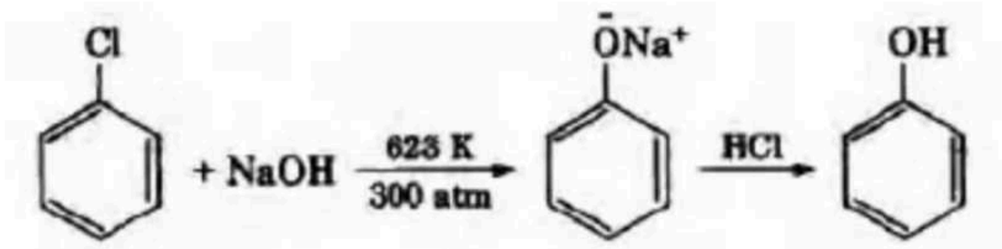
(vii) By alcoholic fermentation

$$\underset{\text{sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{\text{Invertase}} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

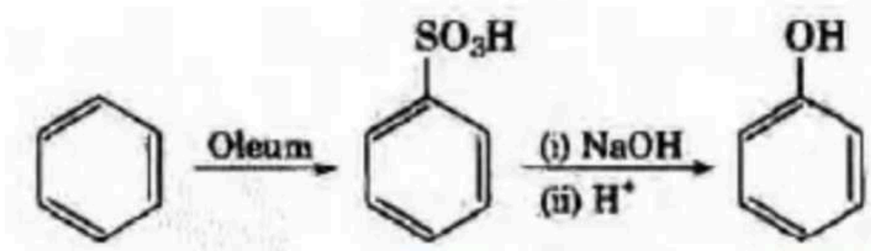
$$\underset{\text{glucose and fructose}}{C_6H_{12}O_6} \xrightarrow{\text{Zymase}} \underset{\text{ethyl alcohol}}{2C_2H_5OH} + 2CO_{2(g)}$$

Preparation of Phenols

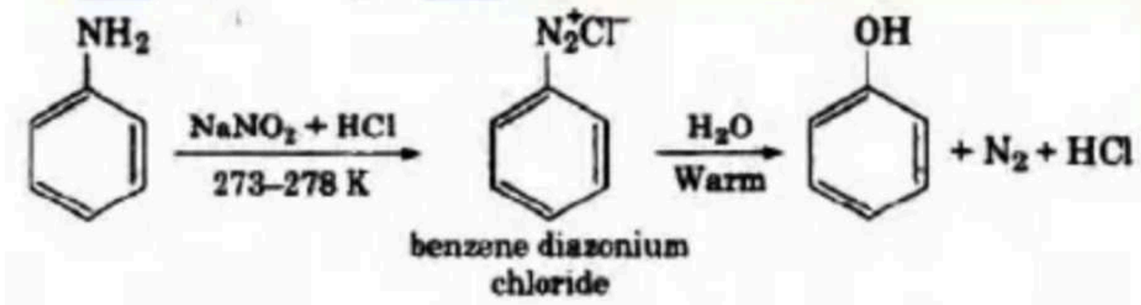
(i) From haloarenes



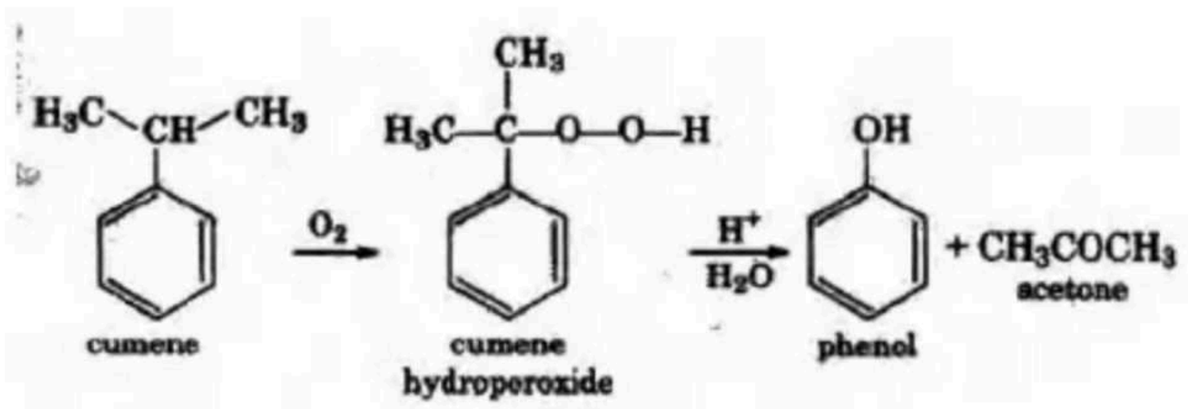
(ii) From benzene sulphonic acid



(iii) From diazonium salts



(iv) From cumene



Chemical properties of alcohols :

1. Reaction involving cleavage of O-H bond :

Monksays

Alcohols react as nucleophiles :

(a) Reaction with metals

$$2R-O-H+2Na \rightarrow \underset{\text{Sodium alkoxide}}{2R-O-Na}+H_2$$

(b) Esterification reaction

$$\underset{\text{Alcohol}}{RO-H}+R'-COOH \overset{H^+}{\Leftrightarrow} ROCOR'+H_2O$$

$$\underset{\text{Alcohol}}{RO-H}+(R'CO_2)O \overset{H^+}{\Leftrightarrow} ROCOR'+R'COOH$$

$$\underset{\text{Alcohol}}{RO-H}+R'COCl \xrightarrow{\text{Pyridine}} R-OCOR'+HCl$$

(b) Reaction with phosphorus halides

$$ROH+PCl_5 \longrightarrow RCl+POCl_3+HCl$$

$$3ROH+PBr_3 \xrightarrow{P/Br_2} 3RBr+H_3PO_3$$

$$3ROH+PI_3 \xrightarrow{P/I_2} 3RI+H_3PO_3$$

(c) Reaction with thionyl chloride

$$ROH+SOCl_2 \longrightarrow RCl+SO_2\uparrow+HCl\uparrow$$

(c) Dehydration reaction

$$\text{Alcohol} \xrightarrow[\text{Or}\\\text{Catalysis (anhyd. } ZnCl_2 \text{ or alumina)}]{\text{Protic acids (conc. } H_2SO_4 \text{ or } H_3PO_4)} C=C+H_2O$$

(d) Oxidation reaction

$$\text{Alcohol} \xrightarrow{\text{Acidified potassium permanganate}} \text{Carboxylic acid}$$

$$\text{(i) Primary Alchol} \xrightarrow[\text{Or}\\CrO_3\\\text{Or}\\\text{PCC}]{\text{Cu, 573k}} \text{Aldehyde}$$

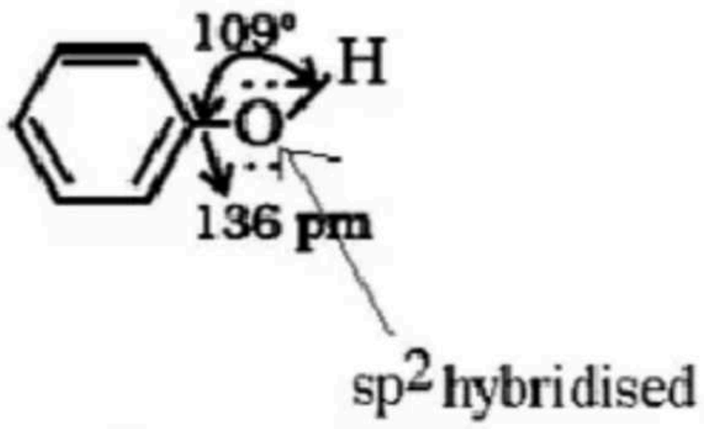
$$\text{(ii) Secondary Alcohol} \xrightarrow[\text{Or}\\CrO_3]{\text{Cu, 573k}} \text{Ketone}$$

$$\text{(iii) Tertiary Alcohol} \xrightarrow[\text{Or}\\KMnO_4]{\text{Cu, 573k}} \text{Alkene}$$

Monksays

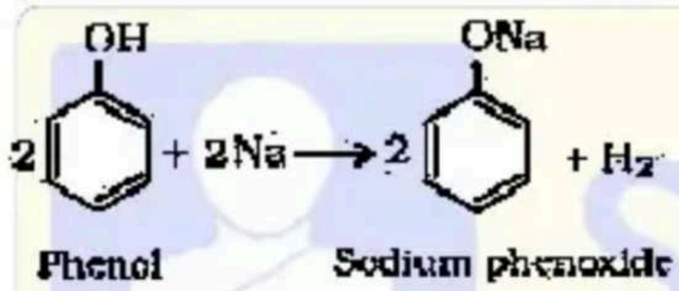
Phenols

Structure of Phenols :



Chemical properties of phenols :

(a) Reaction with metals



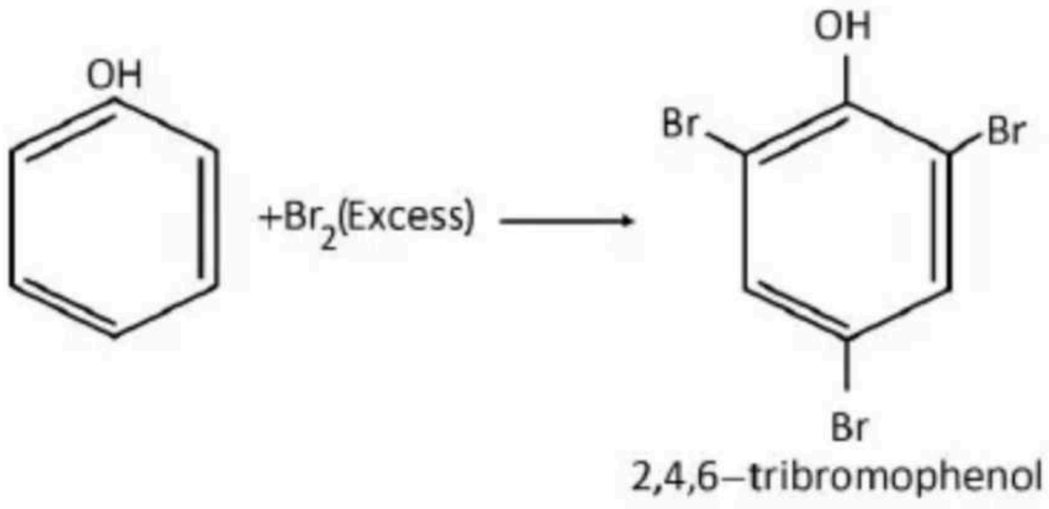
(b) Esterification reaction

$$\underset{\text{Phenol}}{Ar-OH} + R'-COOH \overset{H^+}{\Leftrightarrow} Ar-OCOR' + H_2O$$

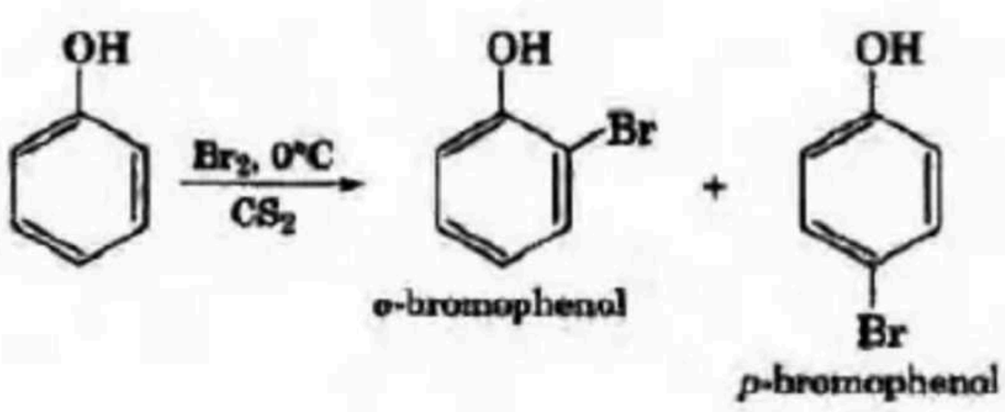
$$\underset{\text{Phenol}}{Ar-OH} + (R'CO)_2O \overset{H^+}{\Leftrightarrow} Ar-OCOR' + R'COOH$$

$$\underset{\text{Phenol}}{Ar-OH} + R'COCl \xrightarrow{\text{Pyridine}} ArOCOR' + HCl$$

(a) Halogenation



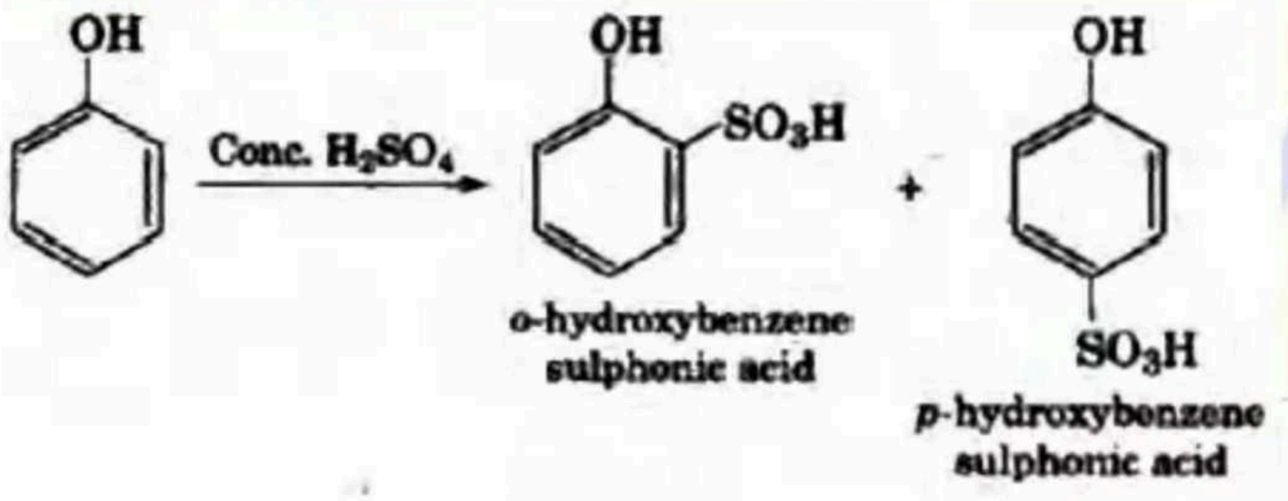
2,4,6–tribromophenol



o-bromophenol

p-bromophenol

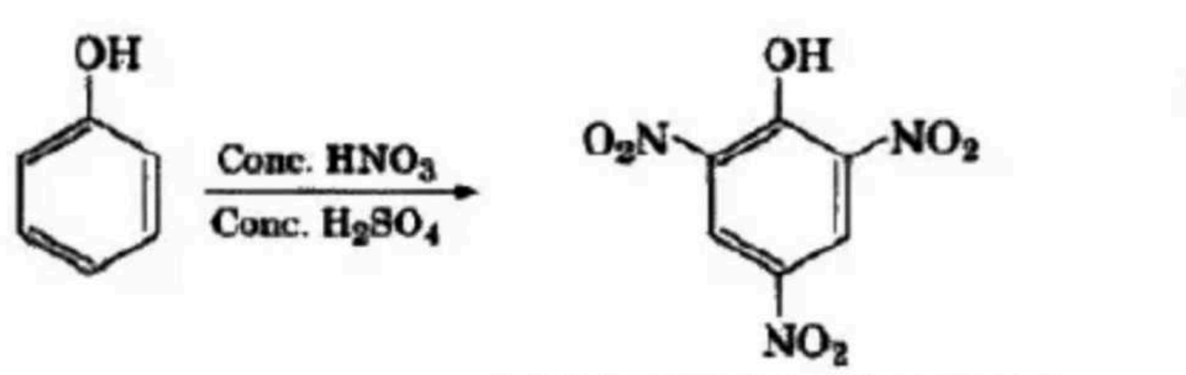
(b) Sulphonation



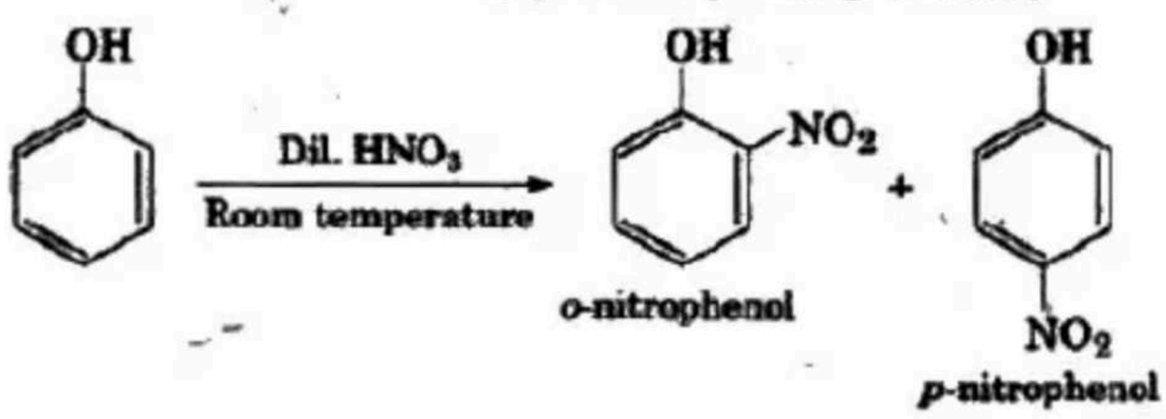
o-hydroxybenzene sulphonic acid

p-hydroxybenzene sulphonic acid

(c) Nitration



2,4,6-trinitrophenol (picric acid)

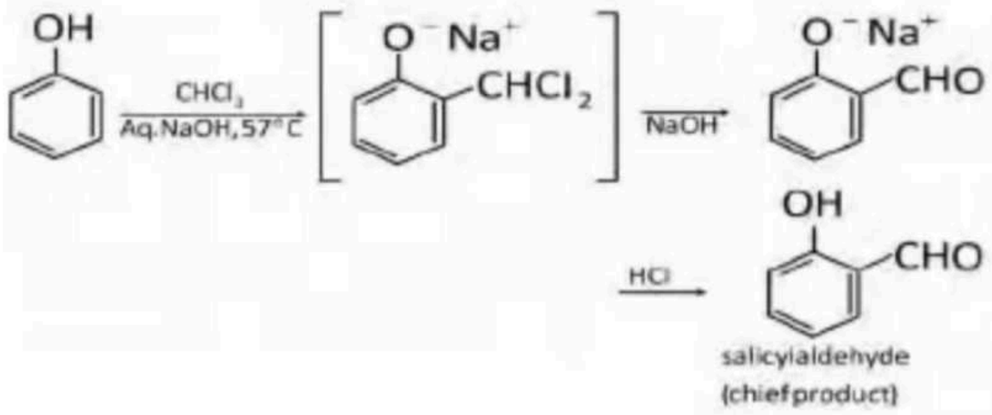


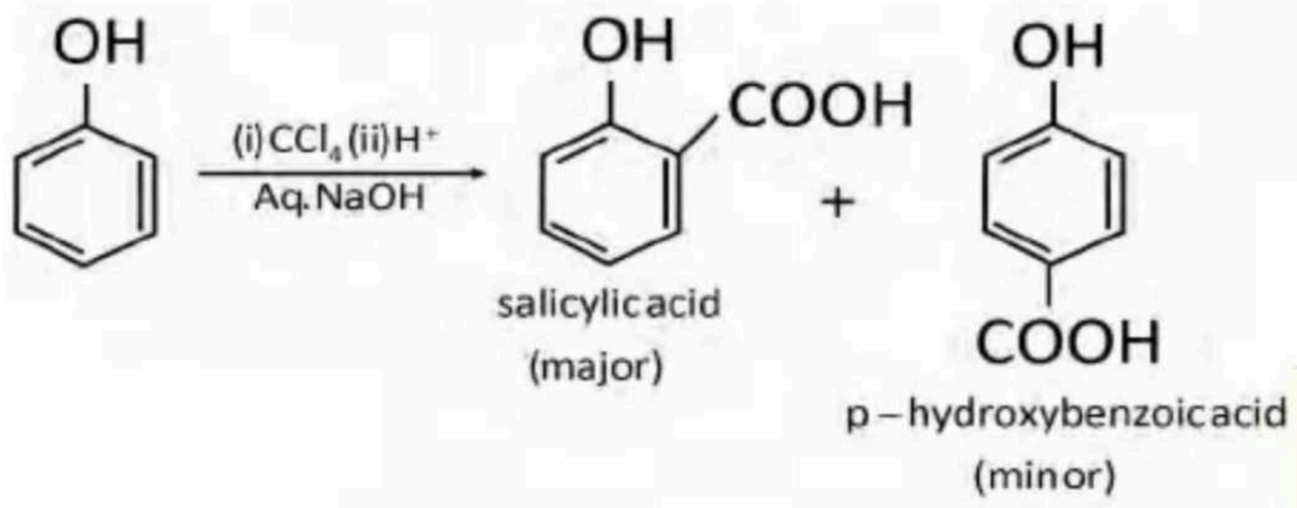
o-nitrophenol

p-nitrophenol

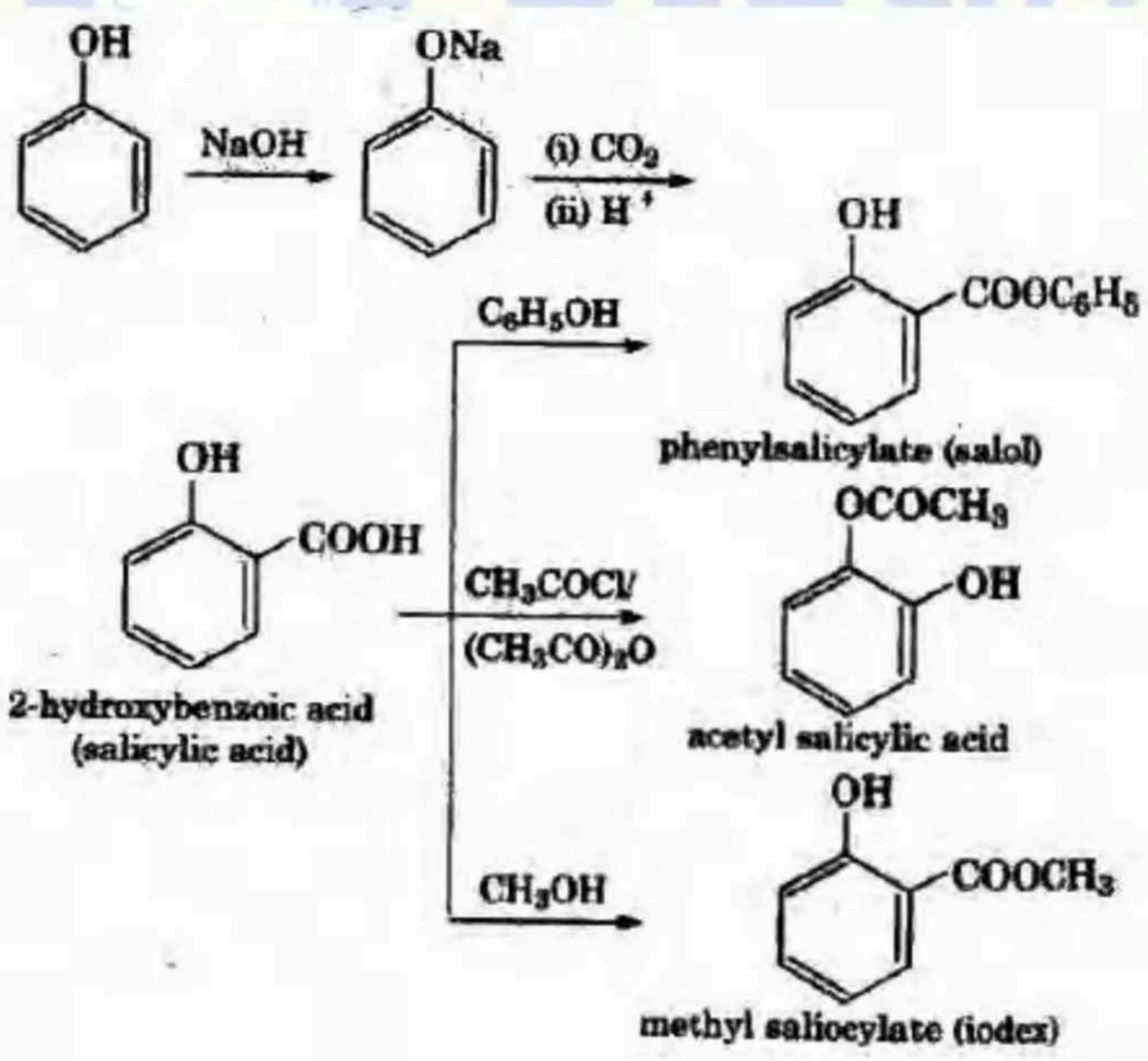
Monksays

(d) Reimer-Tiemann reaction



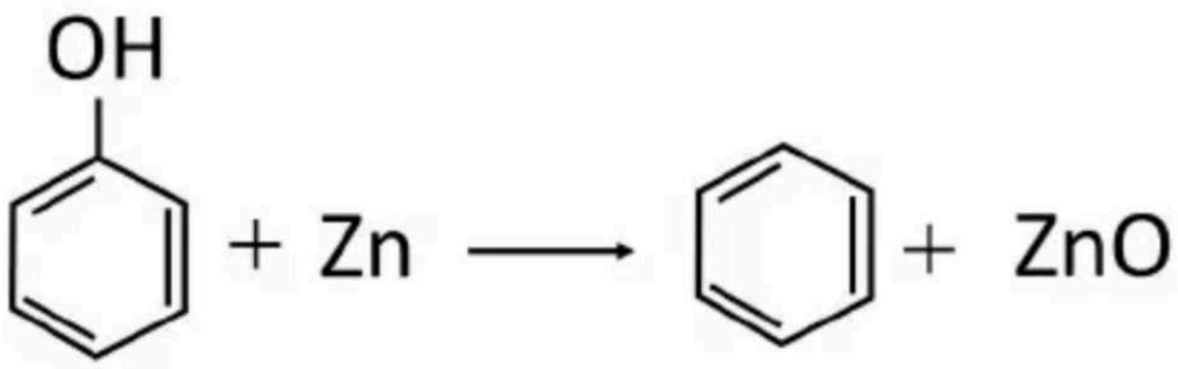


(ii) Kolbe's reaction

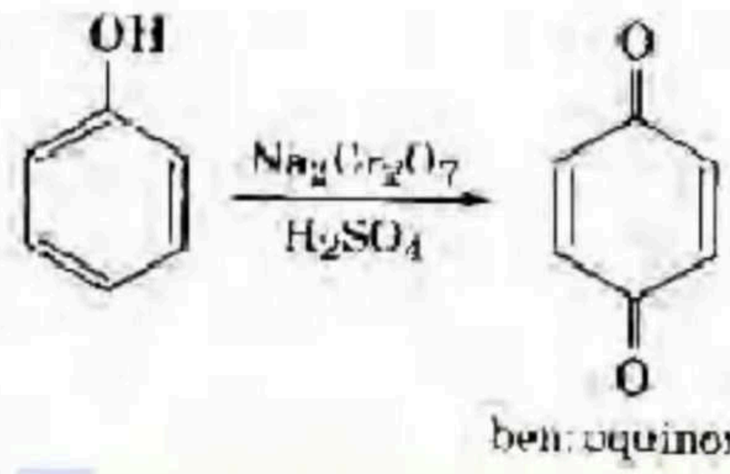


Monksays

iii) Reaction with zinc dust

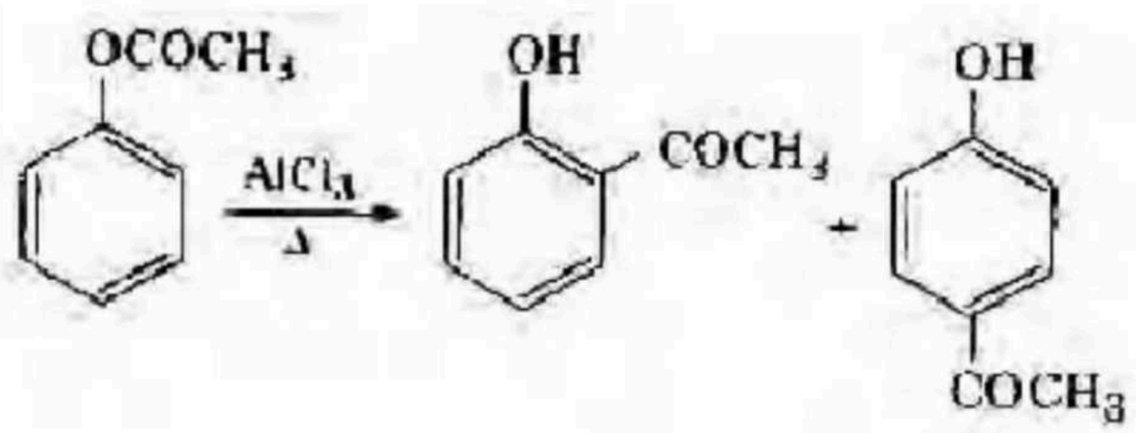


(iv) Oxidation



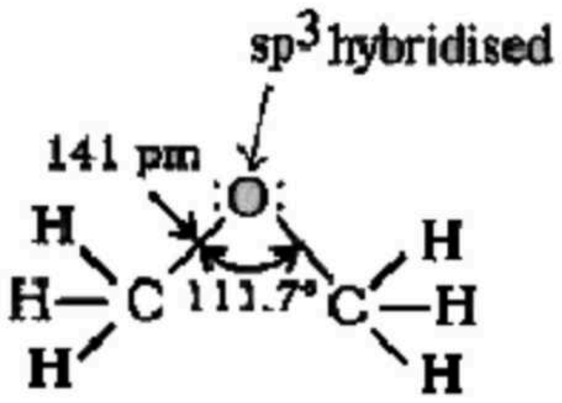
ben: oquinone

(v) Fries rearrangement



Ethers

Structure of Ethers



Methoxymethane

(Ether)

Preparation of Ethers

(i) By dehydration of alcohols

$$\underset{\text{excess}}{2CH_3CH_2-OH} \xrightarrow[418K]{H_2SO_4\,(conc.)} CH_3-CH_2-\ddot{O}-CH_2-CH_3+H_2O$$

Williamson's Synthesis Only primary alkyl halides when react with sodium alkoxide give ether while tertiary alkyl halides give alkene due to steric hindrance.

$$CH_3CH_2Br + CH_3CH_2ONa \longrightarrow CH_3CH_2OCH_2CH_3 + NaBr$$

Chemical Reactions of Ether

$$R-O-R' + HX \longrightarrow RX + R'OR$$

The order of reactivity of hydrogen halides is as follows

HI > HBr > HCl

(ii) Halogenation

$$CH_3CH_2OCH_2CH_3 \xrightarrow[Dark]{Cl_2} \underset{(\alpha-\text{monochloro diethyl ether})}{CH_3CHClOCH_2CH_3}$$

$$C_2H_5OC_2H_5 + \underset{(excess)}{10Cl_2} \xrightarrow[(light)]{h\nu} \underset{(perchlorodiethyl\ ether)}{C_2Cl_5OC_2Cl_5} + 10HCl$$

(iii) Reaction with PCl_5

$$R-O-R + PCl_5 \xrightarrow{\Delta} 2RCl + POCl_3$$

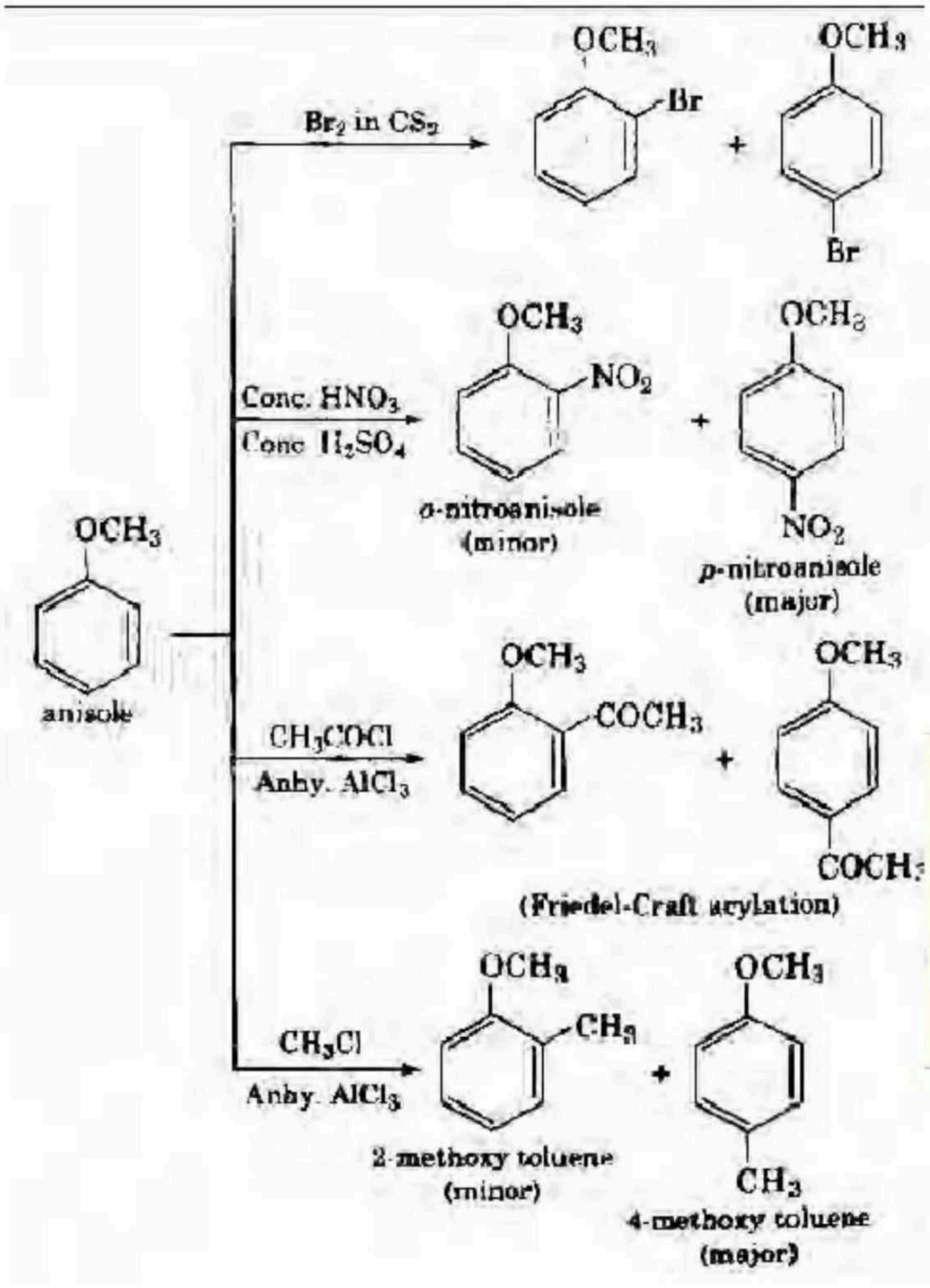
Monksays

(iv) Reaction with CO

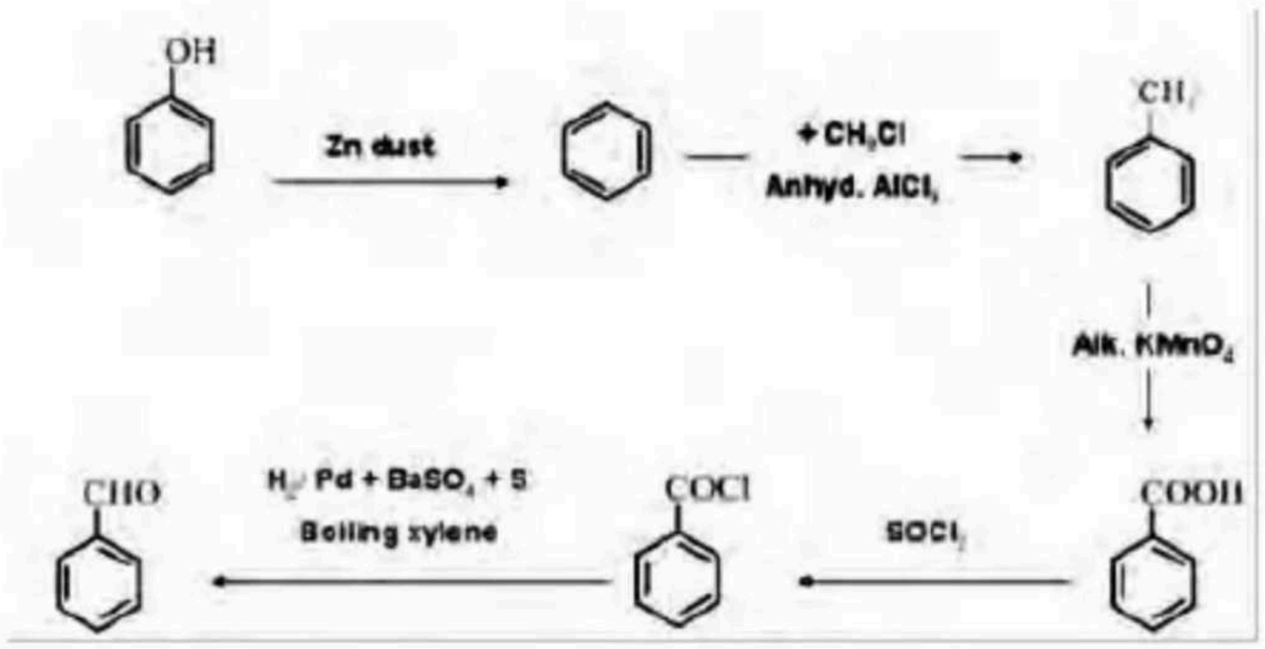
$$ROR + CO \xrightarrow[500\ atm]{BF_3/150^oC} RCOOR$$

Electrophilic Substitution reactions

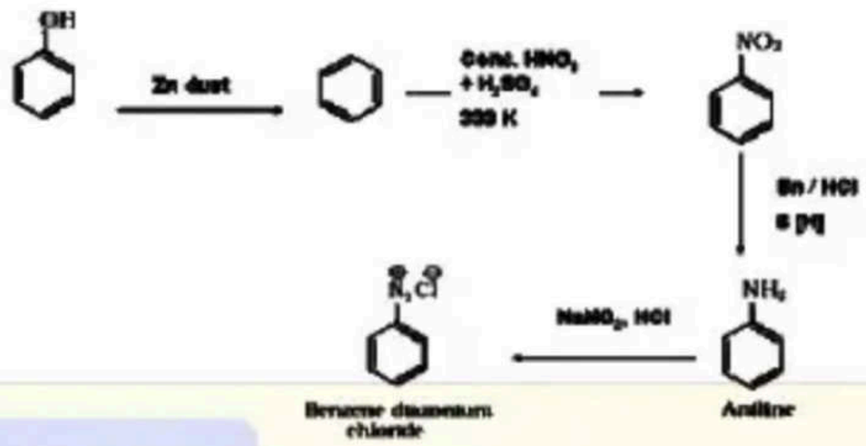
StudMonk



- Other conversion reactions :
  (a) Phenol to salicylaldehyde



(b) Phenol to benzene diazonium chloride



Differentiate between organic compounds :

(a) Alcohols and phenols

Phenol on reaction with neutral $FeCl_3$ gives purple colour whereas alcohols do not give purple colour.

$$6C_6H_5OH + Fe^{3+} \longrightarrow \underset{\text{Purple colour}}{[Fe(OC_6H_5)_6]^{3-}} + 6H^+$$

(b) Primary, secondary and tertiary alcohols

Lucas reagent test :

$$ROH + HCl \xrightarrow{\text{conc. } HCl+ZnCl_2 \text{ /lucas reagent}} RCl + H_2O$$

If it is a primary alcohol, no turbidity appears at room temperature. Turbidity appears only on heating. If it is a secondary alcohol, turbidity appears in 5 minutes. If it is a tertiary alcohol, turbidity appears immediately.

(c) Methanol and ethanol

Iodoform test : Ethanol when reacted with (I_2 and NaOH) or NaOI gives yellow ppt of iodoform since it has the presence of CH_3-CH (OH)-group.

$$C_2H_5OH + 4I_2 + 6NaOH \rightarrow \underset{\text{Yellow ppt}}{CHI_3} + 5\,NaI + 5H_2O + HCOONa$$

$$CH_3OH + I_2 + NaOH \rightarrow \text{No yellow ppt}$$

Monksays

17. ALDEHYDES, KETONES AND CARBOXYLIC ACIDS

Preparation of Aldehydes and Ketones

(i) By oxidation of alcohols

Aldehydes and ketones are generally prepared by oxidation of primary and secondary alcohols, respectively.

$$RCH_2OH \xrightarrow{CrO_3} RCHO$$

$$R-\underset{\underset{OH}{|}}{CH}-R' \xrightarrow{CrO_3} R-\underset{\underset{O}{||}}{C}-R'$$

By dehydrogenation of alcohols

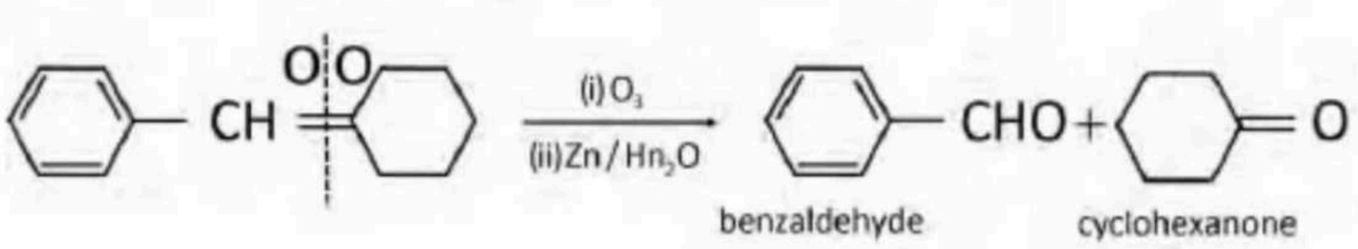
Primary and secondary alcohols give aldehydes and ketones, respectively.

$$R-CH_2-OH \xrightarrow[573\ K]{Cu} RCHO$$

$$R-\underset{\underset{OH}{|}}{CH}-R' \xrightarrow[573\ K]{Cu} R-\underset{\underset{O}{||}}{C}-R'$$

(iii) By ozonolysis of alkenes

$$R-CH=CH-R \xrightarrow[(ii)Zn/Hn_2O]{(i)O_3} 2RCHO$$



(iv) By hydration of alkynes

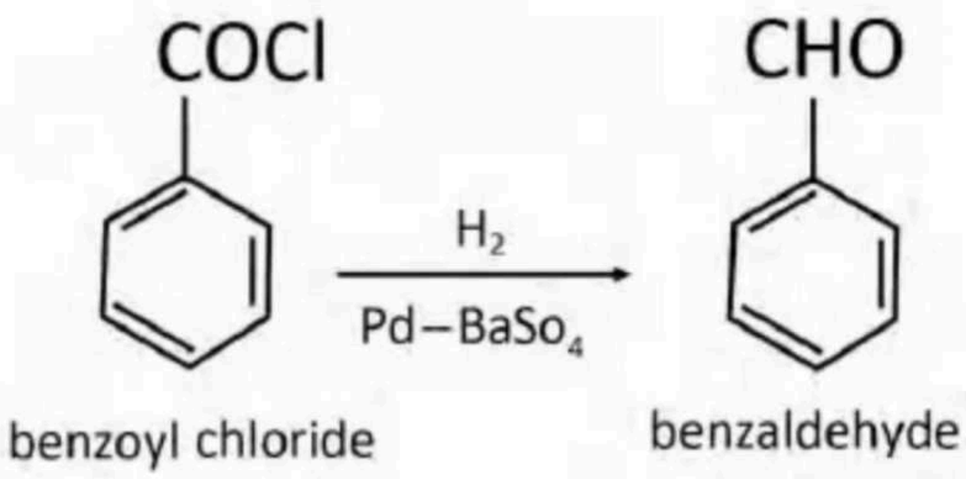
$$\underset{\text{acetylene}}{CH\equiv CH} + H_2O \xrightarrow[H_2SO_4]{HgSO_4} \underset{\text{acetaldehyde}}{CH_3-CHO}$$

$$R-C\equiv CH + H_2O \xrightarrow[H_2SO_4]{HgSO_4} \underset{\text{ketone}}{R-CO-CH_3}$$

Monksays

Preparation of Aldehydes

(i) Rosenmund reduction



benzoyl chloride benzaldehyde

Formaldehyde cannot be prepared by this method as HCOCl is highly unstable.

(ii) From nitriles

$$RCN + SnCl_2 + HCl \longrightarrow RCH = NH \xrightarrow{H_3O^+} RCHO$$

This reaction is called **Stephen reaction.**

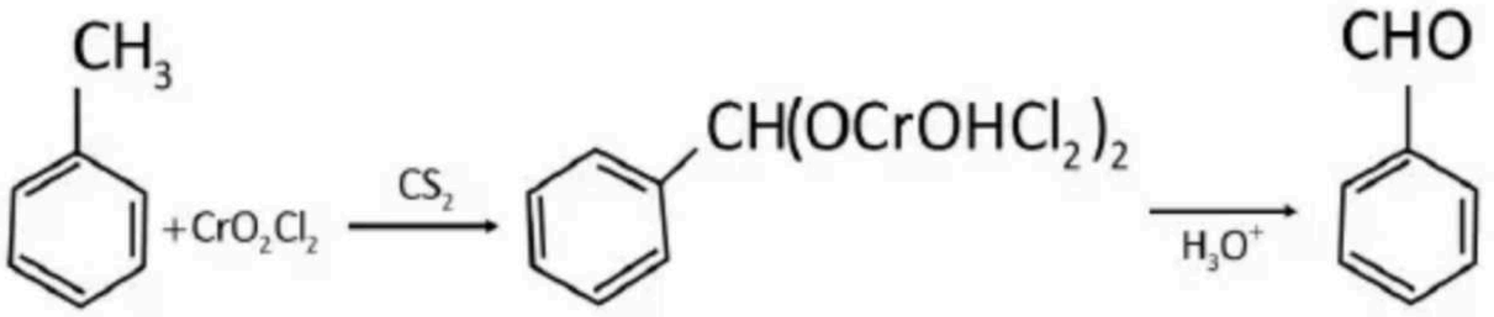
Alternatively, nitriles are selectively reduced by diisobutylaluminium hydride, [DiBAL-H] to imines which on hydrolysis give aldehydes.

$$RCN \xrightarrow[(ii)\ H_2O]{(i)\ AlH(iBu)_2} RCHO$$

Similarly, esters can also be reduced to aldehydes with DiBAL-H.

$$CH_3(CH_2)_3 - COOC_2H_5 \xrightarrow[(ii)\ H_2O]{(i)\ DiBAL-H} CH_3(CH_2)_3 - CHO$$

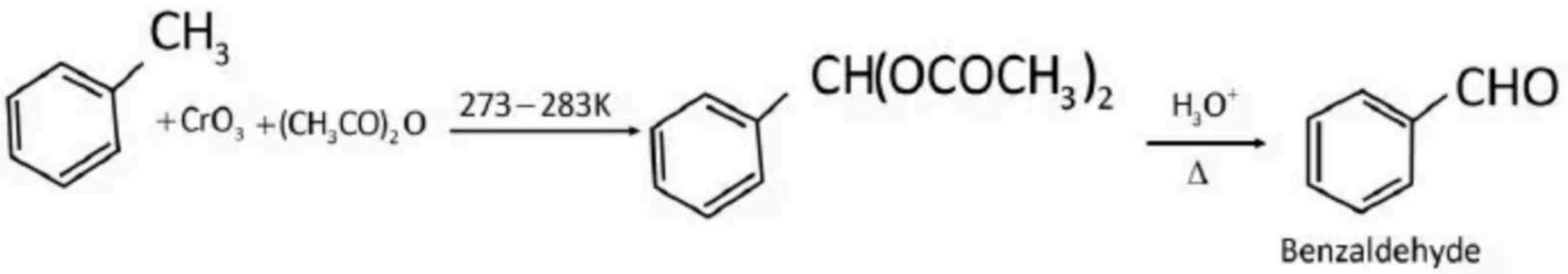
(iii) Etard reaction



toluene benzaldehyde

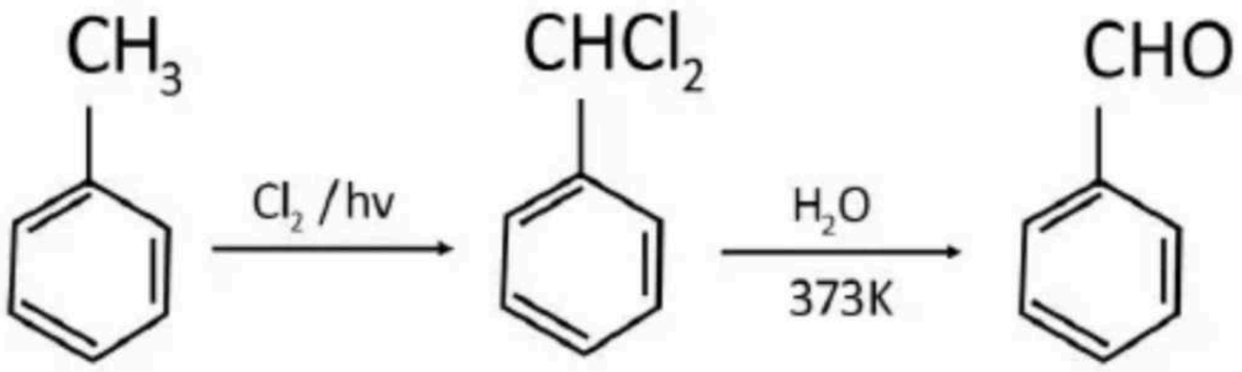
Using chromium oxide (CrO_3) : Toluene or substituted toluene is converted to benzaldehyde in presence of chromic oxide in acetic anhydride.

Monksays

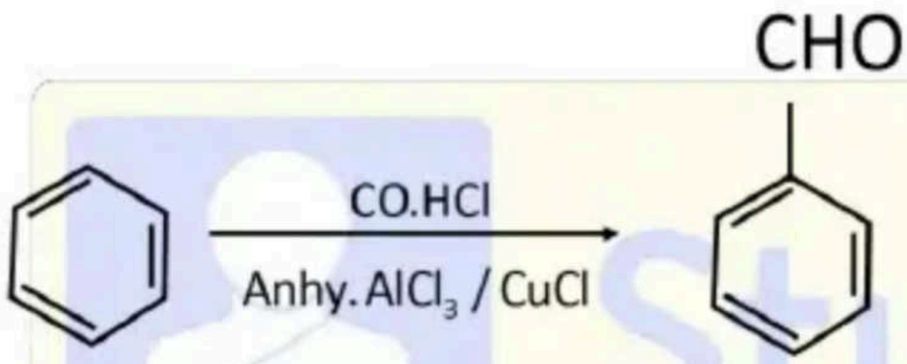


Benzaldehyde

(iv) Side chain halogenation is followed by hydrolysis of toluene

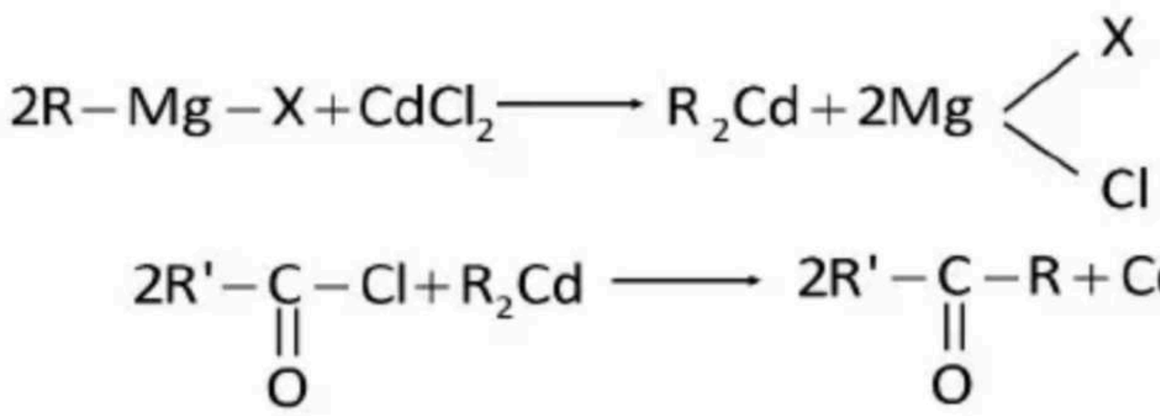


(v) Gattermann-Koch synthesis



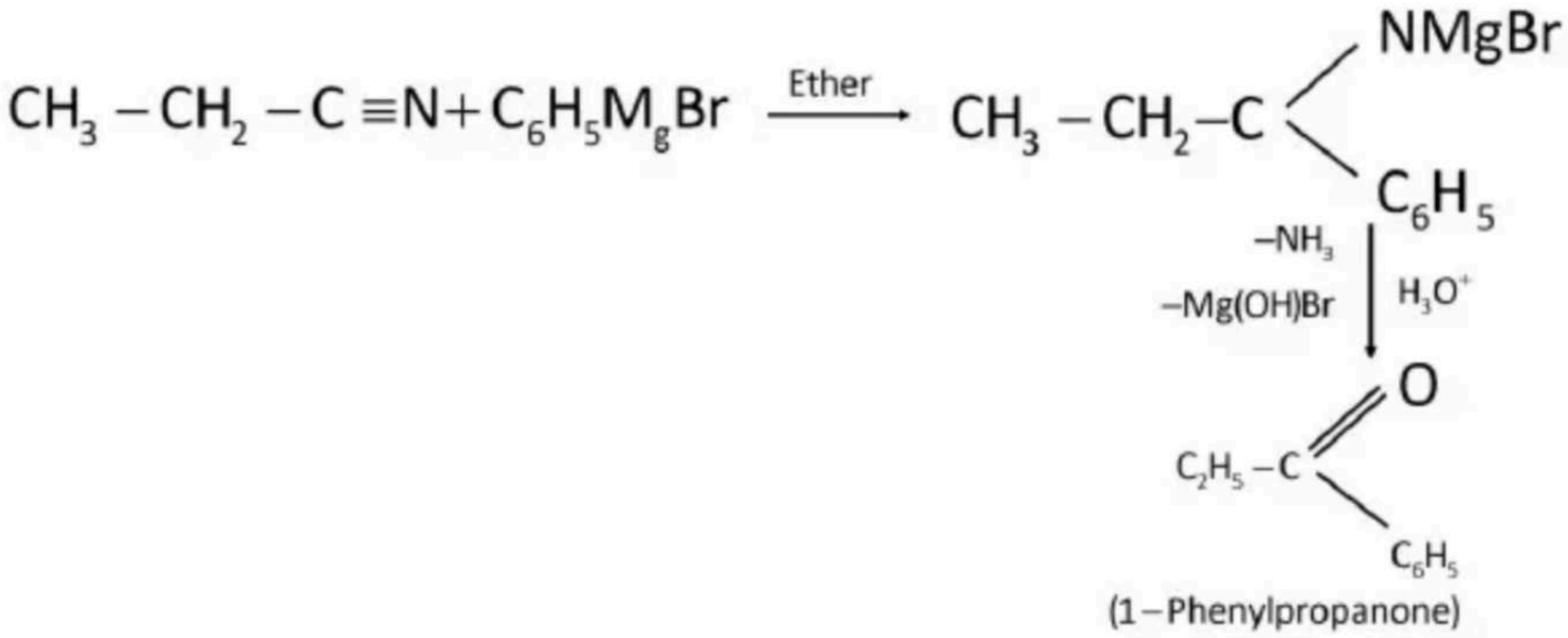
Preparation of Ketones

(i) From acyl chlorides

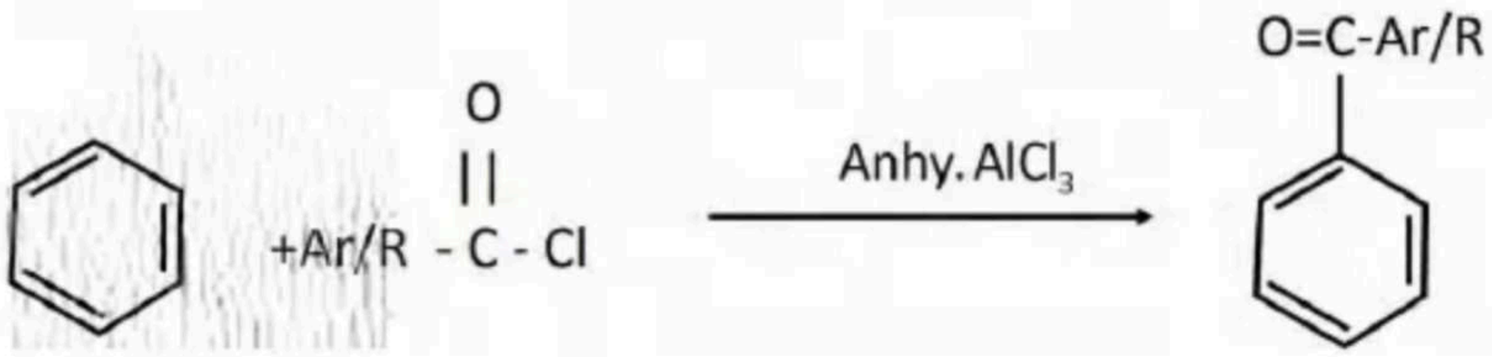


Monksays

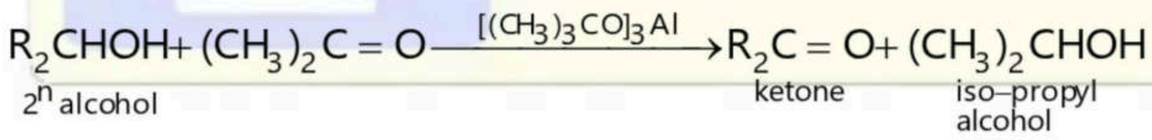
(ii) From nitriles



(iii) Friedel-Crafts acylation

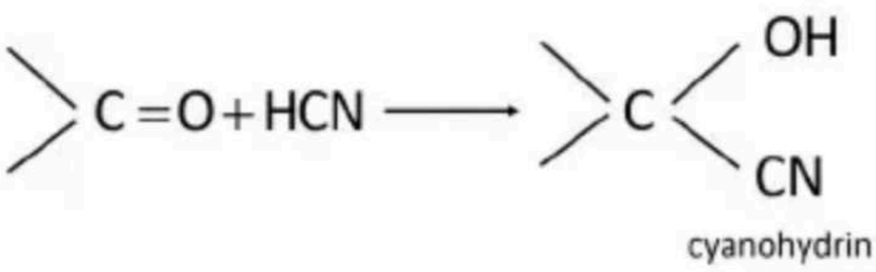


(iv) Oppenauer oxidation



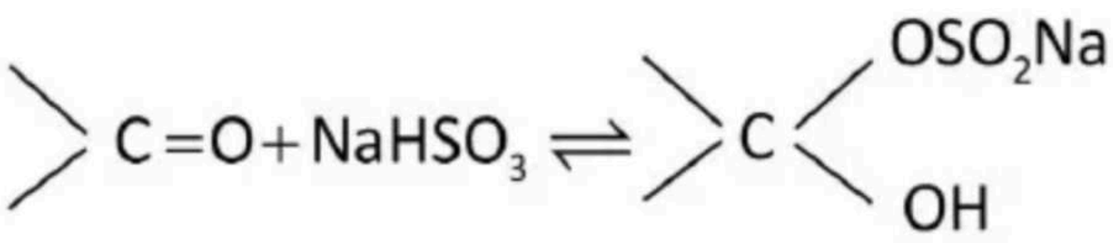
Chemical Reactions of Aldehydes and Ketones

(i) Addition of hydrogen cyanide

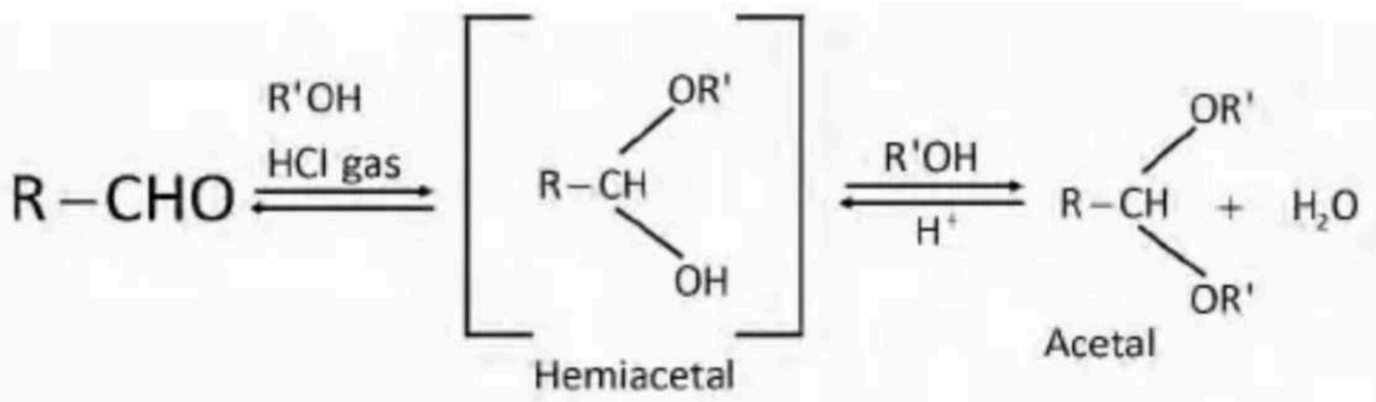


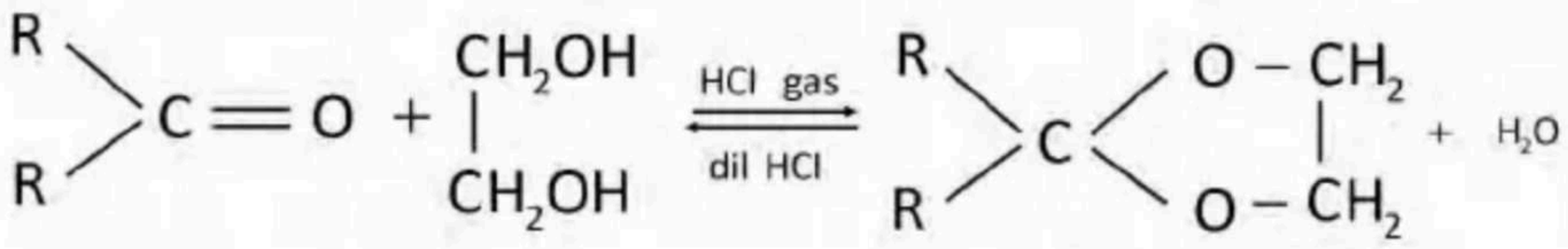
Monksays

(ii) Addition of sodium hydrogen sulphite



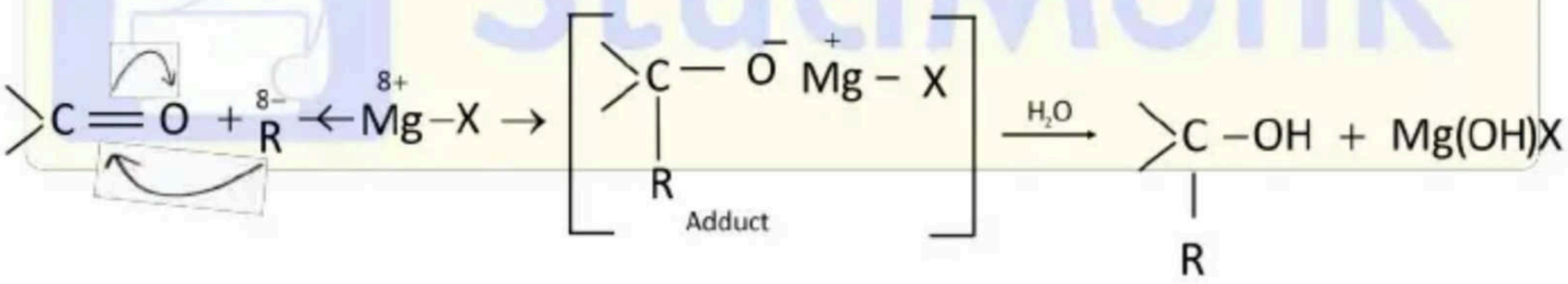
white crystalline solid



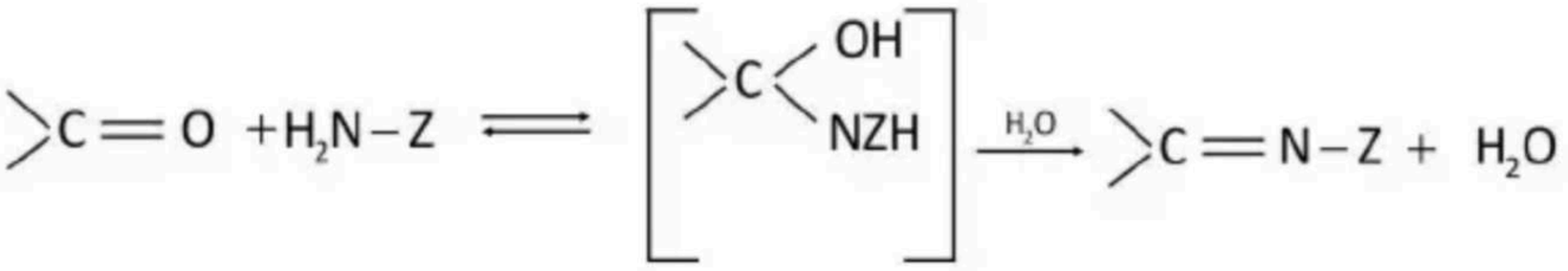


Ethylene glycol ketal

(iii) Addition of Grignard reagent (RMgX) to form alcohol

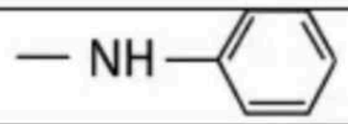
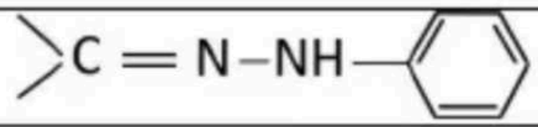
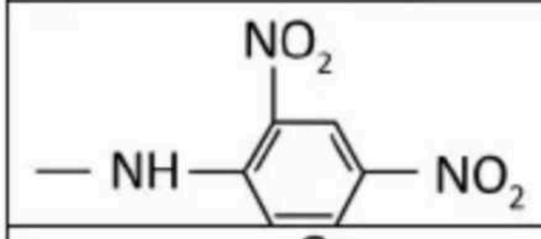
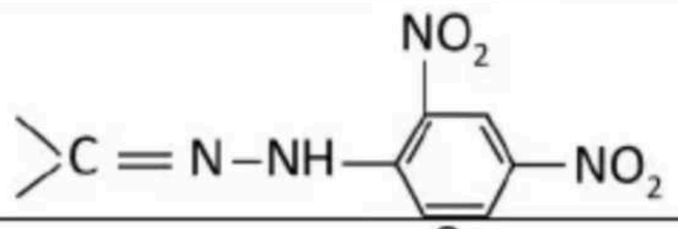
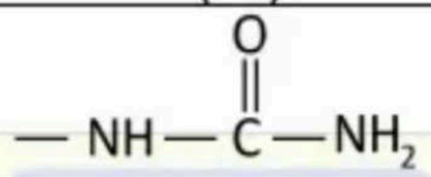
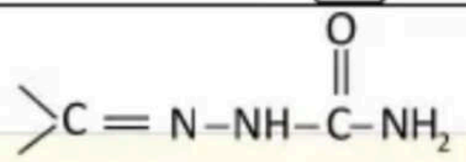


(iv) Addition of ammonia and its derivatives



Z = Alkyl, aryl, OH, NH_2, C_6H_5NH, $NHCONH_2$, etc

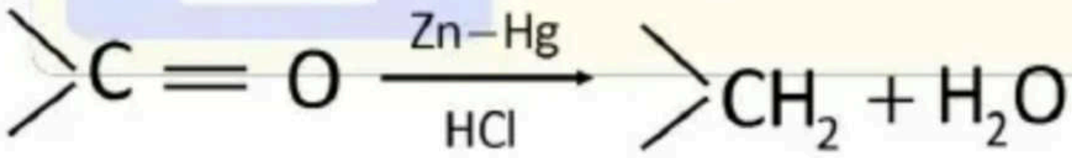
Monksays

Z	Reagent name	Carbonyl derivative	Product Name
-R	Amine	$-C=NR$ |	Substituted imine (Schiff's base)
-OH	Hydroxyl amine	$-C=N-OH$ |	Oxime
-NH_2	Hydrazine	$-C=N-NH_2$ |	Hydrazone
	Phenyl-hydrazine		Phenylhydrazone
	**2, 4-** dinitrophenyl hydrazine		2, 4-dinitrophenyl hydrazone
	Semi-carbazide		Semi-carbazone

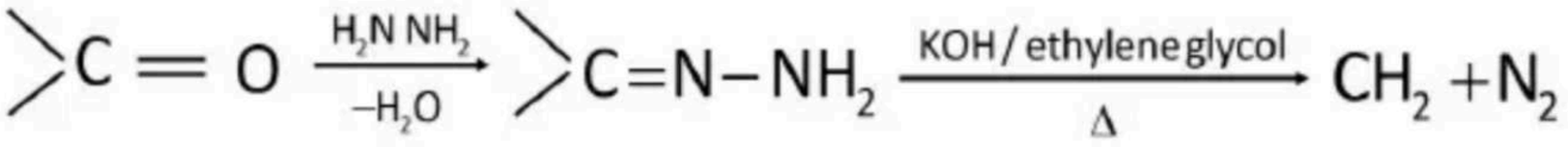
StudMonk

Reduction

(1). Clemmensen reduction

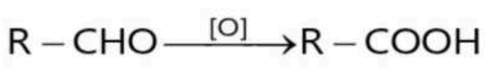


(2). Wolff-Kishner reduction

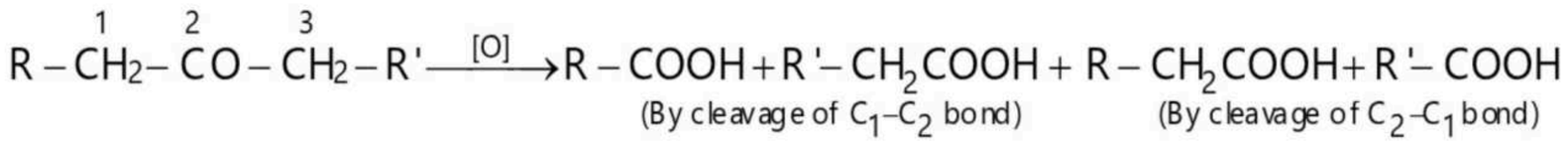


(3). Oxidation

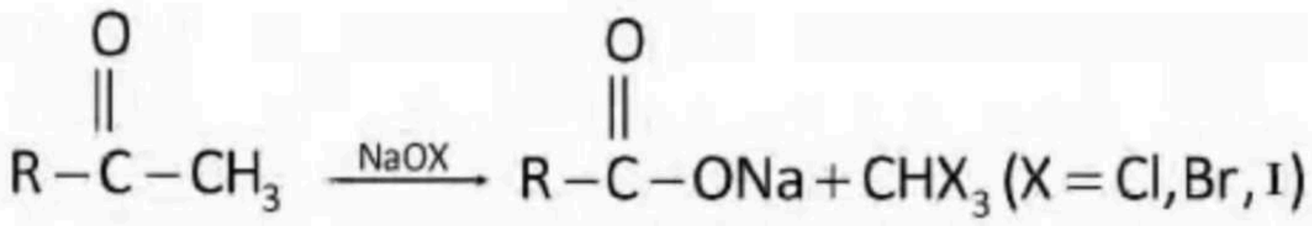
(i) Aldehydes are oxidized to acids in presence of mild oxidizing agents HNO_3, $K_2Cr_2O_7$, $KMnO_4$



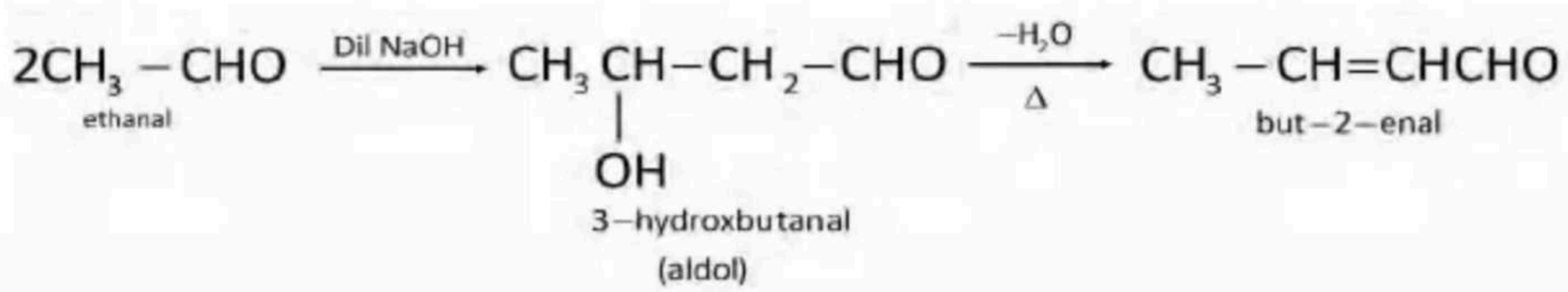
(ii) Ketones are oxidized under drastic conditions i.e. with powerful oxidizing agents like HNO_3, $K_2Cr_2O_7/H_2SO_4$, $KMnO_4/H_2SO_4$ at higher temperature.

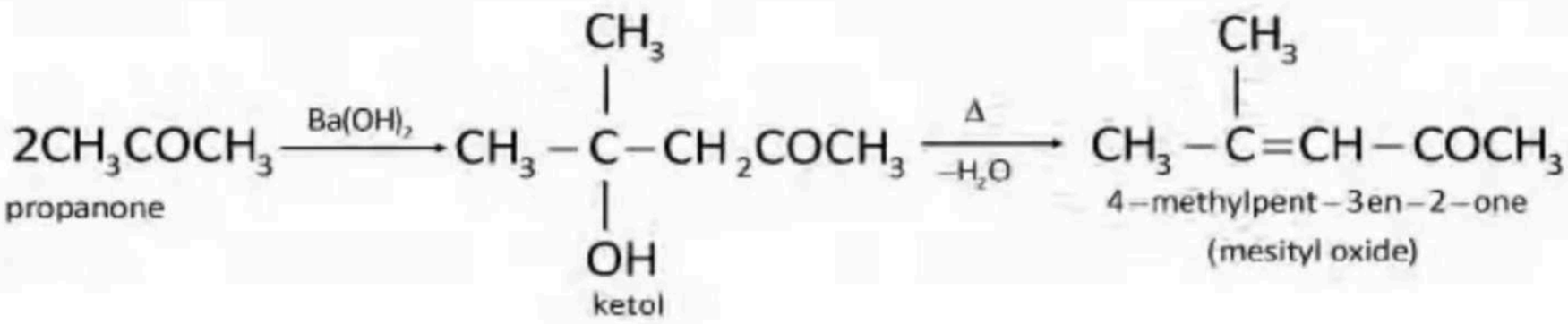


(iii) Haloform reaction :



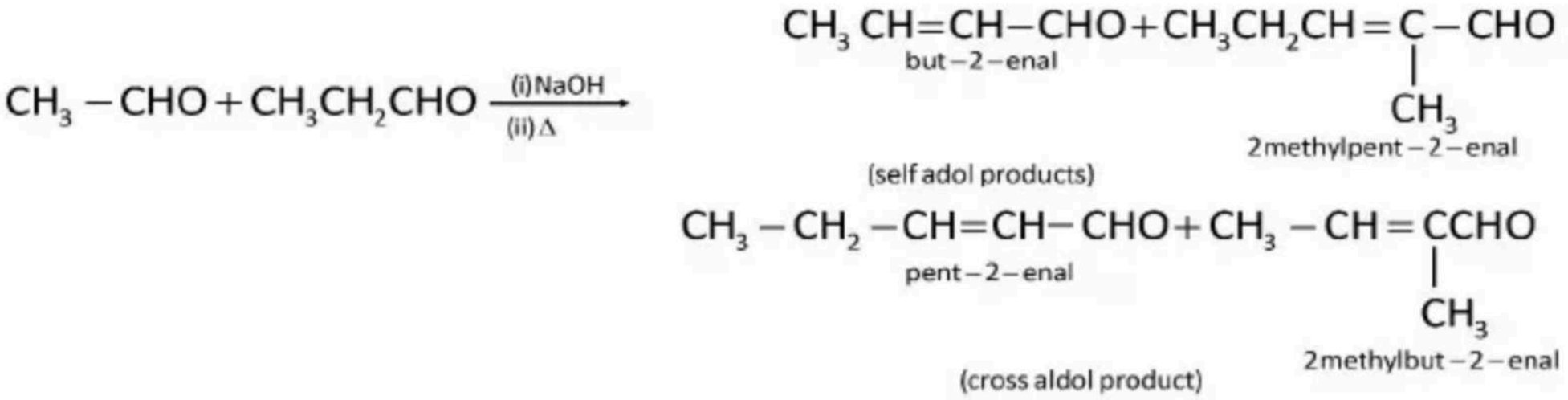
(v) Aldol condensation



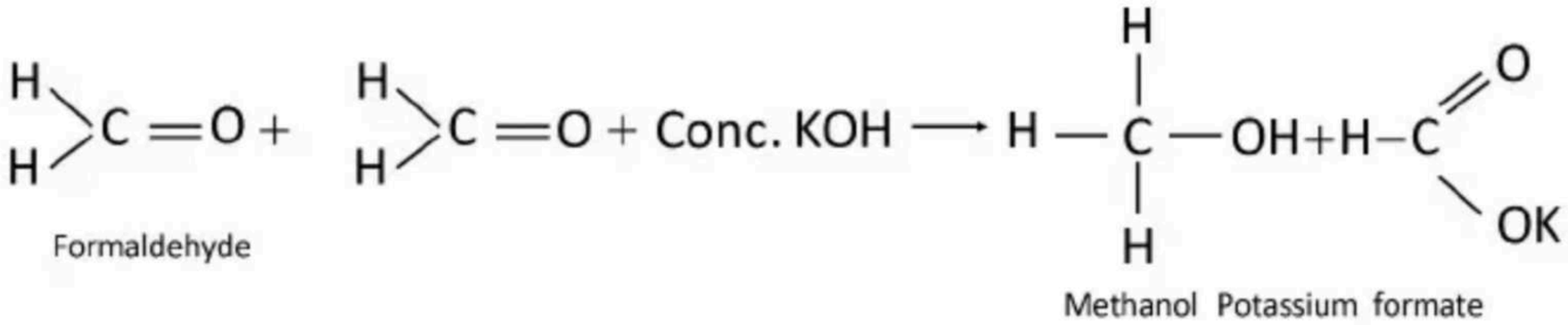


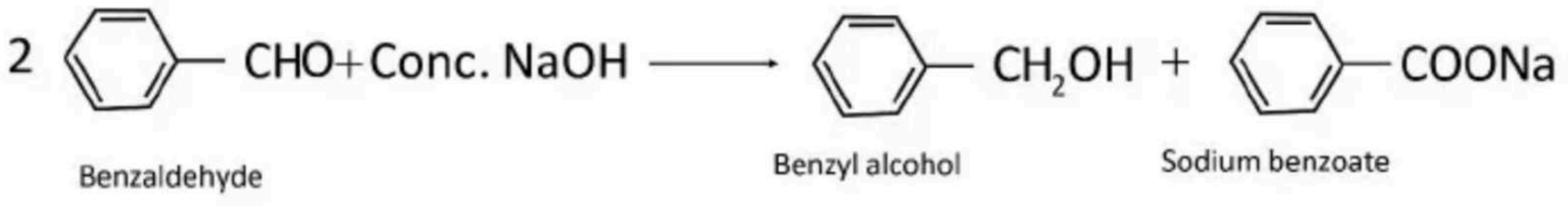
Cross aldol condensation

Base catalysed crossed aldol condensation between an aromatic aldehyde and an aliphatic aldehyde or ketone is called **Claisen-Schmidt condensation or Claisen reaction**.



Cannizzaro reaction





Test to distinguish aldehydes and ketones :

(1) Tollen's test : When an aldehyde is heated with Tollen's reagent it forms silver mirror.

Tollen's reagent is ammoniacal solution of $AgNO_3$

$$RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \longrightarrow RCOO^- + 2Ag + 2H_2O + 4NH_3$$

Ketones do not form silver mirror and hence do not give this test.

(2) Fehling's test : When an aldehyde is heated with Fehling's reagent it forms reddish brown precipitates of cuprous oxide.

Fehling's reagent : Fehling solution A (aqueous solution of $CuSO_4$) + Fehling solution B (alkaline solution of sodium potassium tartarate)

$$R - CHO + 2Cu^{2+} + 5OH^- \rightarrow RCOO^- + \underset{\text{Red–Brown ppt}}{Cu_2O} + 3H_2O$$

Ketones do not give this test.

Carboxylic Acids

Methods of Preparation of carboxylic Acids

From primary alcohols and aldehydes

(a) $RCH_2OH \xrightarrow{\text{alkaline } KMnO_4/H_3O^+} RCOOH$

(b) $RCH_2OH \xrightarrow{CrO_3 - H_2SO_4} RCOOH$

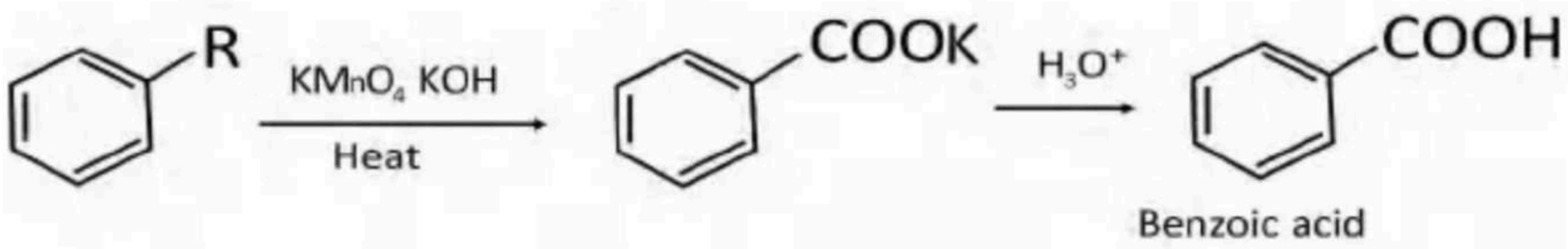
Monksays

From aldehydes

(a) $RCHOH + 2[Ag(NH_3)_2]^+ + 3OH^- \rightarrow RCOO^- + 2Ag + 2H_2O + 4N$

(b) $R-CHO + 2Cu^{2+} + 5OH^- \rightarrow RCOO^- + Cu_2O + 3H_2O$

From alkyl benzenes



From acid derivatives

All acid derivatives like amides ($RCONH_2$), acid halides (RCOCl), esters (RCOOR'), acid anhydrides (RCO-O-COR) on hydrolysis give carboxylic acids. All acid derivatives break from RCO^+.

$$RCOZ \xrightarrow[\text{or dil. NaOH}]{\text{Dil. HCl}} RCOOH$$

$Z = -NH_2, -X(X = Cl, Br, I), OR', RCOO-$ etc.

Ease of hydrolysis : $RCOCl > (RCO)_2O > RCOOR' > RCONH_2$

From nitriles and amides

$$R-CN \xrightarrow{H^+ \text{ or } OH^-/H_2O} R-\underset{\underset{O}{\|}}{C}-NH_2 \xrightarrow{H^+ \text{ or } OH^-/\Delta} RCOOH$$

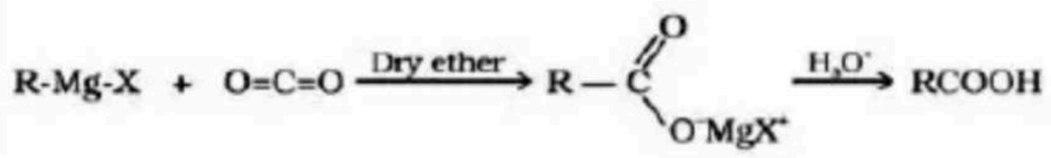
From alkenes

Suitably substituted alkenes are oxidized to carboxylic acids on oxidation with acidic potassium permanganate or acidic potassium dichromate.

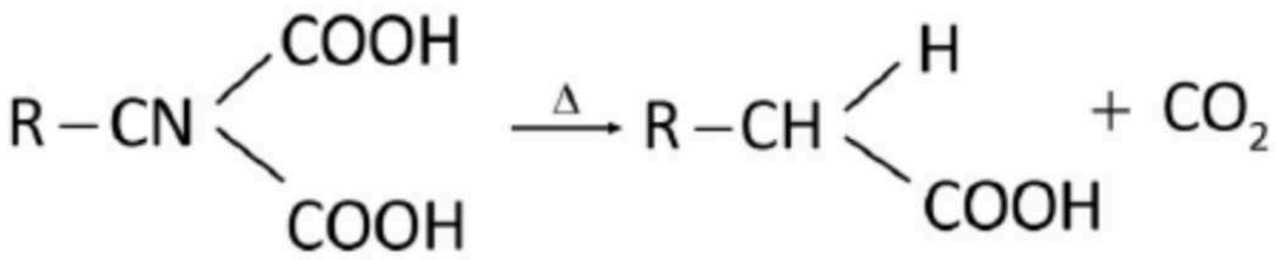
(a) $R-CH=CH-R \xrightarrow{KMnO_4/H^+} 2R-COOH$

(b) $R-CH=CH-R_1 \xrightarrow{KMnO_4/H^+} R-COOH + R_1-COOH$

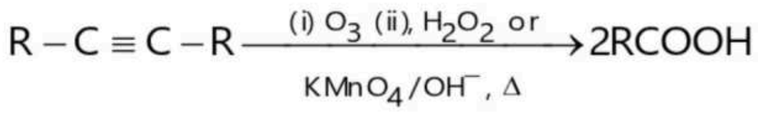
From Grignard reagents



By heating germinal dicarboxylic acids

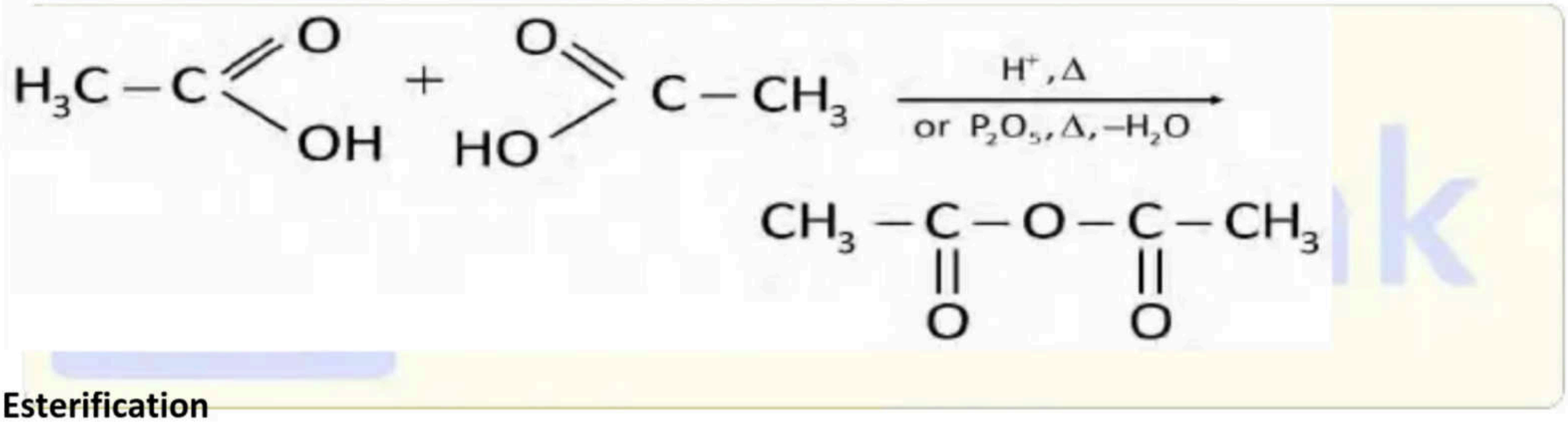


From alkynes

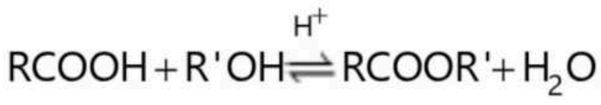


Reactions of Carboxylic acids

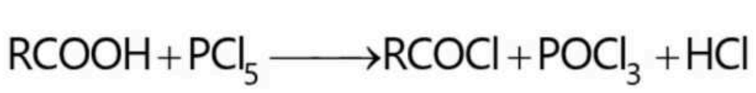
Formation of anhydride

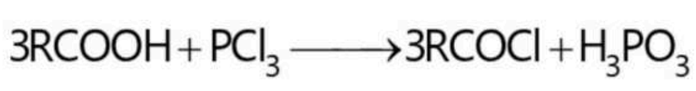


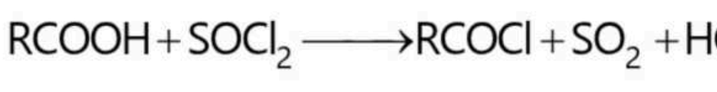
Esterification



Reactions with PCl_5, PCl_3 and $SOCl_2$





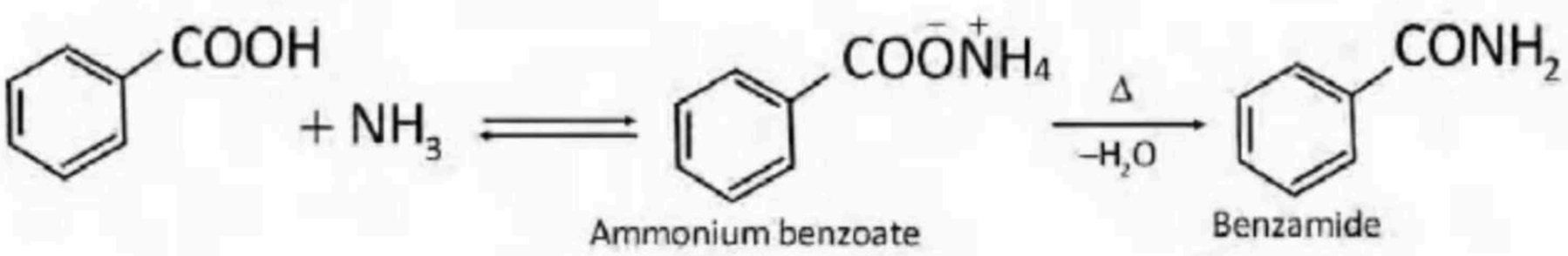


Reaction with ammonia (NH_3)

Carboxylic acids react with ammonia to give ammonium salt which on further heating at high temperature gives amides.

(a) $CH_3COOH + NH_3 \Leftrightarrow \underset{\text{Ammonium acetate}}{CH_3COO^-NH_4^+} \xrightarrow{\Delta/-H_2O} \underset{\text{Acetamide}}{CH_3CONH_2}$

(b)



Chemical reactions involving – COOH group

(a) Reduction

$$RCOOH \xrightarrow[\text{or } B_2H_6 \text{ (ii) } H_3O^+]{\text{(i) } LiAlH_4 \text{/ether}} R - CH_2 - OH$$

(b) Decarboxylation

$$RCOONa \xrightarrow[\Delta]{NaOH,\ CaO \text{ (Ratio 3:1)}} R - H + Na_2CO_3$$

Substitution reactions in the hydrocarbon part

Hell-Volhard-Zelinsky reaction

$$R - CH_2 - COOH \xrightarrow{X_2/\text{Red Phosphorus}/H_2O} \underset{\underset{X}{|}}{R - CH - COOH}$$

$X = Cl, Br$

α–Halocarboxylicacid

Arndt-Eistert reaction

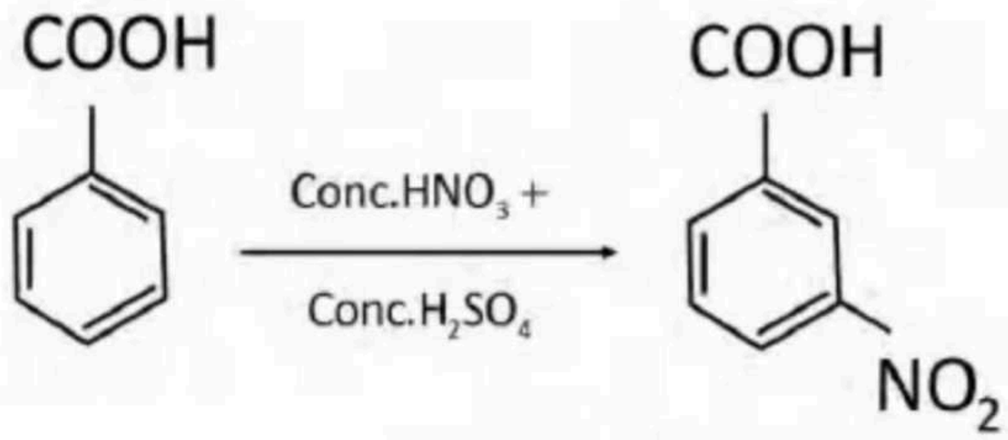
$$RCOOH \xrightarrow{PCl_5} RCOCl \xrightarrow{CH_2N_2} \underset{\text{diazo ketone}}{RCOCHN_2} \xrightarrow{HOH} RCH_2COOH$$

Reducing property

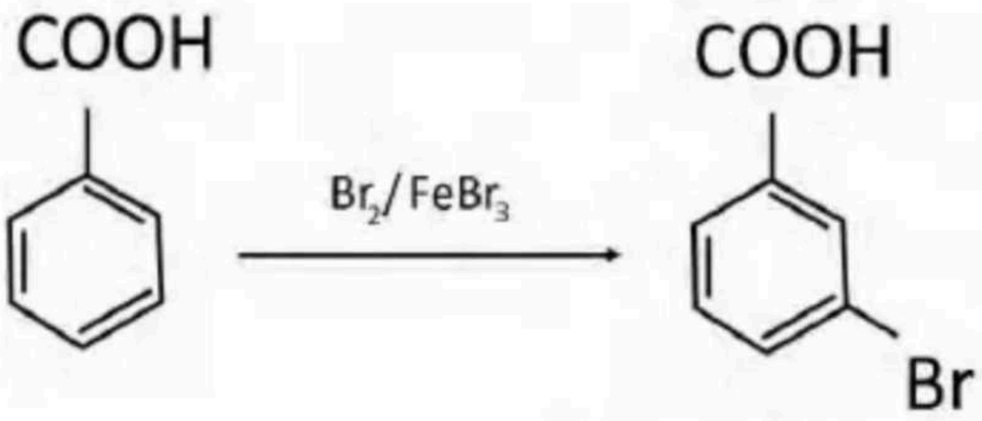
$$HCOOH + HgCl_2 \rightarrow Hg + 2HCl + CO_2$$

Electrophilic substitution reactions of aromatic acids

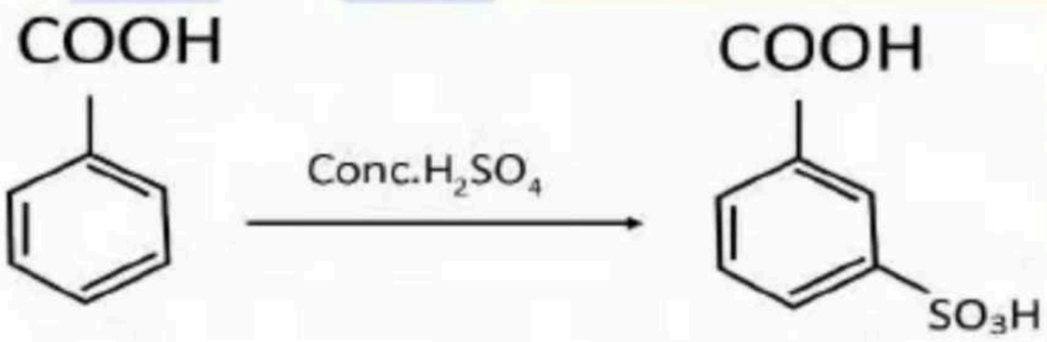
Monksays



m – Nitrobenzoic acid



m – Bromobenzoic acid



3-sulphonic benzoic acid

18. AMINES

Methods of Preparation of Amines

(i) By reduction of nitro compounds :

Nitro compounds can be catalytically reduced by passing hydrogen gas in presence of Raney Ni, finely divided Pt or Pd as catalyst at room temperature.

(a) $R-NO_2+3H_2 \xrightarrow{\text{Ni, Pt or pd}} R-NH_2+2H_2O$

(b) $Ar-NO_2+3H_2 \xrightarrow{Ni,\ Pt\ or\ pd} Ar-NH_2+2H_2O$

Nitro compounds can also be reduced with active metals such as Fe, Sn, Zn etc. with conc. HCl.

(a) $R-NO_2+3H_2 \xrightarrow{Sn/HCl\ or\ Fe/HCl} R-NH_2+2H_2O$

(b) $Ar-NO_2+3H_2 \xrightarrow{Sn/HCl\ or\ Fe/HCl} Ar-NH_2+2H_2O$

(ii) By Hoffmann's method (Ammonolysis of alkyl halides):

$$\underset{(1^o)}{RNH_2} \xrightarrow{RX} \underset{(2^o)}{R_2NH} \xrightarrow{RX} \underset{(3^o)}{R_3N} \xrightarrow{RX} \underset{\text{Quaternary ammonium salt}}{R_4\overset{+}{N}X^-}$$

- The free amine can be obtained from the ammonium salt by treatment with a strong base :

(a) $NH_3+RX \rightarrow RNH_3^+X^- \xrightarrow{NaOH} \underset{(1^o\ \text{amine})}{RNH_2} +H_2O+Na^+X^-$

(b) $RNH_2+RX \rightarrow R_2NH_2^+X^- \xrightarrow{NaOH} \underset{(2^o\ \text{amine})}{R_2NH} +H_2O+Na^+X^-$

(c) $R_2NH+RX \rightarrow R_3NH^+X^- \xrightarrow{NaOH} \underset{(3^o\ \text{amine})}{R_3N} +H_2O+Na^+X^-$

- Order of reactivity of halides is : RI > RBr > RCl

(iii) By reduction of nitriles :

Nitriles can be reduced to amines using H_2/Ni, $LiAlH_4$ or $Na(Hg)/C_2H_5OH$

$$R-C\equiv N \xrightarrow[\text{Or}\ Na(Hg)/C_2H_5OH\ \text{Or}\ LiAlH_4]{H_2/Ni} R-CH_2-NH_2$$

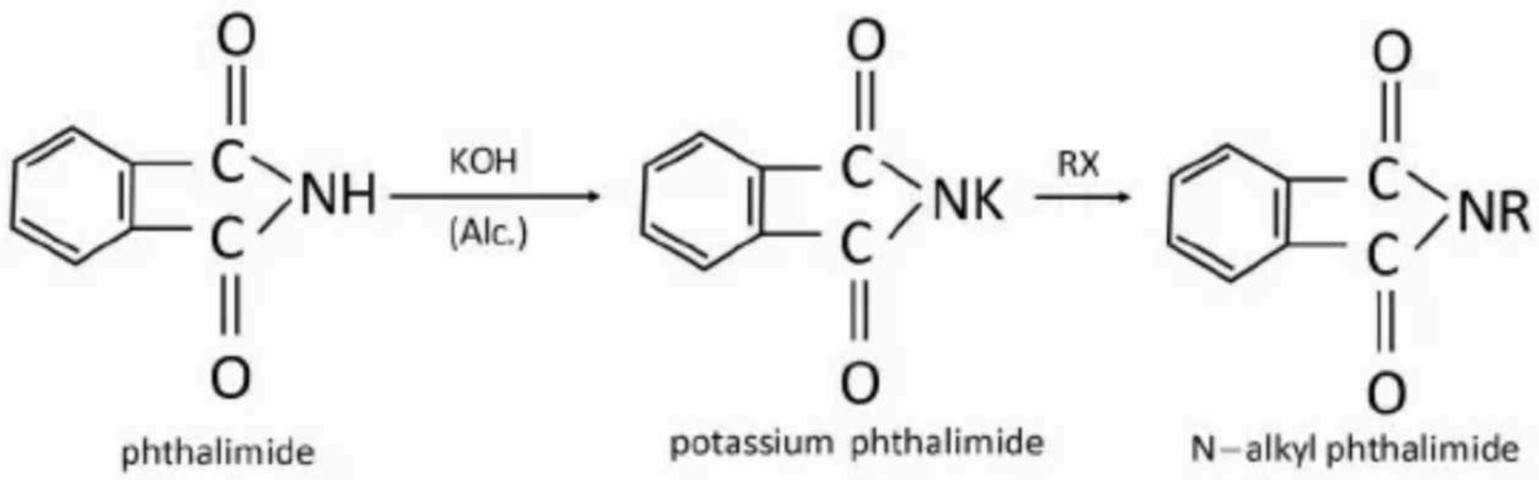
(iv) By reduction of amides :

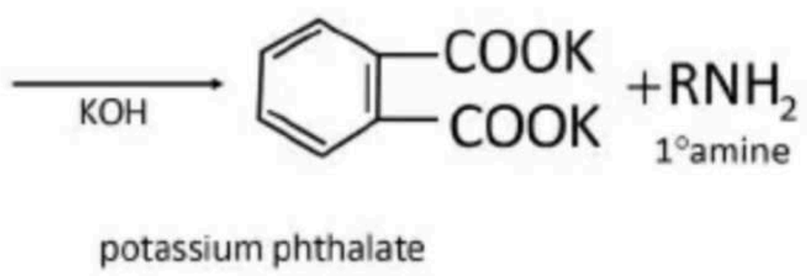
Amides are reduced to corresponding amines by $LiAlH_4$

$$R-\underset{\underset{O}{\|}}{C}-NH_2 \xrightarrow{LiAlH_4/H_2O} R-CH_2-NH_2$$

(v) Gabriel's phthalimide reaction

Monksays



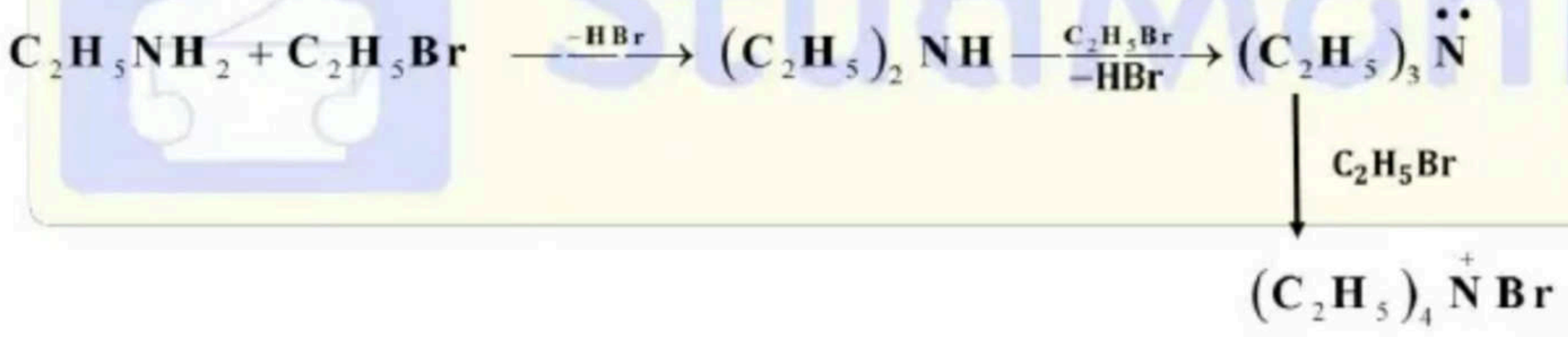


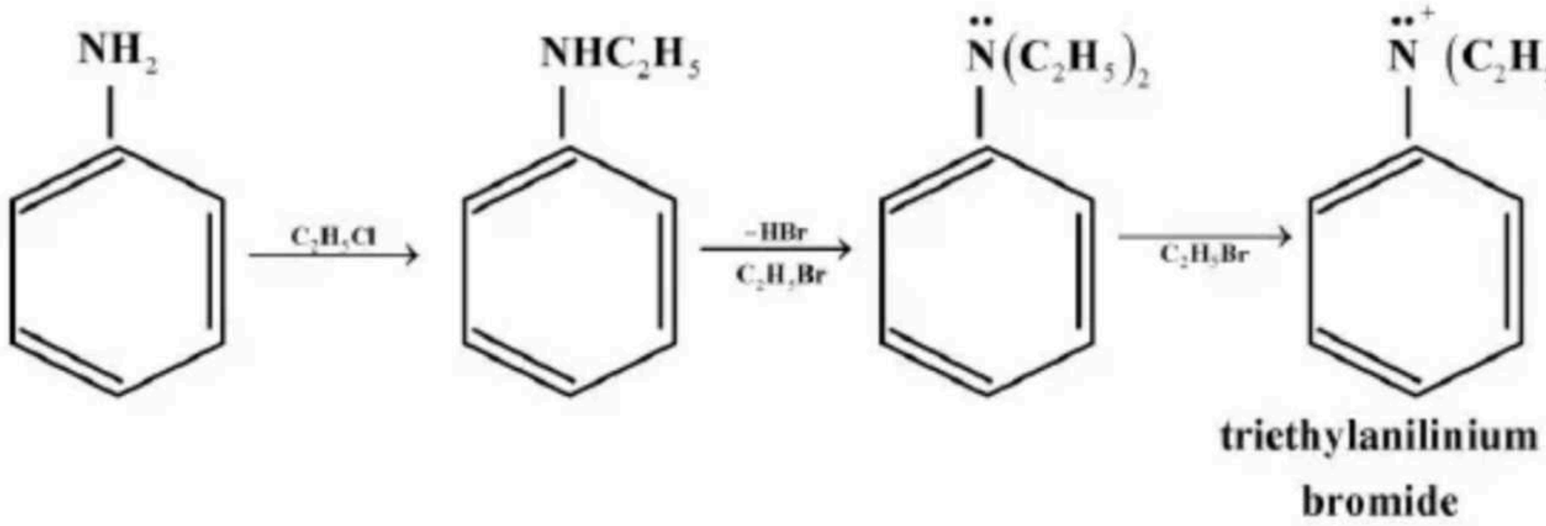
(vi) Hofmann bromamide degradation reaction

$$R-CO-NH_2+Br_2+4NaOH\longrightarrow RNH_2+Na_2CO_3+2NaBr+2H_2O$$

Reactions of Amines

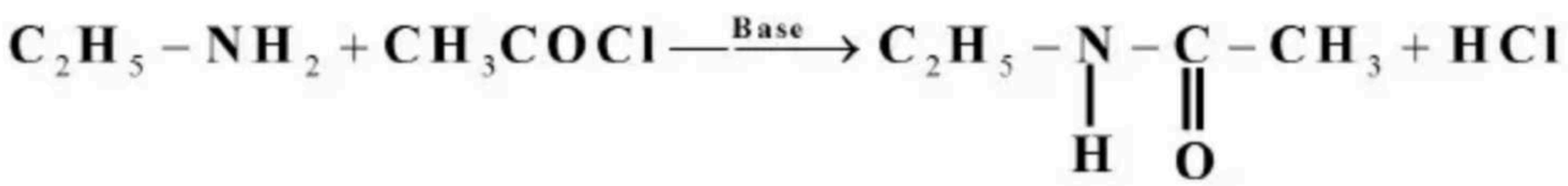
(i) Alkylation

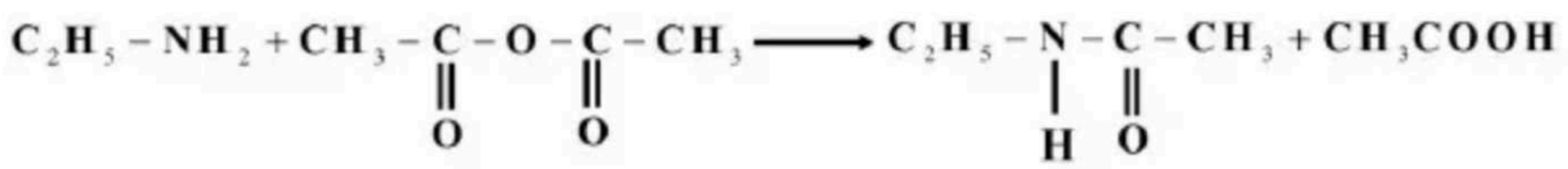




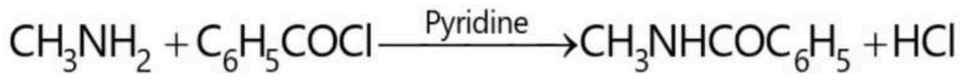
(ii) Acylation

Monksays



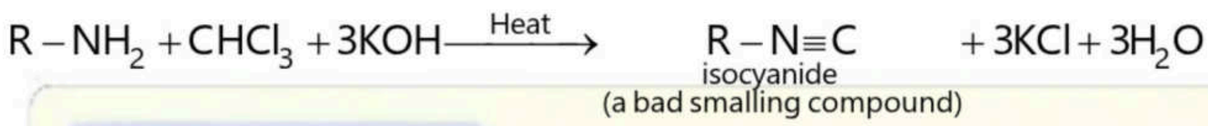


(iii) Benzoylation



Benzoylation of aniline is known as Schotten Baumann reaction.

(iv) Carbylamine reaction [only by 1° amines]

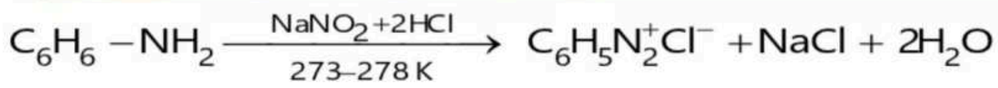


Methyl isocyanate gas $(CH_3 - N = C = O)$ was responsible for Bhopal gas tragedy in December 1984.

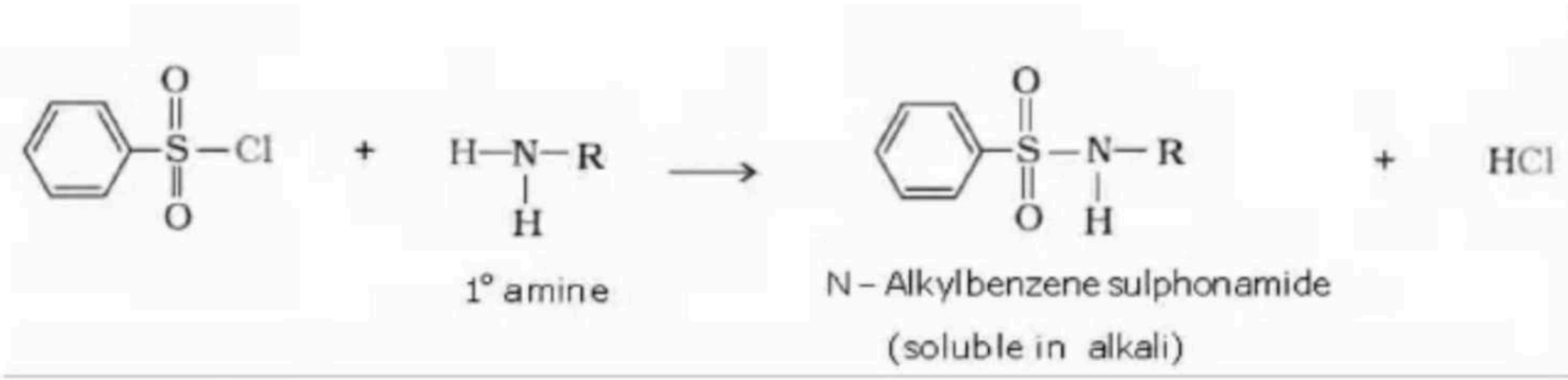
(v) Reaction with nitrous acid



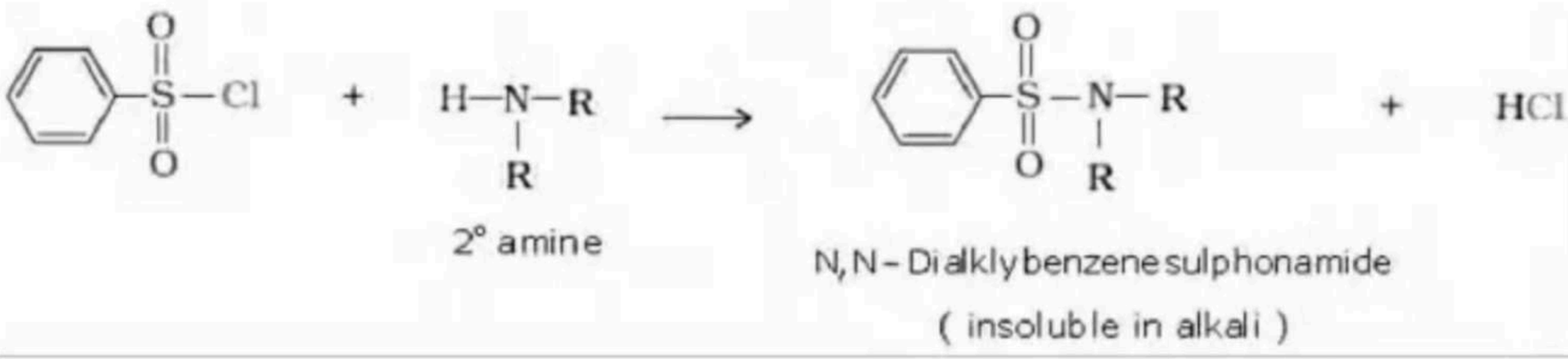
Quantitative evolution of nitrogen is used in estimation of amino acids and proteins.

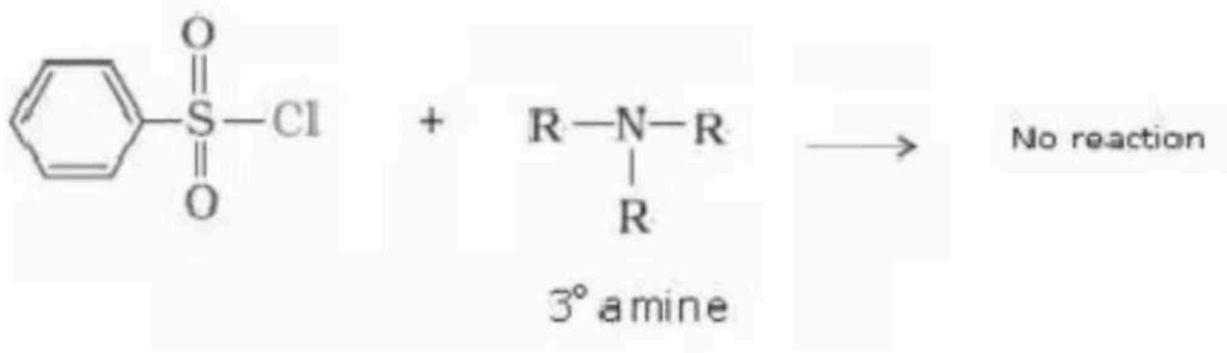


Reaction with aryl sulphonyl chloride [Hinsberg reagent]



Monksays





Reaction with aldehydes

Schiff base is obtained.

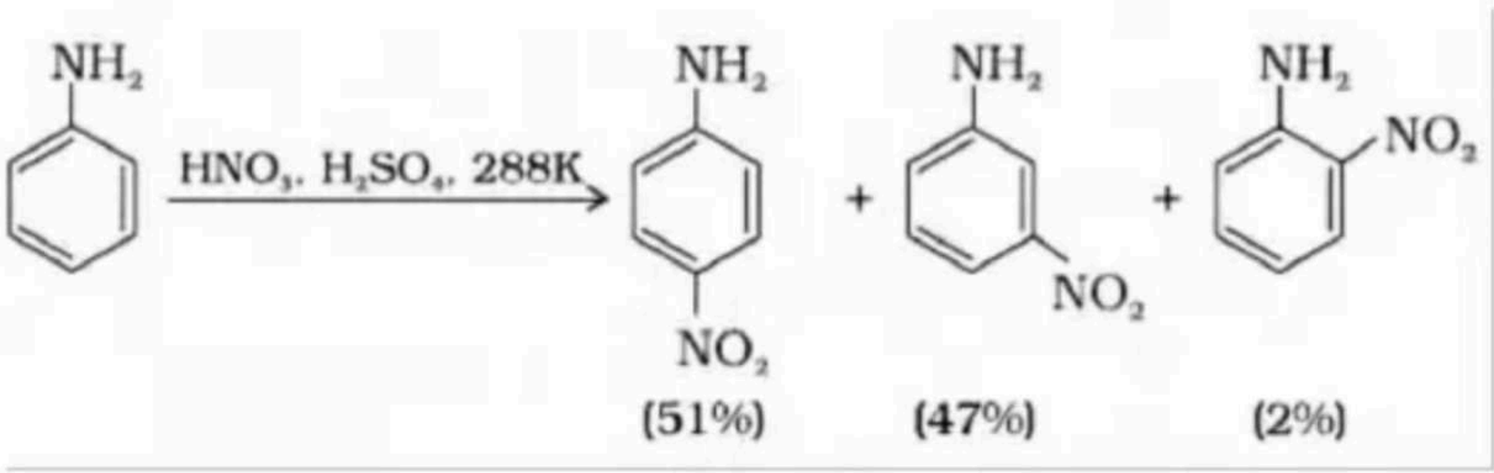
$$C_6H_5NH_2 + \underset{\text{benzaldehyde}}{OHCC_6H_5} \xrightarrow[-H_2O]{\Delta} \underset{\text{benzylidene aniline (Schiff base)}}{C_6H_5N=CHC_6H_5}$$

Electrophilic substitution reactions



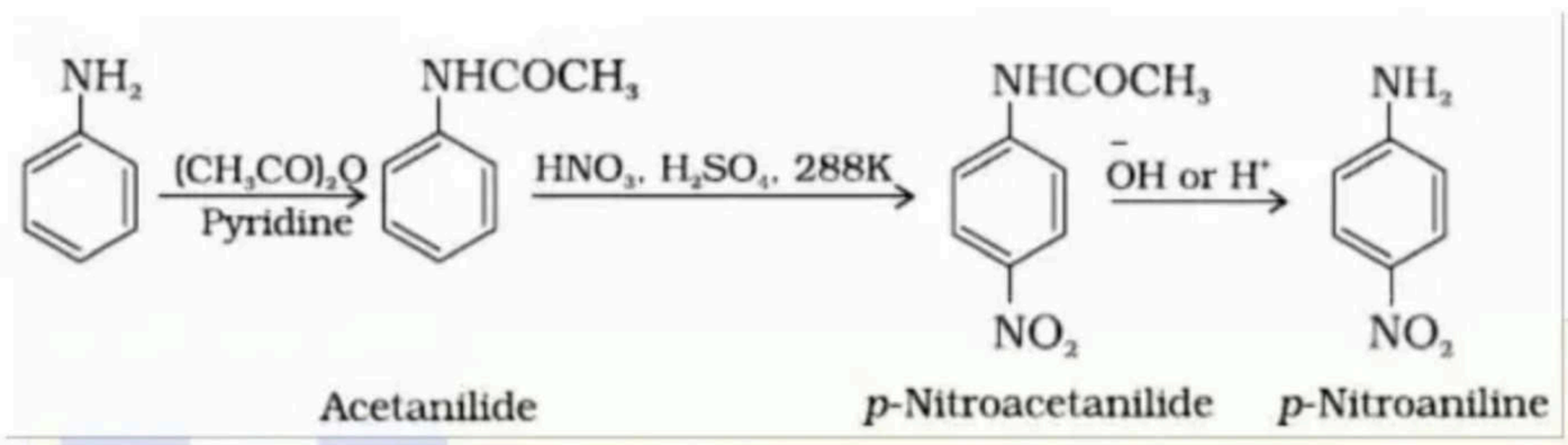
Nitration

(a) Under strongly acidic medium aniline gets protonated to form anilinium ion, which is deactivating group and is meta directing. Hence m-nitroaniline is also formed in 47 % along with ortho and para products.

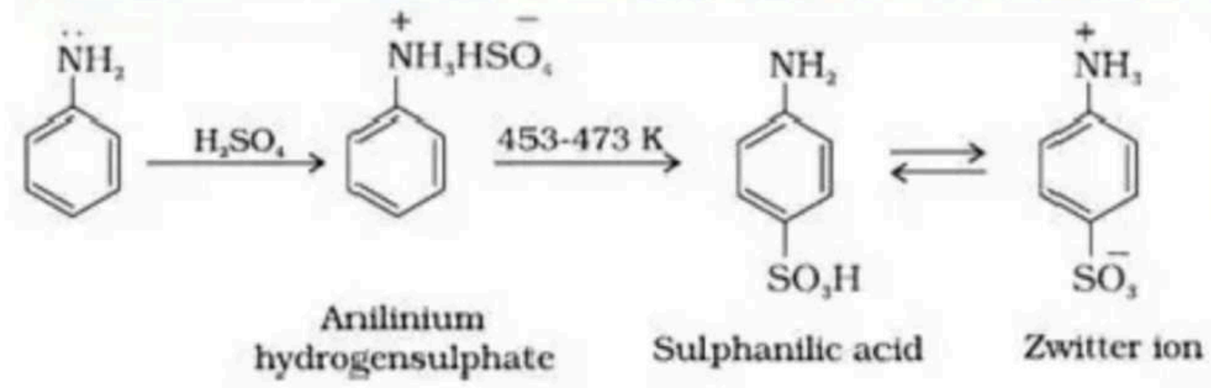


Aromatic amines cannot be nitrated directly because HNO_3 being a strong oxidizing agent oxidizes it forming black mass.

(b) Nitration by protecting the $-NH_2$ group by acetylation reaction with acetic anhydride :



(c) Sulphonation On sulphonation, aniline gives sulphanilic acid, as the major product.



Oxidation

Oxidising agent	**Product**
Acidified $KMnO_4$ (or $Na_2Cr_2O_7 + CuSO_4 + \text{dill acid}$)	Aniline back (a dye)
Chromic acid $(Na_2Cr_2O_7 + \text{Conc } H_2SO_4)$	p-benzaquinone
Caro's acid (H_2SO_5)	Nitrobenzene and nitrosobenzene
Conc. Nitric acid	decomposes

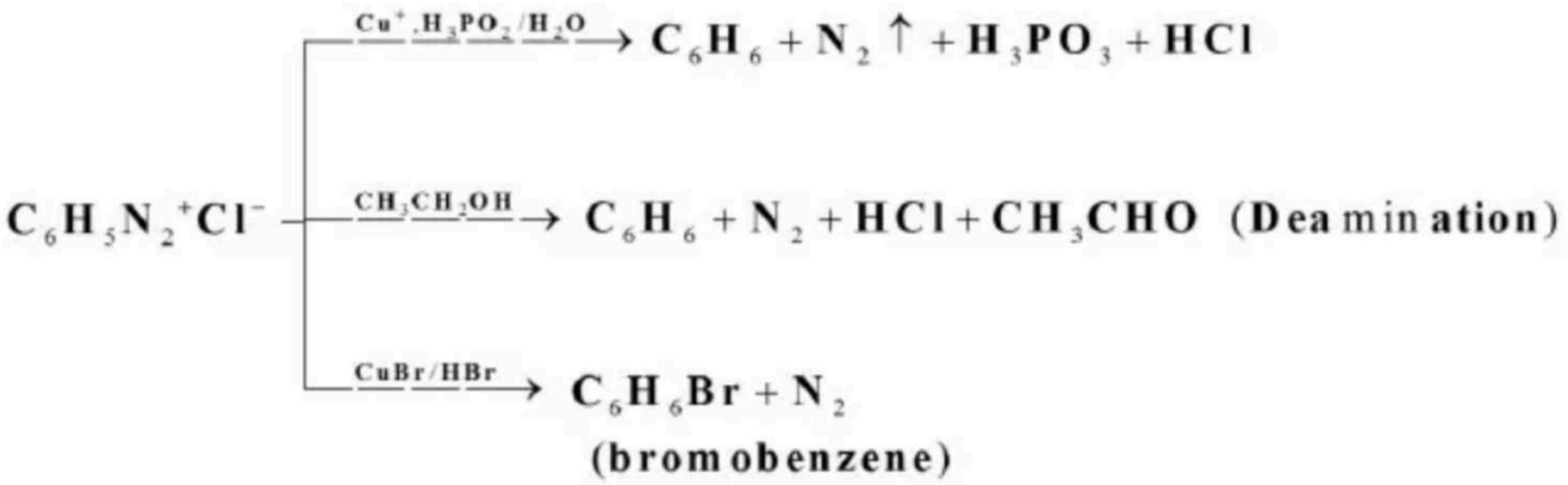
Benzene Diazonium Chloride $(C_6H_5N_2^+ ; Cl^-)$

Preparation (Diazotisation reaction)

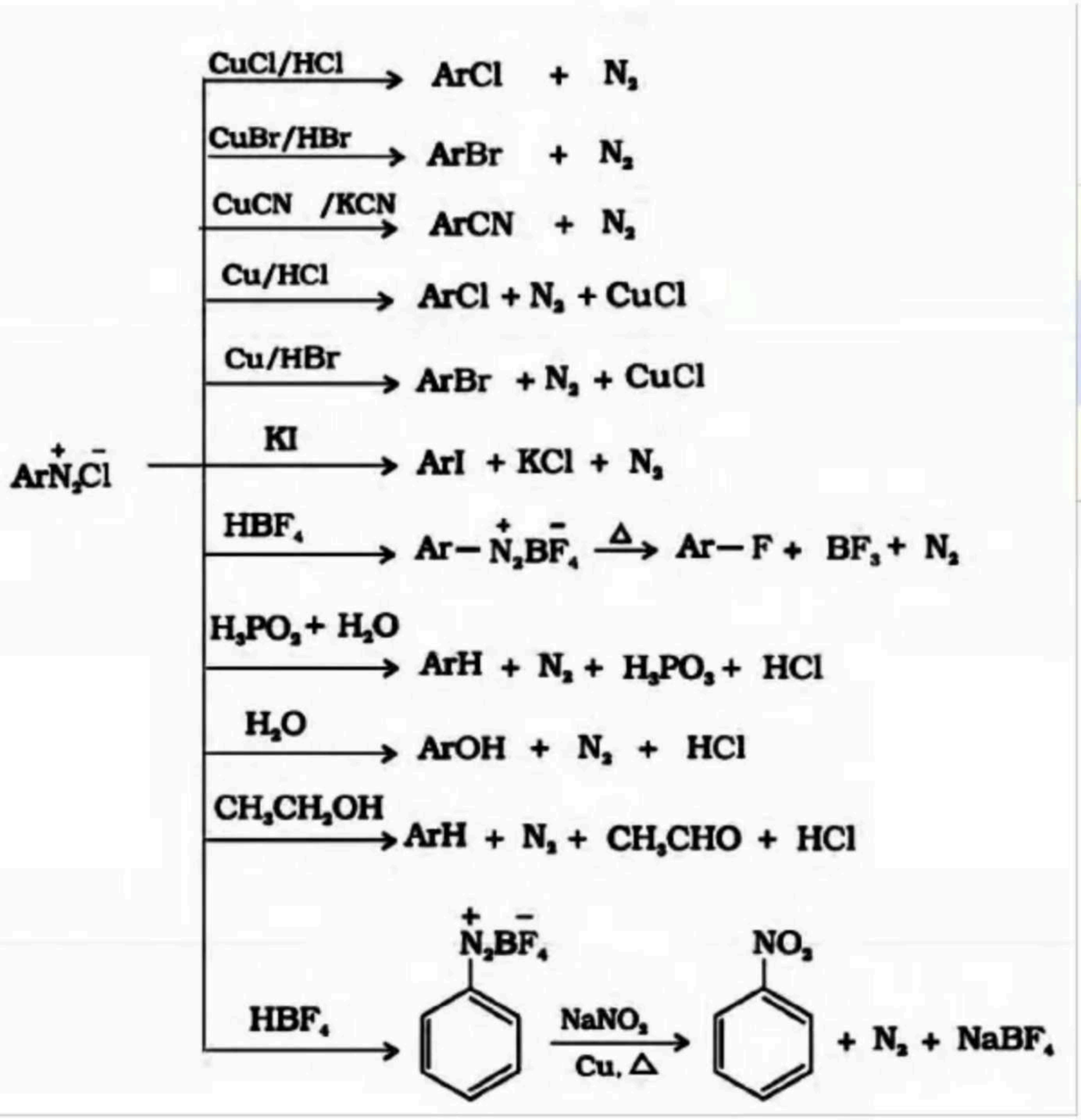
$$C_6H_5NH_2 + NaNO_2 + 2HCl \xrightarrow{273-278\,K} C_6H_5N=N-Cl + NaCl + 2H_2O$$

Monksays

Chemical reactions

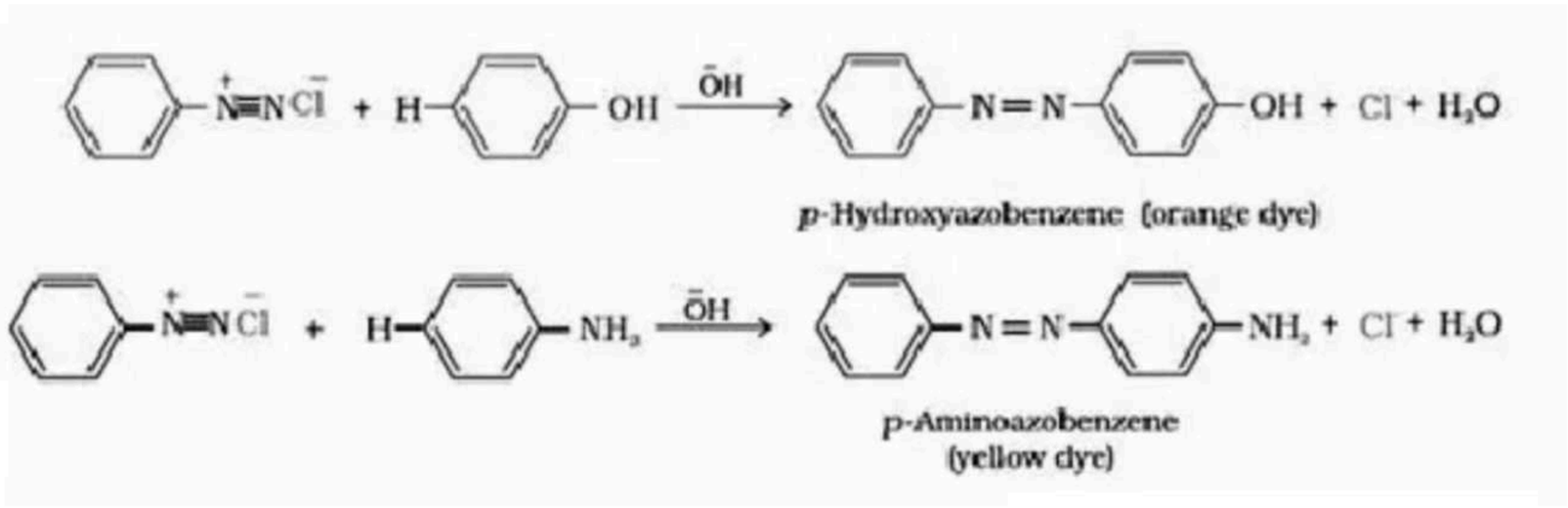


(a) Reactions involving displacement of nitrogen :



Monksays

Azo coupling reactions



Alkyl Cyanides

Methods of Preparation

(i) From alkyl halides

$$RX + KCN_{(alc)} \xrightarrow{100^{o}C} \underset{(major)}{RCN} + RNC$$

(ii) From acid amides

$$RCONH_2 \xrightarrow[-H_2O]{P_2O_5\Delta} RC \equiv N$$

Reactions of Alkyl Cyanides

(i) Hydrolysis

$$RCN \xrightarrow[\substack{Alk\ H_2O_2\\(partiial\ hydrolysis)}]{H_2O} \underset{amide}{RCONH_2} \xrightarrow[H^+]{H_2O} \underset{carboxylic\ acid}{RCOOH} + NH_3$$

(ii) Reduction

$$RCN + 4[H] \xrightarrow{Na+C_2H_5OH} RCH_2NH_2$$

$$RCN + 4[H] \xrightarrow{LiAlH_4} RCH_2NH_2 \qquad \textbf{(Mendius reduction)}$$

$$RCN \xrightarrow{SnCl_2/HCl} \underset{imine\ hydrochloride}{R\,CH = NH \cdot HCl} \xrightarrow{H_2O} \underset{aldehyde}{RCHO} + NH_4Cl$$

(iii) Reaction with Grignard reagent

$$RCN + R'MgX \xrightarrow{Ether} R - \underset{\displaystyle R'}{\underset{|}{C}} = NMgX \xrightarrow{2H_2O} R - \underset{\displaystyle R'}{\underset{|}{C}} = O$$

Monksays

Alkyl isocyanides

Methods of Preparation

(a) From alkyl halides

$$R-X+AgCN \xrightarrow[\Delta]{C_2H_5OH} \underset{\text{(major)}}{RNC} + RCN$$

(b) Carbylamine reaction

$$RNH_2 + CHCl_3 + 3KOH_{(alc)} \xrightarrow{\Delta} RNC + 3KCl + 3H_2O$$

(c) From N-alkyl formamide

$$R-NH-\underset{\text{Pyridine}}{CHO} \xrightarrow{POCl_3} R-N\equiv C+H_2O$$

Reactions of Alkyl isocyanides

Hydrolysis

$$R-N\equiv C+2H_2O \xrightarrow{H^+} RNH_2 + HCOOH$$

(ii) Reduction

$$RN\equiv C+4[H] \xrightarrow[\text{or Ni or Pt}]{Na/C_2H_5OH} \underset{2^o \text{ amine}}{RNHCH_3}$$

(iii) Addition reaction

Due to the presence of unshared pair of electrons on C atom, alkyl isocyanides give addition reaction.

$$RNC + S \longrightarrow RNCS$$

$$RNC + HgO \longrightarrow RNCO + Hg$$

$$RNC + O_3 \longrightarrow RNCO$$

(iv) Isomerisation At 250°C, it isomerizes to nitrile.

$$RNC \xrightarrow{\Delta} RCN$$

Monksays

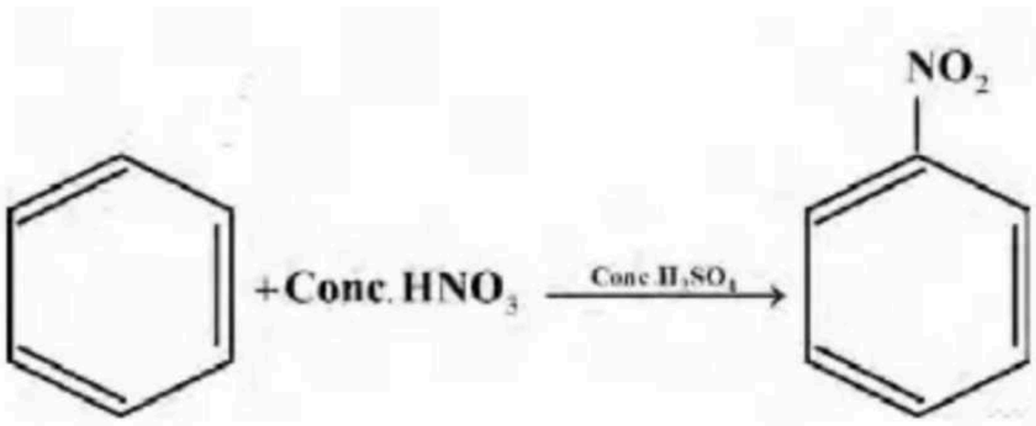
Nitro Compounds

Methods of Preparation

(i) From alkyl halides

$$RX + AgNO_2 \xrightarrow{\Delta} RNO_2 + AgX$$

(ii) Nitration: Nitrating mixture is conc HNO_3 + conc H_2SO_4



Reactions of Nitro Compounds

Reduction

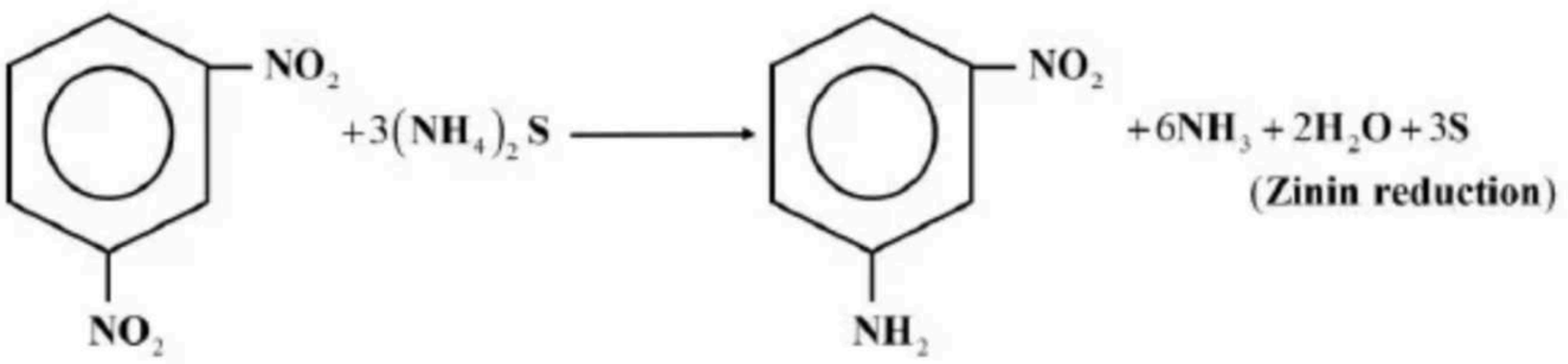
With Sn/HCl or catalytic hydrogenation, nitroalkanes are reduced to amines.

$$RNO_2 + 6[H] \xrightarrow{Sn/HCl} R-NH_2 + 2H_2O$$

If neutral reducing agent like Zn dust + NH_4Cl is used, hydroxylamines are obtained as major product.

$$RNO_2 + 4[H] \xrightarrow{Zn+NH_4Cl} \underset{\text{N–alkylhydroxylamine}}{R-NHOH} + H_2O$$

In the presence of $(NH_4)_2S$ or Na_2S, selective reduction takes place.



Nitrobenzene gives different products with different reagents and in different mediums.

Medium	Reagent	Product
Acid	Sn/HCl	Aniline
Neutral	Zn/NH_4Cl	N-phenyl hydroxylamine
	$Na_3AsO_3/NaOH$	Azoxbenzene $(C_6H_5N=\overset{\overset{O}{\uparrow}}{N}C_6H_5)$

Monksays

Alkaline	Zn/NaOH, CH_3OH Zn/NaOH, C_2H_5OH	Azobenzene hydrazobenzene
Metallic hydride	$LiAlH_4$	aniline
Electrolytic	dil H_2SO_4	p-aminophenol

Action of HNO_2

1° nitroalkane gives nitrolic acid which gives red colour with NaOH.

$$RCH_2NH_2 \xrightarrow[-H_2O]{HNO_2} \underset{\text{nitrolic acid}}{RC(NO_2)=NOH} \xrightarrow{NaOH} \underset{\text{(red)}}{RC(NO_2)=NONa}$$

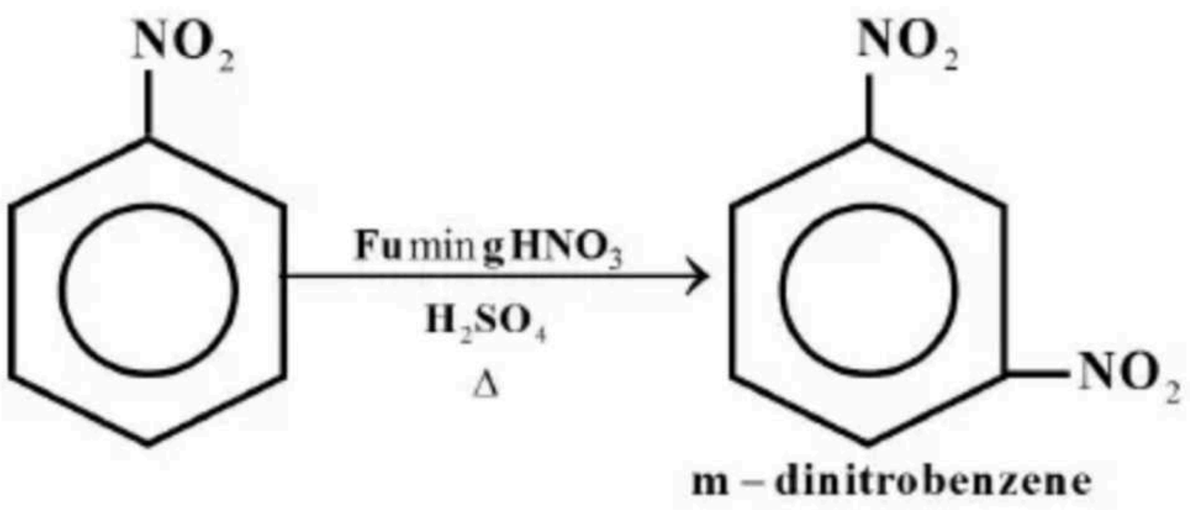
2° nitrolkanes give pseudonitrol with HNO_2

$$R_2CH(NO_2) \xrightarrow[-H_2O]{HNO_2} \underset{\underset{\text{Pseudonitrol}}{|\ NO_2}}{R_2C-NO} \xrightarrow{NaOH} \text{Blue}$$

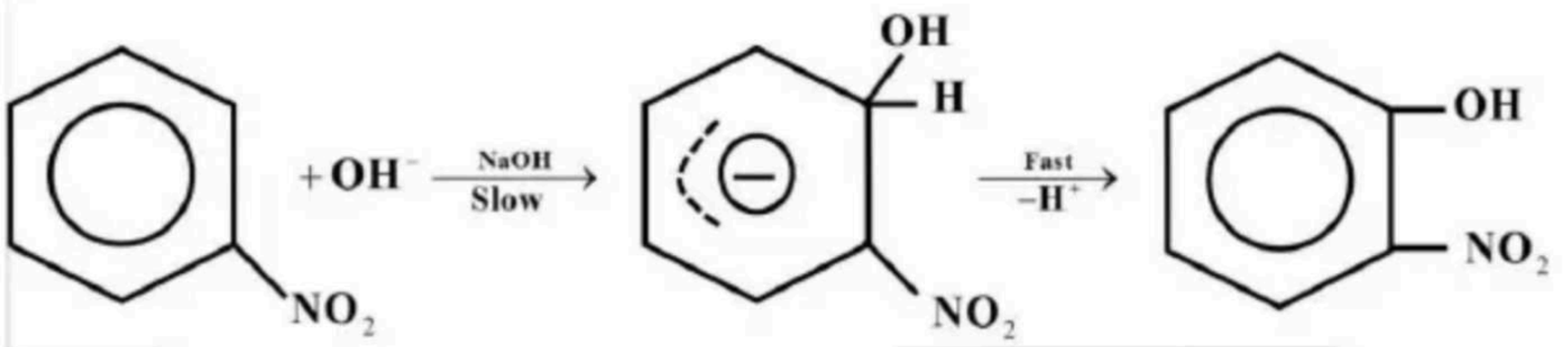
Nef carbonyl synthesis

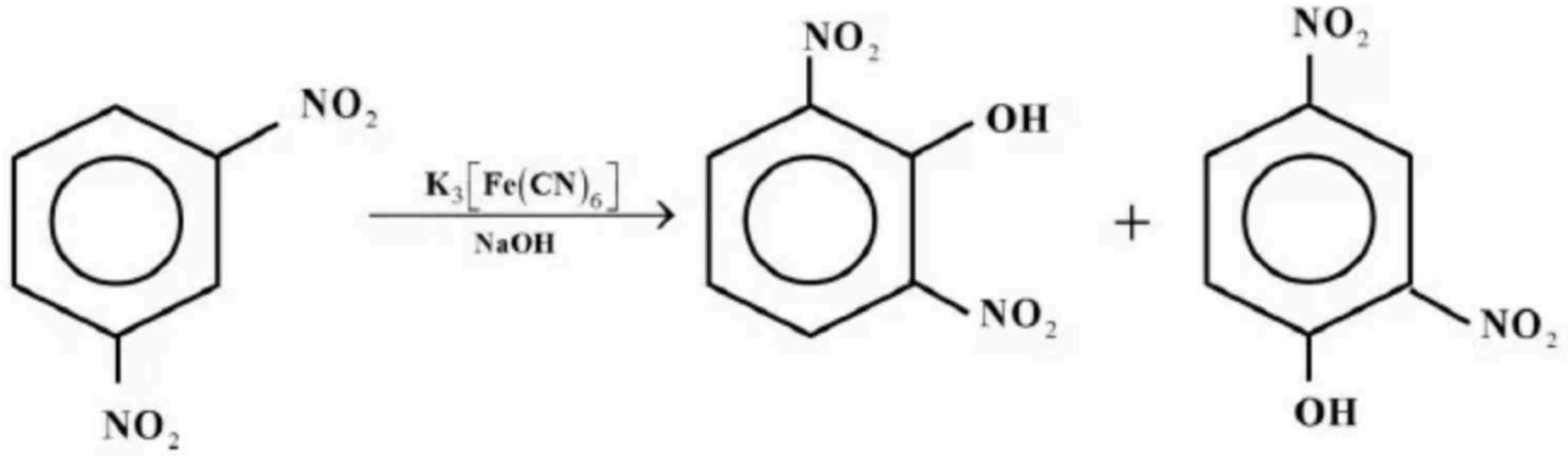
$$R-CH_2NO_2 \xrightarrow{NaOH} R-CH=\underset{\underset{O^-}{|}}{\overset{+}{N}-O^-Na^+} \xrightarrow{50\%\ H_2SO_4} \underset{\text{aldehyde}}{R-CH=O}$$

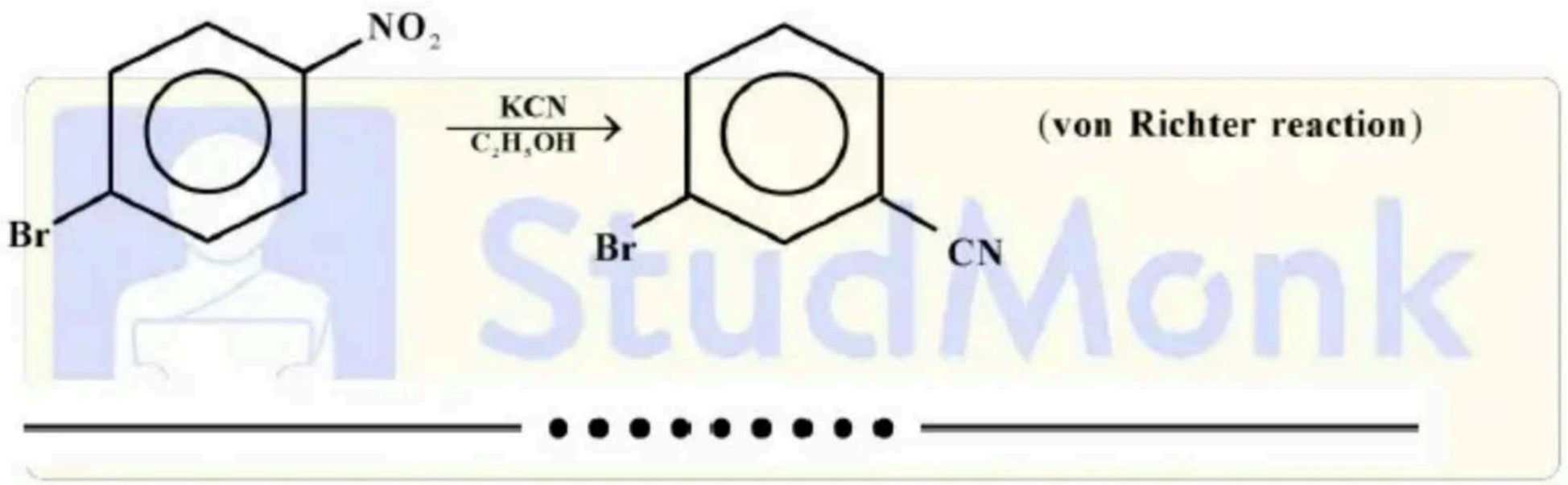
Electrophilic substitution On nitration



Nucleophilic substitution reaction -NO_2 group activates the ring towards nucleophilic substitution.



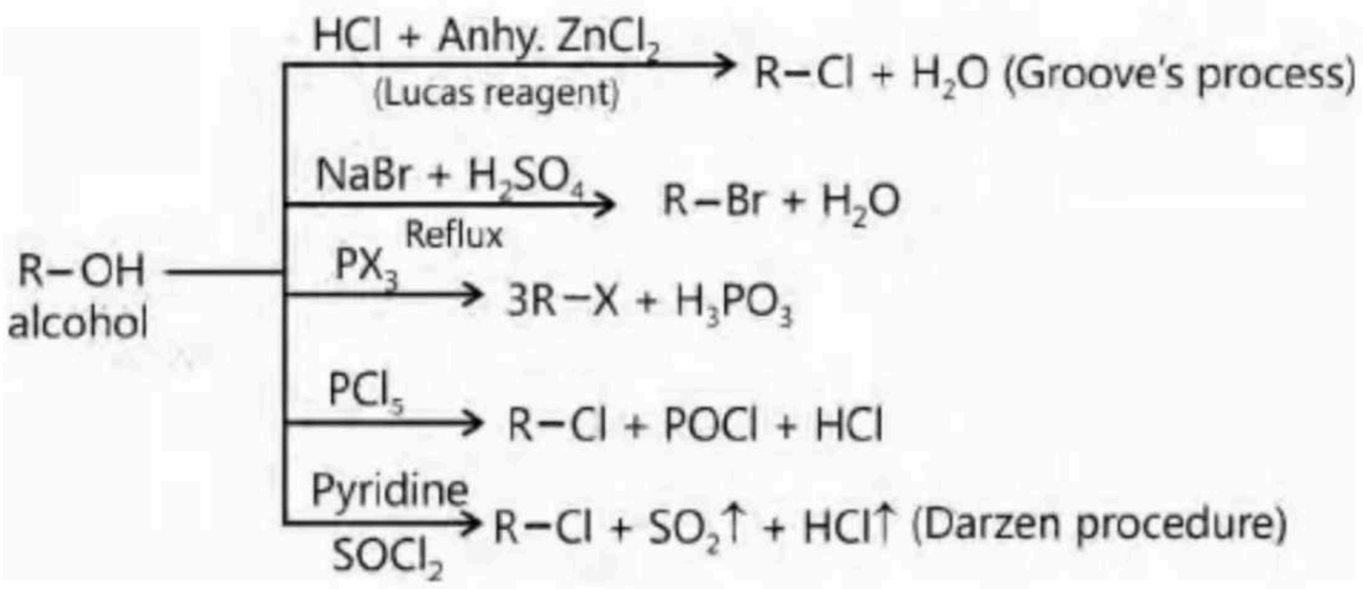




19. HALOALKANES AND HALOARENES

General Methods of Preparation of Haloalkanes

From Alcohols



Monksays

Free Radical Halogenation of Alkanes

$$CH_3CH_2CH_2CH_3 \xrightarrow[\text{UV light or Heat}]{Cl_2} CH_3CH_2CH_2CH_2Cl + CH_3CH_2\underset{\displaystyle Cl}{\underset{|}{C}}HCH_3$$

Addition of hydrogen Halides on Alkenes

$$-\overset{|}{C}-\overset{|}{C}=\overset{|}{C}- + HBr \longrightarrow -\overset{|}{\underset{|}{C}}-\overset{|}{\underset{|}{C}}-\overset{|}{\underset{|}{C}}-$$

2–bromopropane

$$-\overset{|}{C}-\overset{|}{C}=\overset{|}{C}- + HBr \xrightarrow{\text{Organic peroxide}} -\overset{|}{\underset{|}{C}}-\overset{|}{\underset{|}{C}}-\overset{|}{\underset{|}{C}}-Br$$

(i) Halogen Exchange method :

$R-X + NaI \rightarrow R-I + NaX$ **(Finkelstein reaction)**

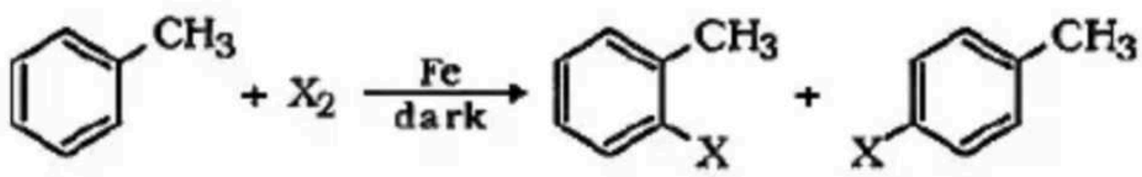
$R-Br + AgF \rightarrow R-F + AgBr$ **(Swarts reaction)**

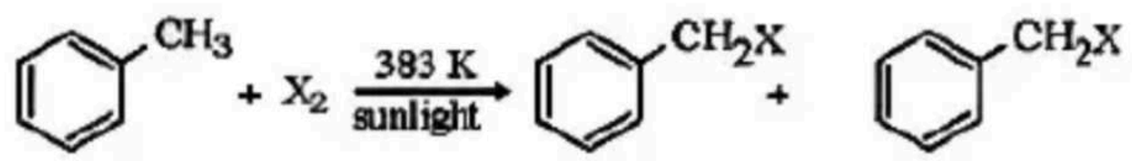
Hunsdiecker Reaction

$$CH_3COOAg + Br_2 \xrightarrow[CCl_4]{} CH_3Br + AgBr + CO_2$$

- **Preparation of haloarenes :**

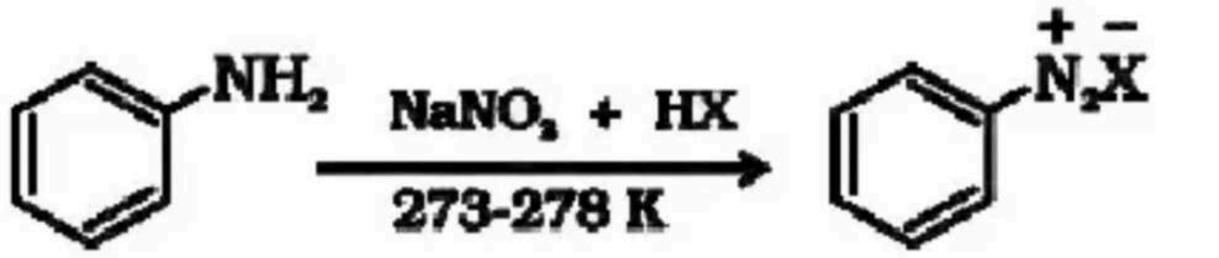
(a) By electrophilic substitution reaction :



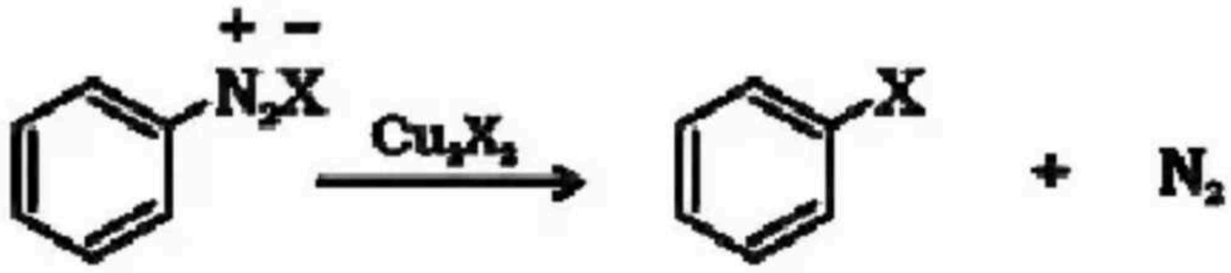


(b) Sandmeyer's reaction :

Monksays



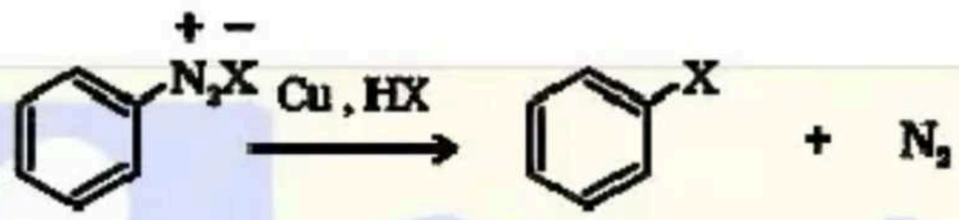
Benzene diazonium halide



Aryl halide

X = Cl, Br

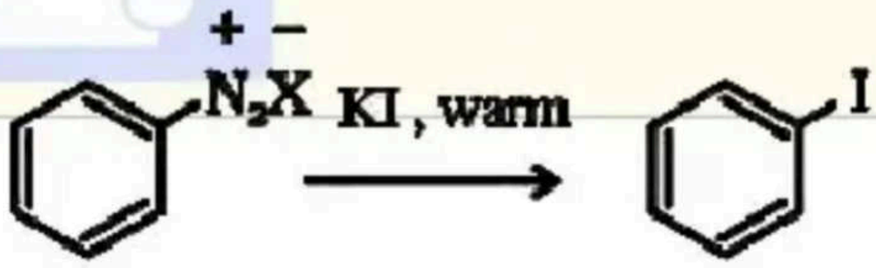
(c) Gattermann reaction :



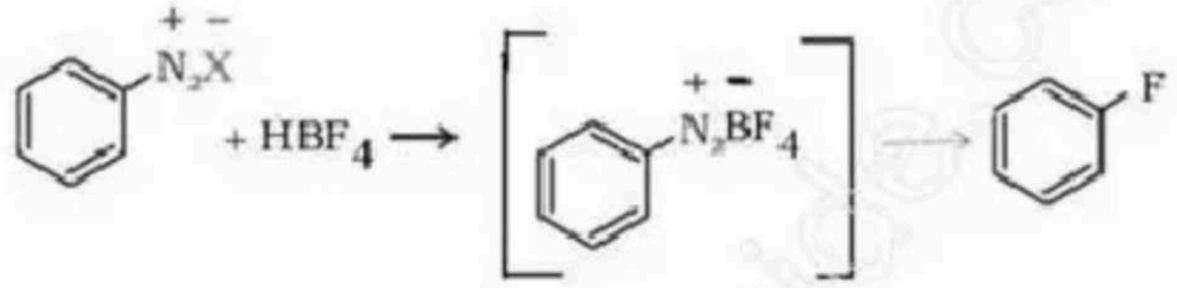
Aryl halide

X = Cl, Br

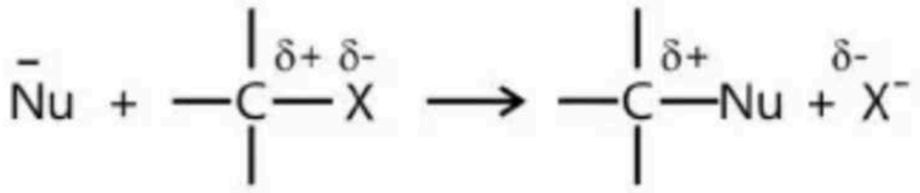
(d) From Diazonium Chloride :

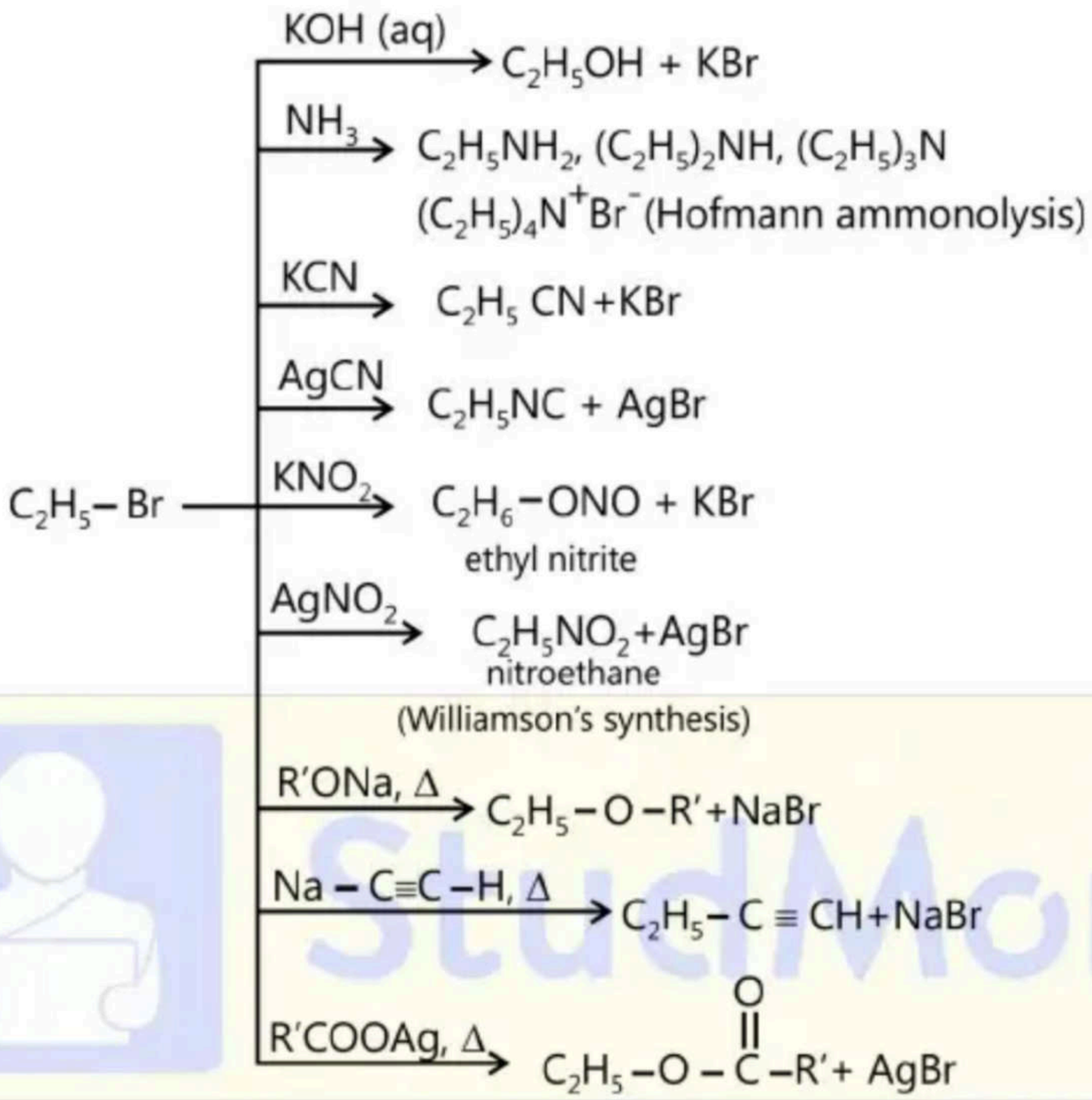


(e) Balz-Schiemann reaction :



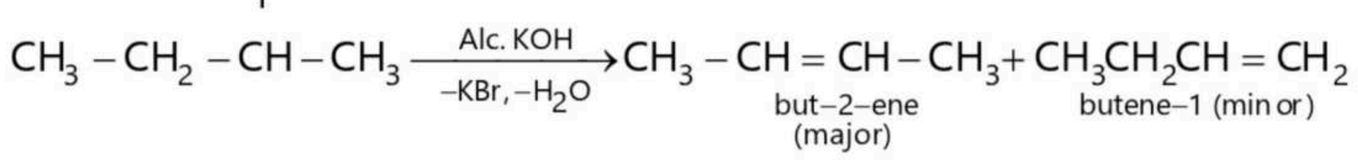
Nucleophilic Substitution Reactions





2. Elimination Reactions

Dehydrohalogenation is a β-elimination reaction in which halogen is from α-carbon atom and the hydrogen from the α-carbon according to Saytzeff rule, e.g.



Reduction

$$C_2H_5 - Br + H_2 \xrightarrow{\text{Ni, 575 K}} C_2H_6 + HBr$$

$$C_2H_5I + HI \xrightarrow{\text{Red P, 420 K}} C_2H_6 + I_2$$

Monksays

Reaction with Metals

(i) Wurtz reaction

$$RX + 2Na + XR \xrightarrow{\text{Dry ether}} R - R(\text{alkane}) + 2NaX$$

(ii) Wurtz-Fitting reaction

$$C_6H_5\text{-}Cl + 2Na + Cl\text{-}CH_3 \xrightarrow{\text{Dry ether}} C_6H_5 - CH_3 + 2NaCl$$

(iii) Reaction with Mg

$$C_2H_5Br + Mg \xrightarrow{\text{Dry ether}} \underset{\text{Grignard's reagent)}}{C_2H_5 - Mg - Br}$$

Isomerisation

$$CH_3CH_2CH_2 - Cl \xrightarrow[\text{or anhy. } AlCl_5]{573\ K} CH_3 - \underset{|}{\overset{Cl}{CH}} - CH_3$$

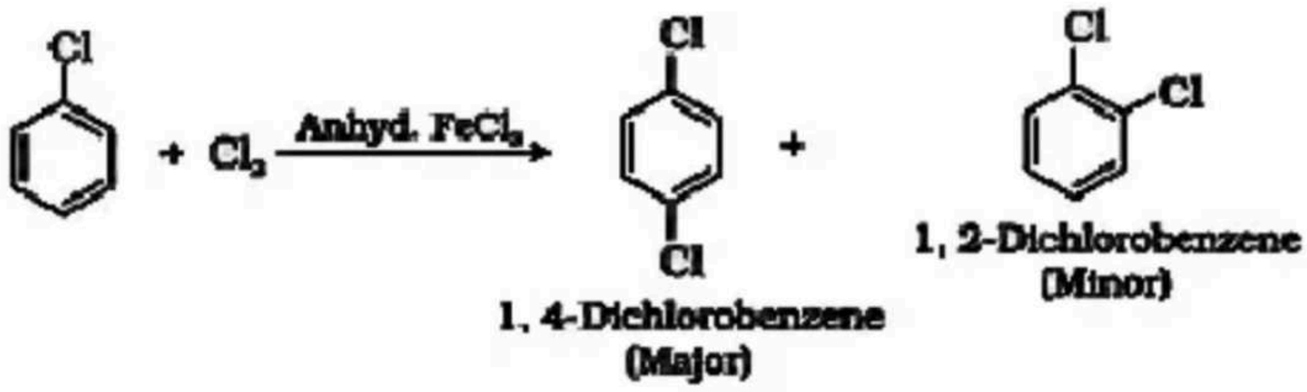
Chemical properties of haloarenes :

(a) Dow's Process



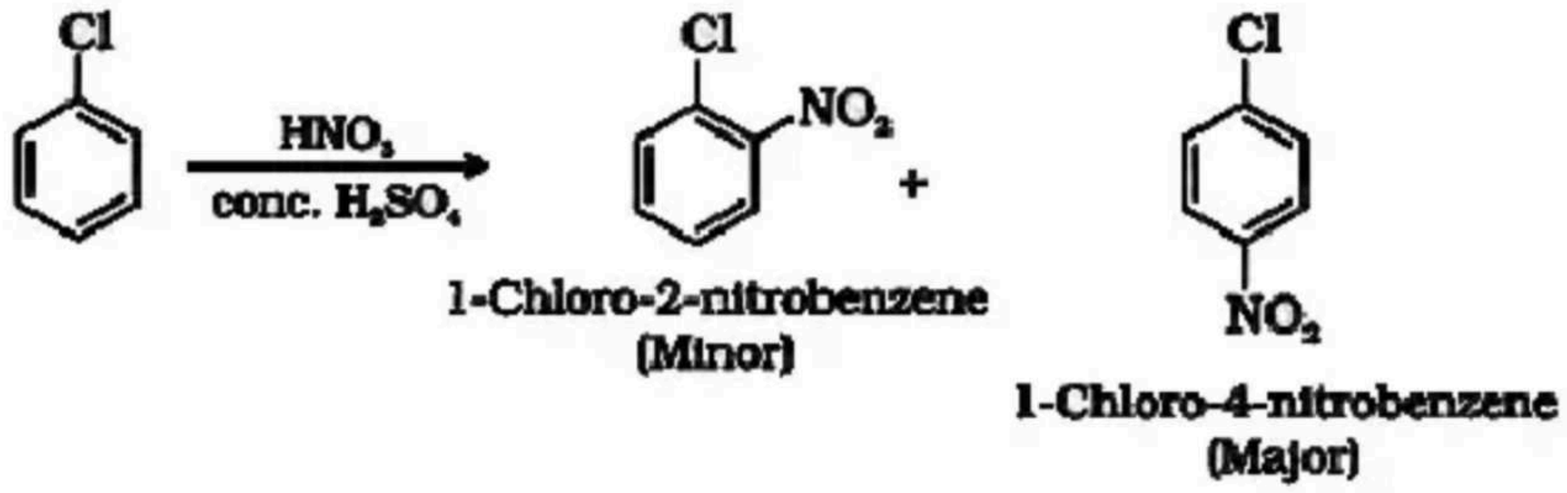
Electrophilic substitution Reactions

(b) With halogens

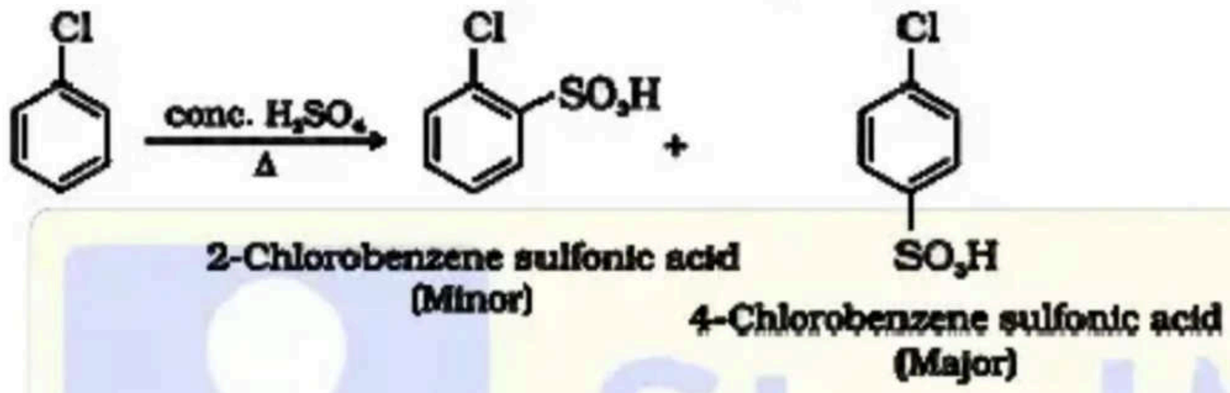


Monksays

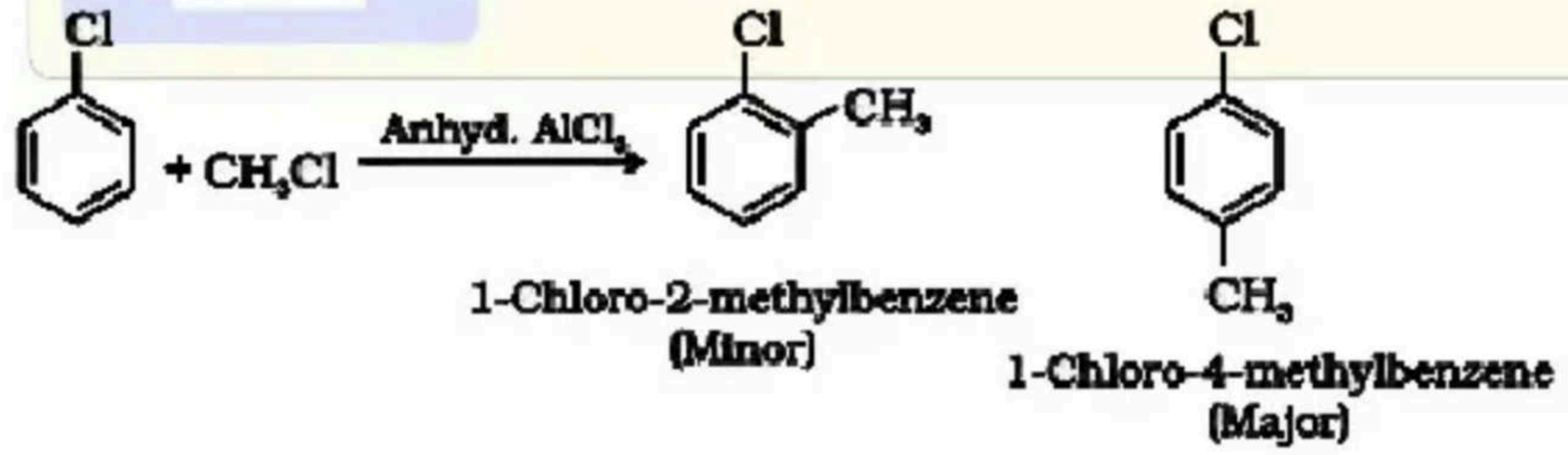
(c) With conc. sulphuric acid and nitric acid



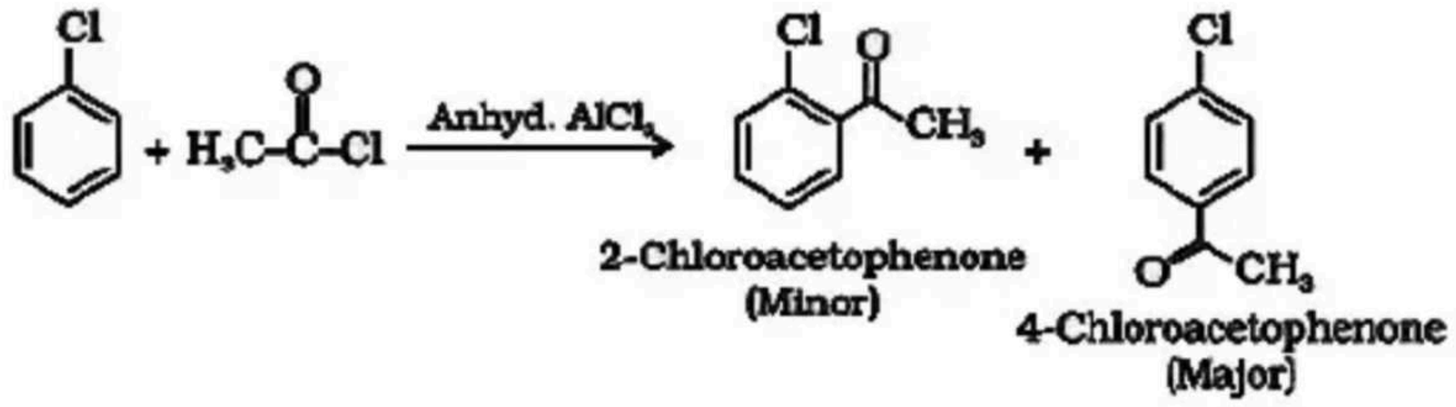
(d) On heating with conc. sulphuric acid



(e) With methyl chloride



(f) With acetyl chloride



Monksays

(i) Other conversions :

$$R-X+KCN \rightarrow R-CN+KX$$

$$R-CN \xrightarrow{Na,C_2H_5OH \text{ or } LiAH_4 \text{ or } Ni/H_2} R-CH_2NH_2 \xrightarrow{HONO} RCH_2OH$$

$$R-CN \xrightarrow{H_3O^+} R-COOH \xrightarrow{NH_3} R-CONH_2$$

$$R-COOH \xrightarrow{LiAlH_4} RCH_2OH$$

$$R-COOH \xrightarrow{PCl_5 \text{ or } PCl_3 \text{ or } SOCl_2} R-COCl$$

$$R-X+Mg \xrightarrow{\text{Dry ether}} RMgX$$

$$RMgX+CO_2 \rightarrow RCOOH$$

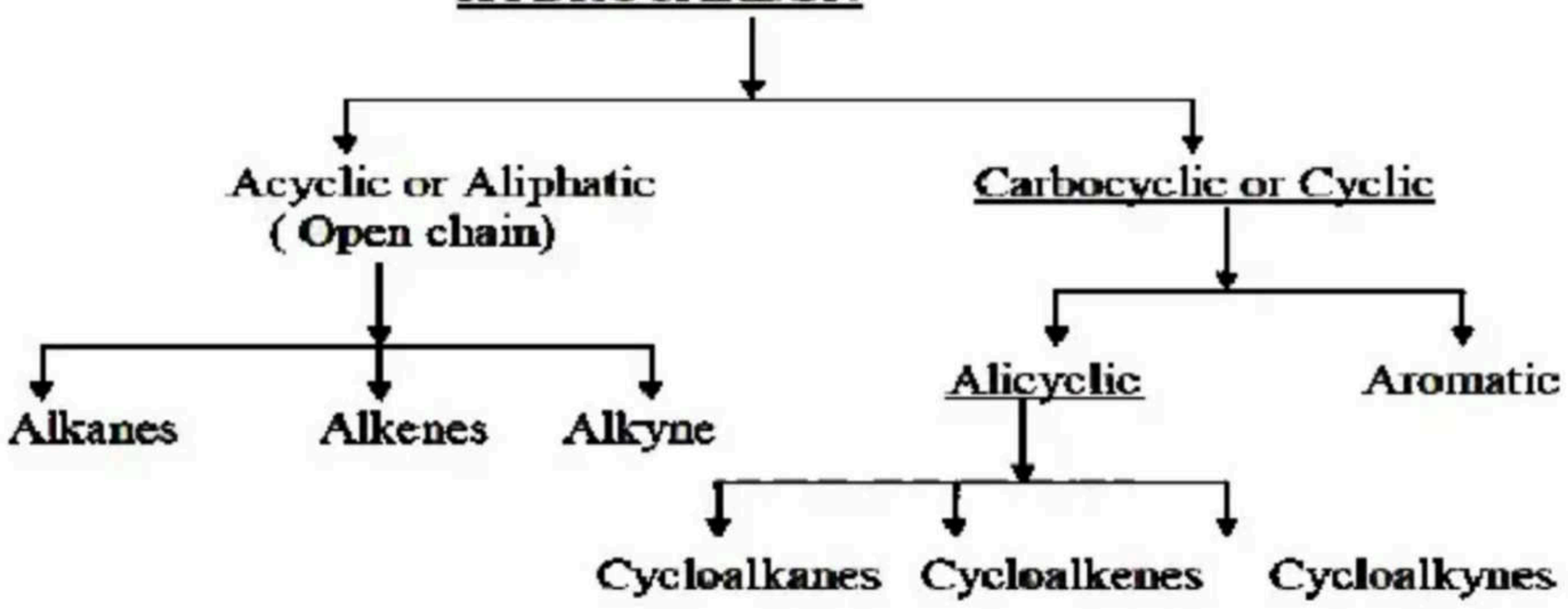
$$RMgX+HCHO \rightarrow \text{Primary alcohol}$$

$$RMgX+RCHO \rightarrow \text{Secondary alcohol}$$

$$RMgx+R-CO-R \rightarrow \text{Tertiary alcohol}$$

20. HYDROCARBONS

CLASSIFICATION OF HYDROCARBONS:



Monksays

Alkanes

Preparation of Alkanes

(1) Wurtz reaction :

$$2CH_3CH_2Br + 2Na \xrightarrow[\text{ether}]{\text{Dry}} CH_3CH_2CH_2CH_3 + 2NaBr$$

Frankland reaction

$$RX + Zn + Rx \longrightarrow R - R + ZnX_2$$

(2) From Grignard reagent (RMgX)

$$RMgX + HOH \rightarrow RH + Mg(OH)X$$

$$RMgX + R'OH \rightarrow RH + Mg(OR')X$$

$$RMgX + R'NH_2 \rightarrow RH + Mg(NHR')X$$

(3) From unsaturated hydrocarbons :-

Sabatier-Senderens reduction

$$R - CH = CH_2 + H_2 \xrightarrow{Ni/\Delta} R - CH_2 - CH_3$$

$$R - C \equiv CH + H_2 \xrightarrow{Ni/\Delta} R - CH_2 - CH_3$$

4. From carboxylic acids-

Decarboxylation –

$$\underset{\text{Sodium ethanoate}}{CH_3COO^-Na^+} + NaOH \xrightarrow[\Delta]{CaO} CH_4 + Na_2CO_3$$

Kolbe's electrolytic method –

$$\underset{\text{Sodium acetate}}{2CH_3COO^-Na^+} + 2H_2O \xrightarrow{\text{Electrolysis}} CH_3 - CH_3 + 2CO_2 + H_2 + 2NaOH$$

Reactions

Combustion :-

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O, \qquad \Delta H = -217.0 \text{ K cal / mole}$$

Oxidation :-

Monksays

$$CH_4 + O_2 \xrightarrow[573\,K]{Cu} 2CH_3OH$$

$$CH_4 + O_2 \xrightarrow{Mo_2O_3} \underset{\text{Methanal}}{HCHO} + H_2O$$

Substitution :-

Halogenation :-

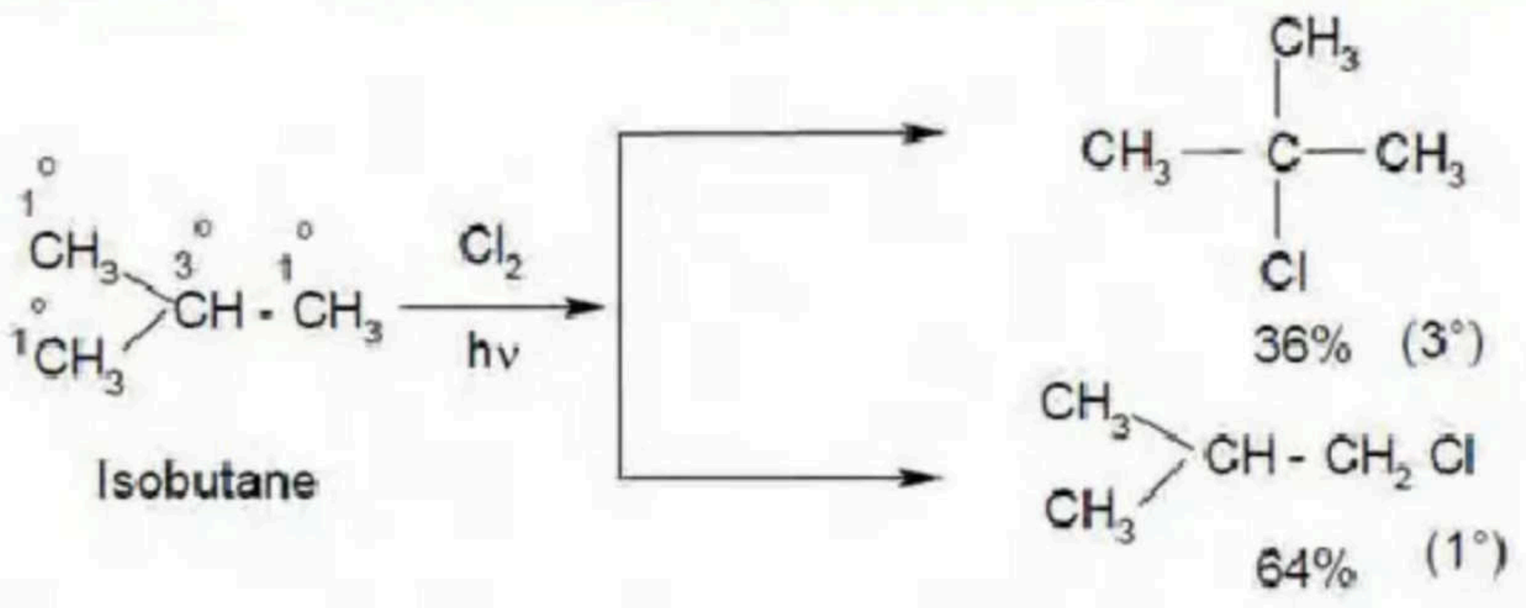
$$CH_4 + Cl_2 \xrightarrow{UV} CH_3Cl + HCl$$

$$CH_3Cl \xrightarrow{UV} CH_2Cl_2 \xrightarrow{UV} CHCl_3 \xrightarrow{UV} CCl_4$$

(i) The reactivity of Halogens : $F_2 > Cl_2 > Br_2 > I_2$

(ii) The rate of replacement of Hydrogens of alkanes is : $3^\circ > 2^\circ > 1^\circ$

$$\underset{\text{n–Butane}}{CH_3CH_2CH_2CH_3} \xrightarrow[hv]{Cl_2} CH_3CH_2CH_2CH_2Cl + CH_3CH_2ClCHCH_3$$



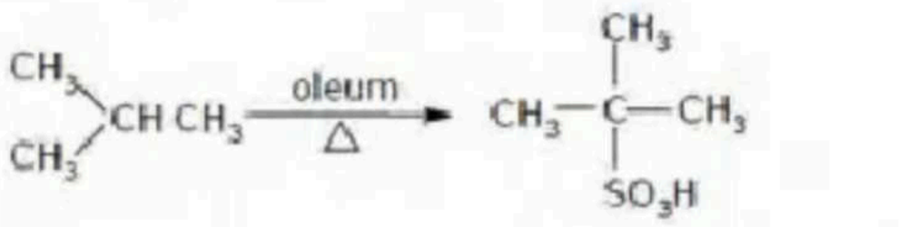
Nitration

$$CH_3CH_2CH_3 \xrightarrow[Conc.\ HNO_3]{450^\circ C} CH_3CH_2CH_2NO_2 + CH_3CHNO_2CH_3 + CH_3CH_2NO_2 + CH_3NO_2$$

25 % 40 % 10 % 25 %

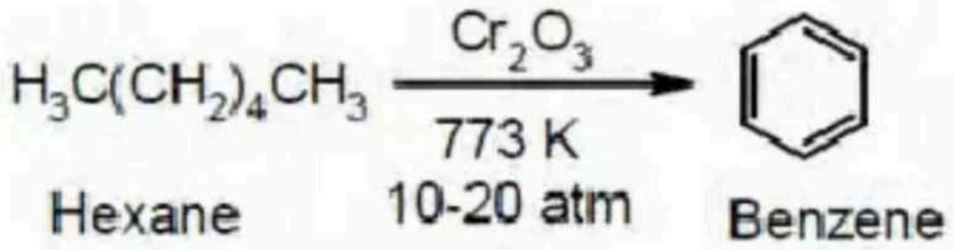
Sulphonation : - replacement of hydrogen atom of alkane by – SO_3H group.

Monksays

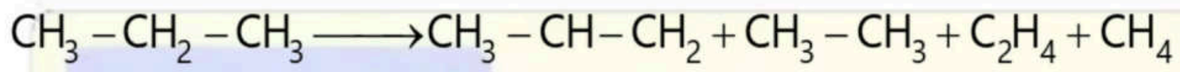


isobutane tert butyl sulphonic acid

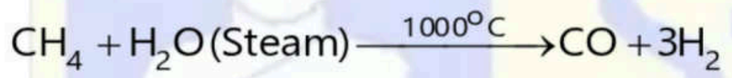
Aromatization :-



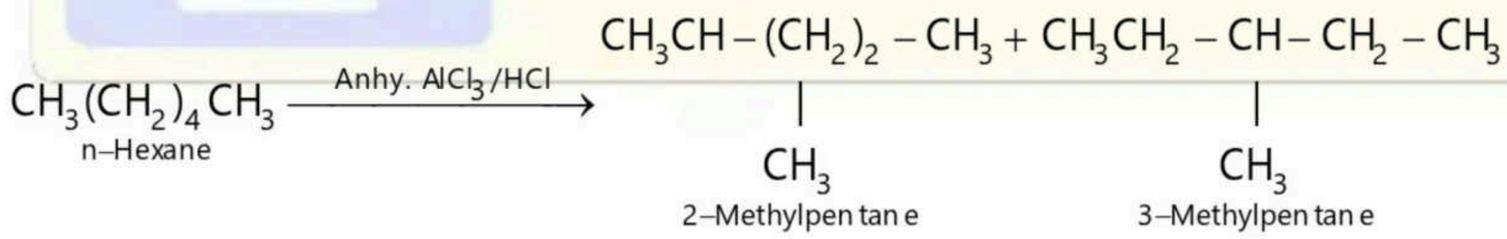
(6) Thermal decomposition or Pyrolysis or cracking or Fragmentation : - when higher alkanes are heated at high temp (about 700 – 800 K) in the presence of alumina or silica catalysts, the alkanes break down to lower alkenes and alkanes.



(7) Action of steam : - Catalyst : nickel, alumina Al_2O_3



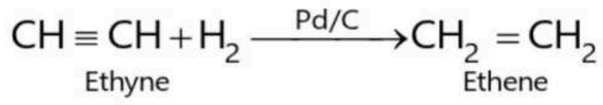
Isomerization



Alkenes

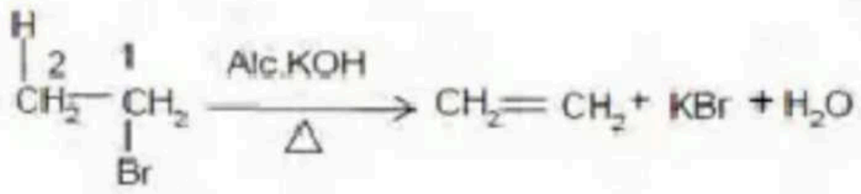
Preparation :

1. From Alkynes : - Alkynes on partial reduction with partially deactivated palladised charcoal known as Lindlar's catalyst give alkynes.

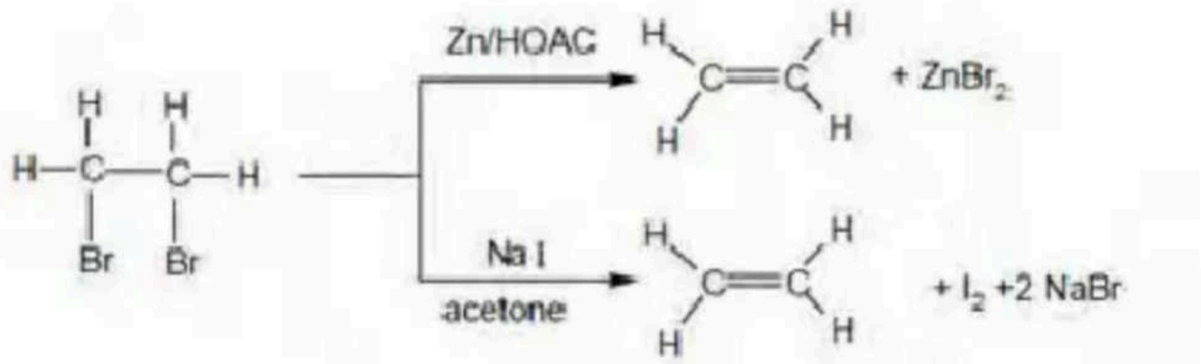


2. From Haloalkanes :- dehydrohalogenation (E_2 or 1, 2-elimination or Beta-elimination)

Monksays



3. From Dihaloalkanes :- Dehalogenation



4. From Alcohols :- Dehydration (E1 – elimination)

$$CH_3CH_2CH_2OH \xrightarrow[160^O]{\text{Conc. } H_2SO_4} CH_3CH=CH_2 + H_2O$$

$$CH_3CH_2CH_2CH_2OH \xrightarrow[300^OC]{Al_2O_3} CH_3CH_2CH=CH_2$$

Reactions of Alkenes

(1) Addition Reaction :- Alkene show electrophilic addition reaction.

1. Addition of Hydrogen :-

$$RCH=CH_2 \xrightarrow{H_2/Ni} RCH_2CH_3$$

2. Addition of Halogens :-

$$CH_2=CH_2 + Br_2 \xrightarrow[\text{Solvent}]{CCl_4} BrCH_2-CH_2Br$$

$$\underset{\text{(Brown colour)}}{CH_2=CH_2 + Br_2} \xrightarrow[\text{Solvent}]{H_2O} Br-CH_2-\underset{\text{(Colourless)}}{CH_2-OH} + HBr$$

3. Addition of hydrogen halides –

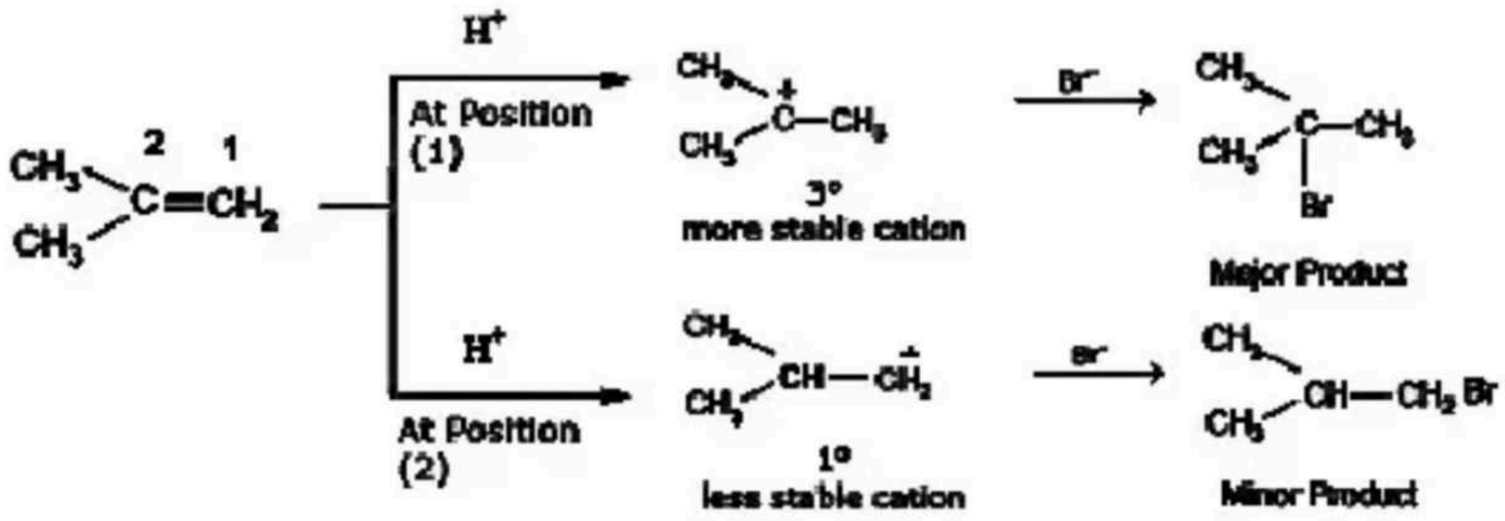
Addition reaction of HBr to symmetrical alkenes

$$CH_2=CH_2 + H-Br \longrightarrow CH_3-CH_2-Br$$

Addition reaction of HBr to unsymmetrical alkenes takes place according to Markovnikov Rule

Monksays

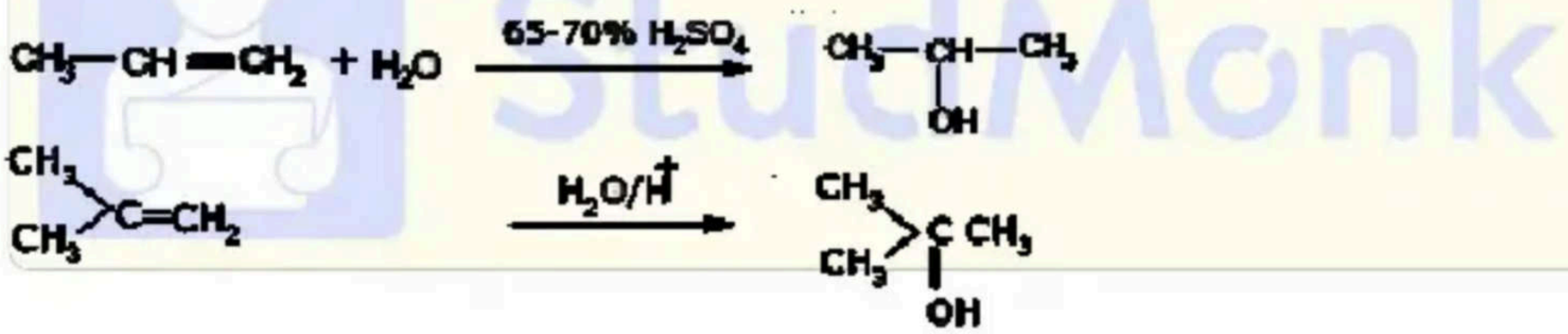
Markownikov rule :- negative part of the addendum (adding molecule) gets attached to that carbon atom which possesses lesser number of hydrogen atoms. E.g.



Peroxide effect or Kharasch (Anti Markownikoff's addition) :- In 1933 Kharasch and Mayo observed that when HBr is added to an unsymmetrical double bond in the presence of organic peroxide, the reaction take places opposite to the markovnikov rule.

$$CH_3 - CH = CH_2 \xrightarrow[\text{Peroxide}]{HBr} \underset{\text{Propyl bromide}}{CH_3 - CH_2 - CH_2Br}$$

4. Addition of water (Hydration):- Acid catalyzed addition of water



2. Oxidation :-

1. Combustion : $CO_2 + H_2O$

2. Hydrocarboration-oxidation:- Alkenes react with diborane to form trialkyl boranes which on oxidation with alkaline H_2O_2 give alcohols.

$$3CH_2 = CH_2 \xrightarrow{BH_3} \underset{\text{Triethylborane}}{CH_3CH_2\ B} \xrightarrow{H_2O_2/OH^-} \underset{\text{Ethyl alcohol}}{3CH_3CH_2OH}$$

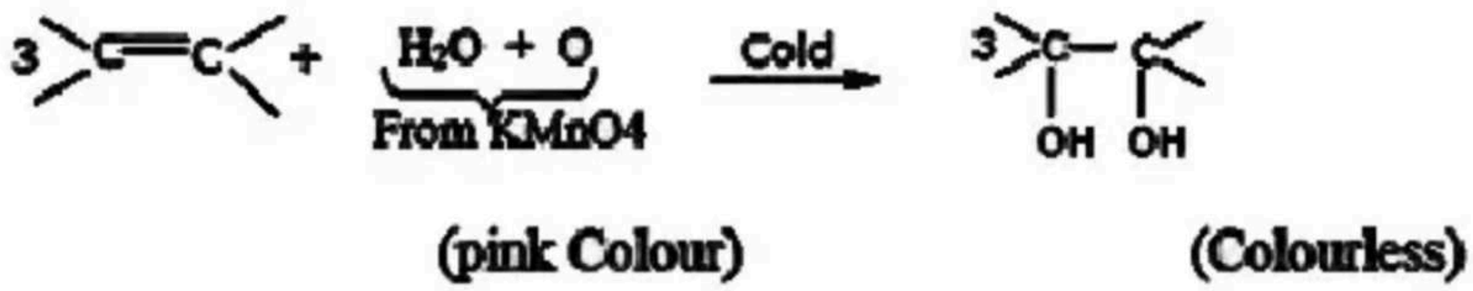
3. Oxymercuration-demercuration :-

$$H_2C = CH_2 \xrightarrow[THF,\ H_2O]{Hg(OAC)_2} OH - CH_2 - CH_2 - Hg\ OAc \xrightarrow[OH^-]{NaBH_4} CH_3CH_2OH + Hg$$

4. Oxidation with potassium permanganate :-

Monksays

$$2KMnO_4 + H_2O \longrightarrow 2KOH + 2MnO_2 + 3[O]$$

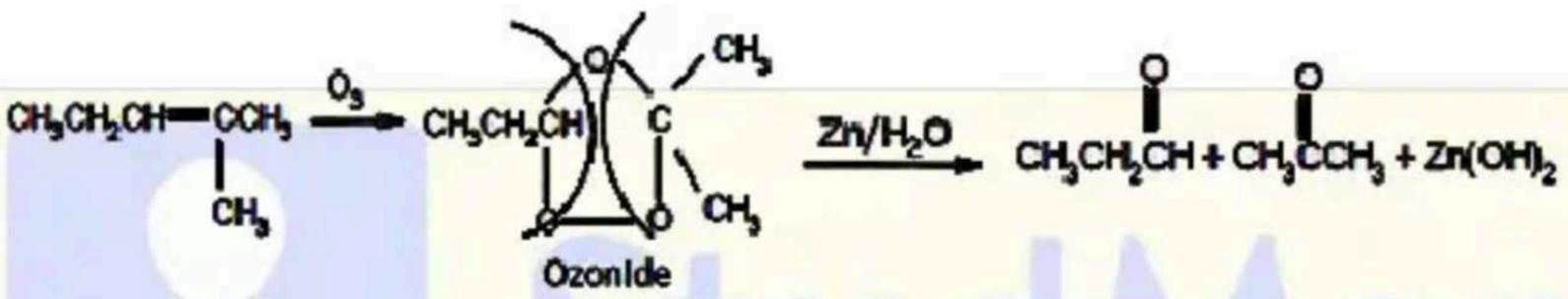


(pink Colour) (Colourless)

Bayer's Test

$$CH_3 - CH = CH_2 \xrightarrow[(ii)\ H^+\Delta]{(i)\ \text{Alk. } KMnO_4} CH_3COOH + CO_2 + H_2O$$

Oxidation with Ozone :- Ozonolysis-give carbonyls compounds



Alkynes

Preparation :-

From vicinal dihalides :- dehalogenation

$$CH_3 - BrCH - CH_2 - Br \xrightarrow[\Delta]{2KOH\ (alc)} CH_3 - C \equiv CH + 2KBr + 2H_2O$$

By the action of water on calcium carbide :-

$$CaC_2 + H_2O \rightarrow HC \equiv CH + Ca(OH)_2$$

Reactions

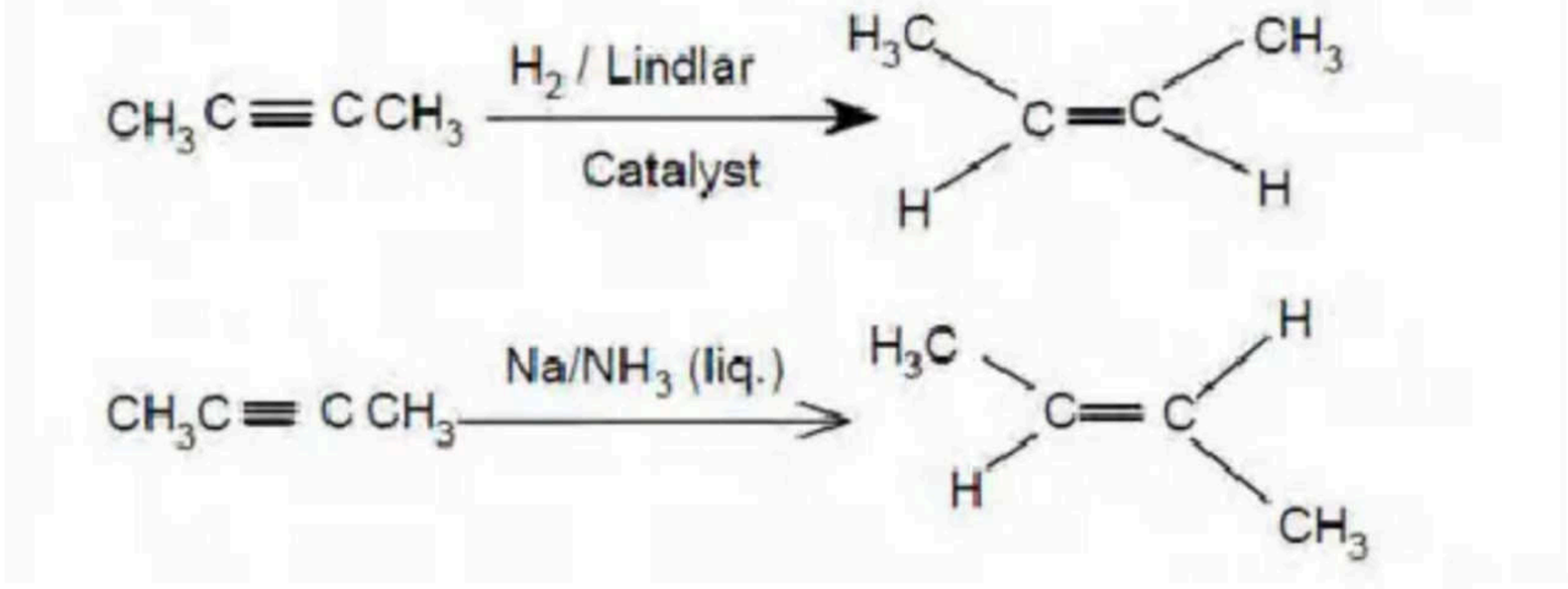
(1) Addition Reaction : Alkyne show electrophilic addition reaction.

1. Addition of Hydrogen – Hydrogenation.

$$\underset{\text{Propyne}}{CH_3C \equiv CH} + 2H_2 \xrightarrow{Ni} CH_3CH_2CH_3$$

Monksays

Birch Reduction



2. Addition of Halogens :-

$$HC \equiv CH \xrightarrow{2Br_2} Br_2HC - CHBr_2$$

3. Addition of hydrogen halides :-

$$HC \equiv CH + 2HBr \longrightarrow CH_3CH\,Br_2$$

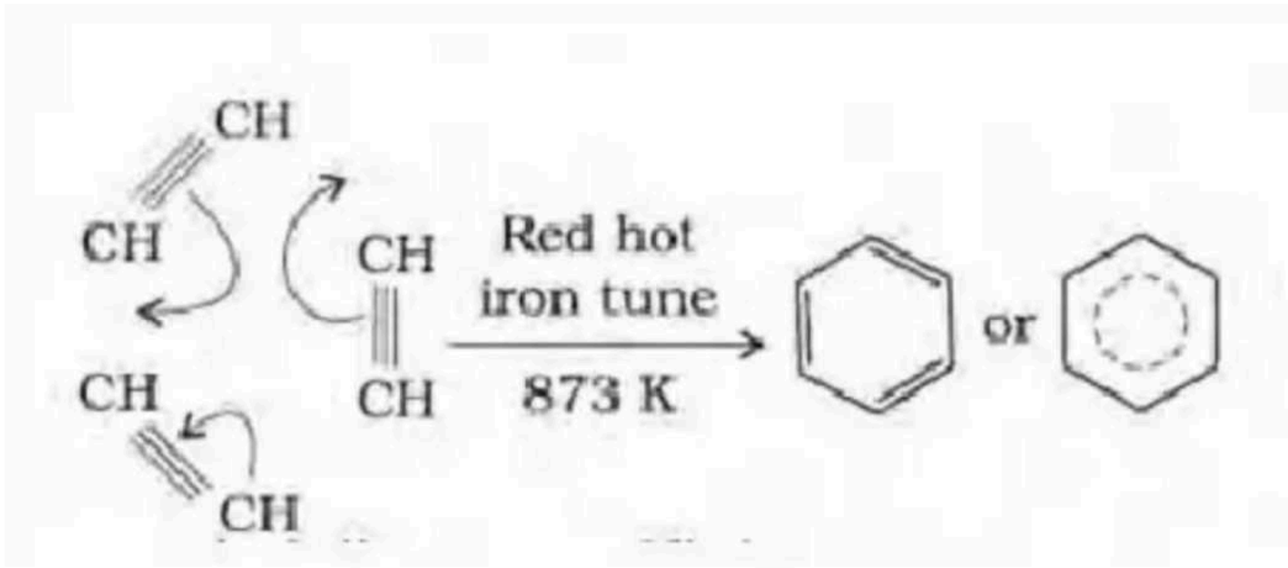
4. Addition of water (Hydration):- Acid catalyzed addition of water

$$HC \equiv CH + H_2O \xrightarrow[H_2SO_4]{HgSO_4} \underset{\text{Unstable}}{[CH_2 = CHOH]} \rightleftharpoons CH_3CHO$$

5. Polymerisation –

(a) Linear polymerization : of ethyne gives polyacetylene or polyethyne which is a high molecular weight polyene containing repeating units of (CH = CH – CH = CH) and can be represented as – (CH = CH – CH= CH)n-

(b) Cyclic polymerization-results in the formation of aromatic compound.

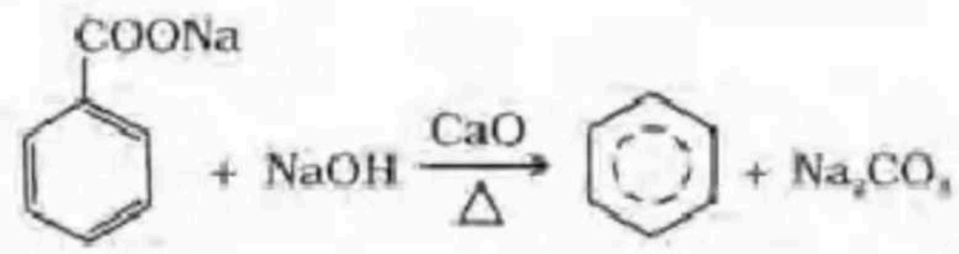


Monksays

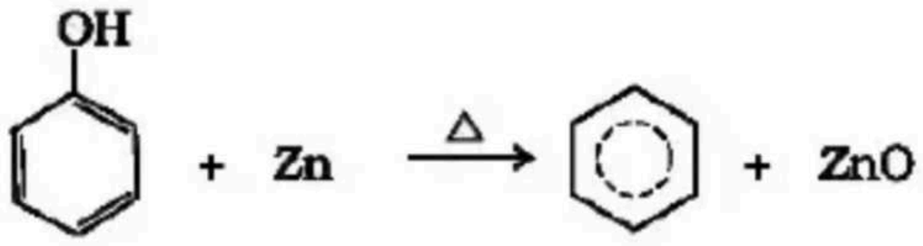
Aromatic Hydrocarbon

Preparation of Benzene

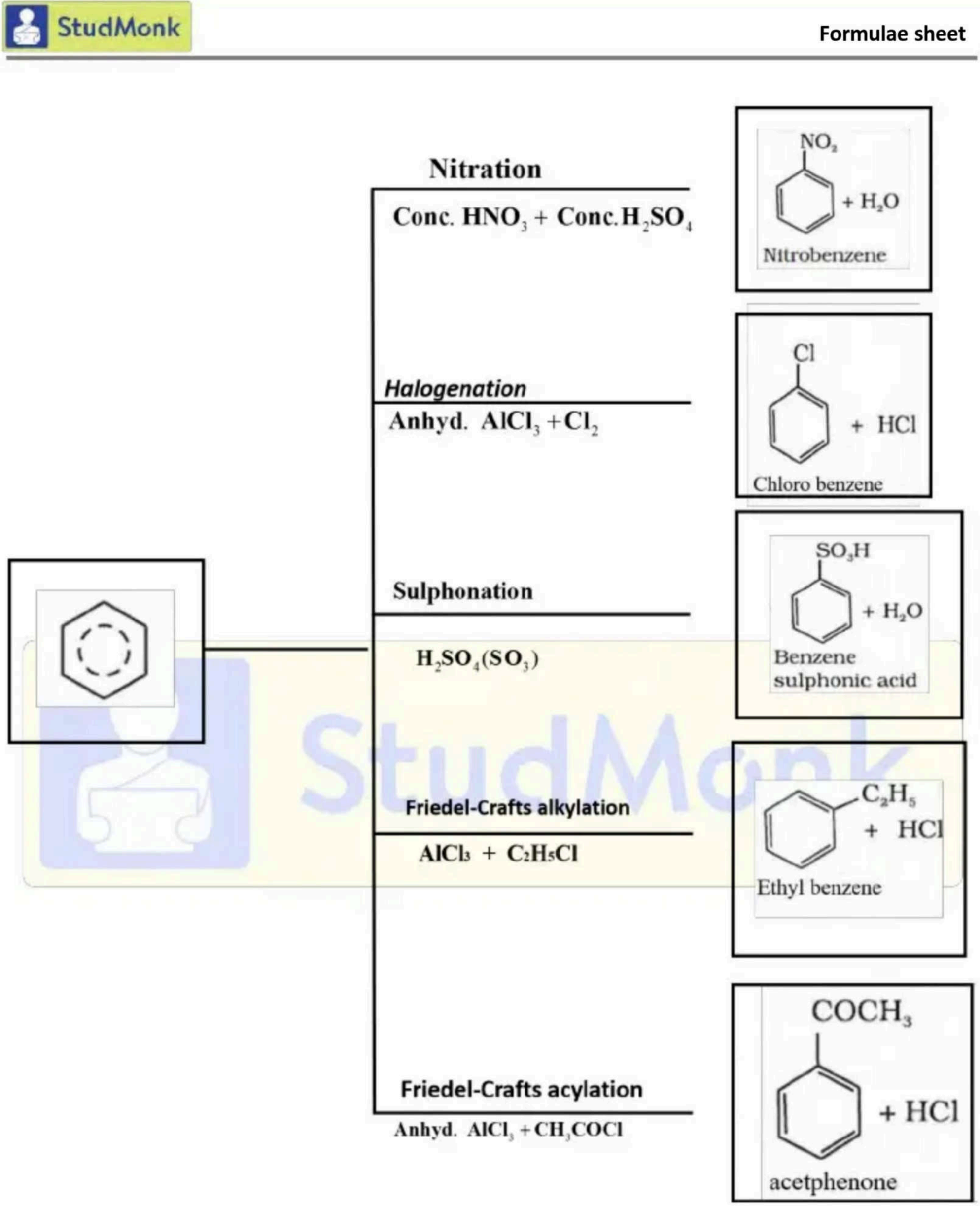
(i) Decarboxylation of aromatic acids.



(ii) Reduction of phenol: Phenol is reduced to benzene by passing its vapours over heated zinc dust

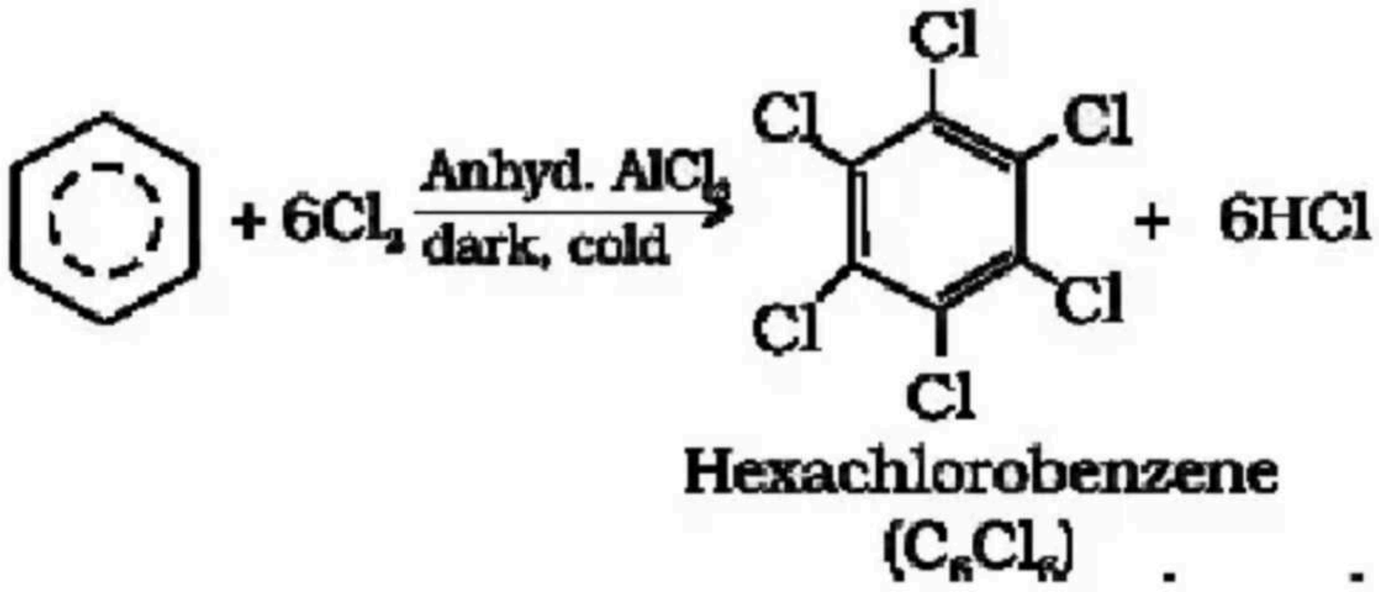


StudMonk



Benzene on treatment with excess of chlorine in the presence of anhydrous $AlCl_3$ in dark yields hexachlorobenzene (C_6Cl_6)

Monksays



Hexachlorobenzene
(C_6Cl_6)

Directive influence of a functional group in monosubstituted benzene :-

1. Ortho and para directing groups and activating

$-OH, -NH_2, -NHR, -NHCOCH_3, -OCH_3, -CH_3, -C_2H_5$ etc.

2. Meta directing group and deactivating :- $-NO_2, -CN, -CHO, -COR, -COOH, -COOR, -SO_3H$,

3. Ortho and para directing groups and deactivating – Halogens because of their strong – I effect, overall electron density on benzene ring decreases. However due to resonance the electron density on o- and p- position is greater than that at the m-position. Hence, they are also o- and p- directing groups.

21. POLYMERS

Important Addition Polymers

Polyolefins

1. Polythene

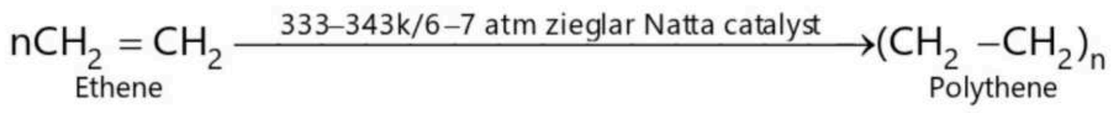
Polymer of ethylene or ethene.

(a) Low density polythene (LDP)

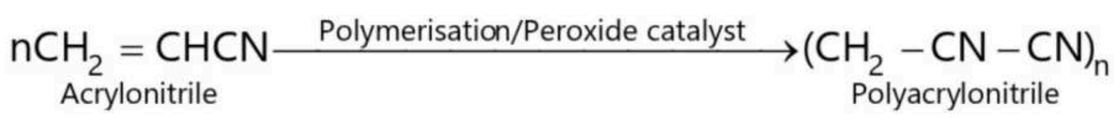
$$\underset{\text{Ethene}}{nCH_2 = CH_2} \xrightarrow{350-750k/1000 \text{ to } 2000 \text{ atm traces of } O_2} \underset{\text{Polythene}}{(CH_2 - CH_2)_n}$$

(b) High density polythene (HDP)

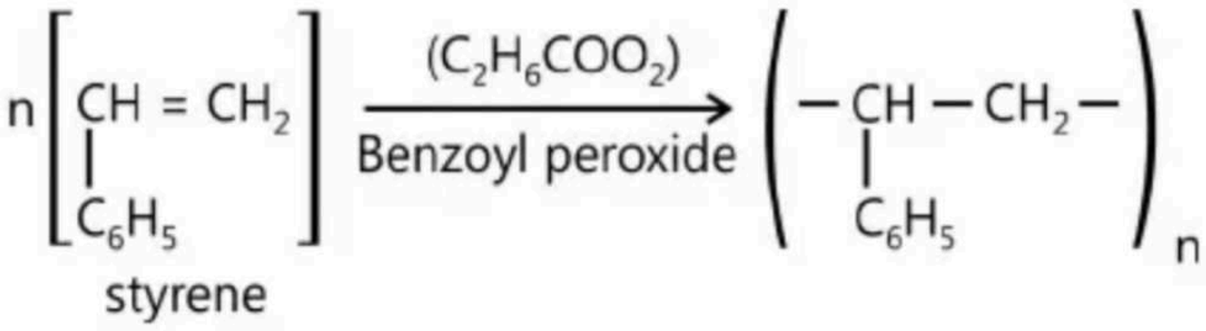
Monksays



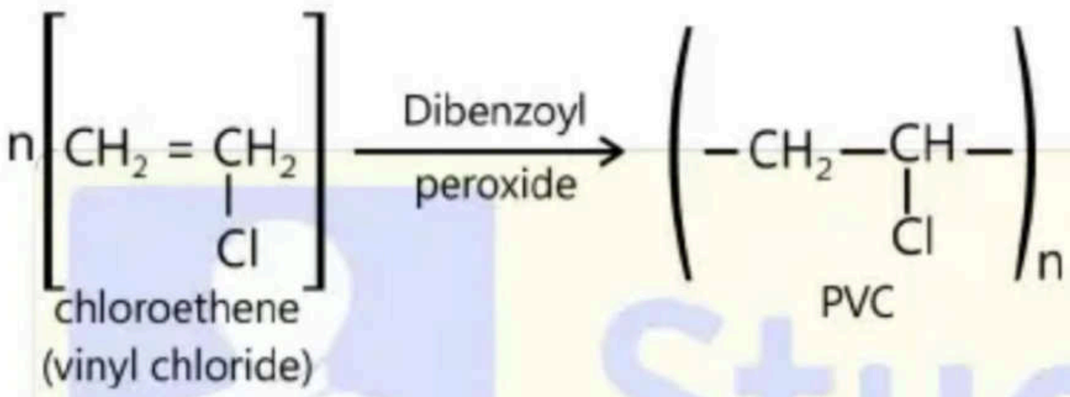
(d) Polyacrylonitrile



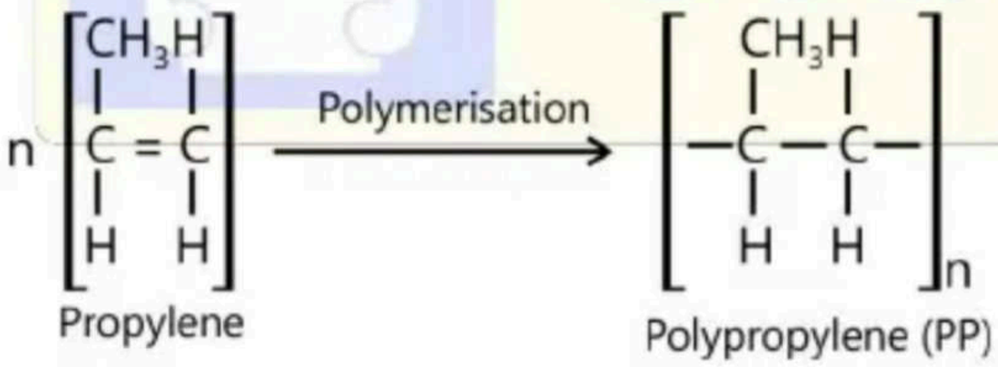
2. Polystyrene (Styrone)



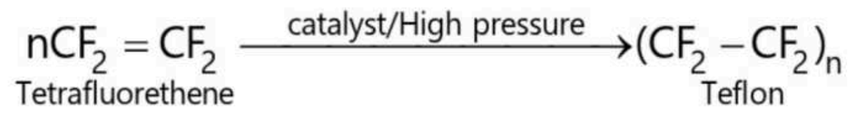
3. Polyvinylchloride (PVC)



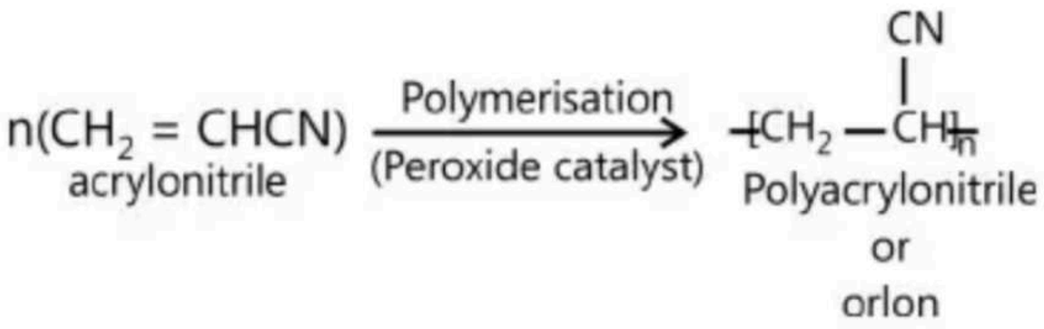
4. Polypropylene (PP)



5. Polytetrafluoroethene (Teflon)



6. Polyacrylonitrile



Monksays

Condensation polymers

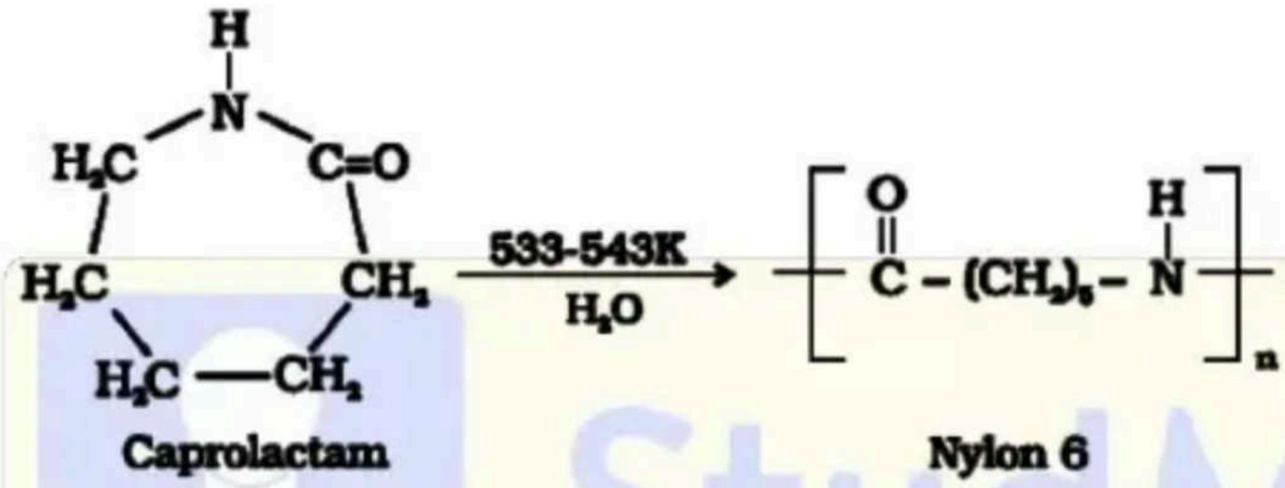
Polyamides

(a) Nylon 6, 6 : It is prepared by the condensation polymerization of hexamethylenediamine with adipic acid under high pressure and at high temperature.

$$nHOOC(CH_2)_4COOH + nH_2N(CH_2)_6NH_2 \xrightarrow{553\,K/\text{High pressure}} [N-(CH_2)_6-\underset{H}{\overset{|}{N}}-\overset{O}{\overset{||}{C}}(CH_2)_4-\overset{O}{\overset{||}{C}}]_n$$

Nylon 6,6

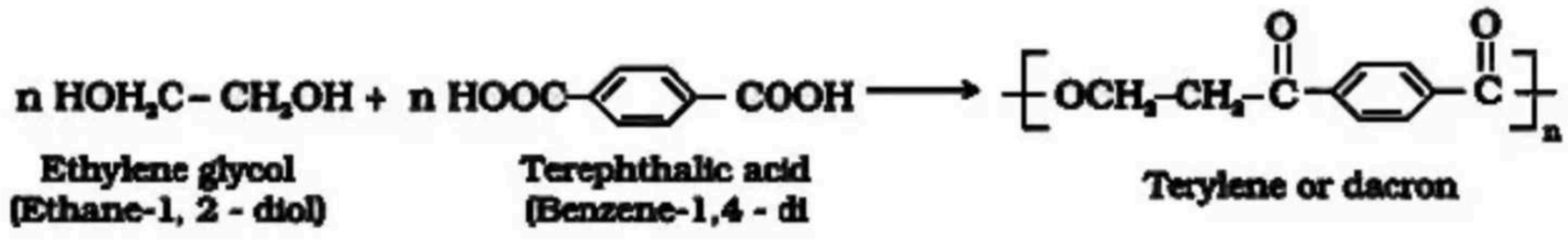
(b) Nylon 6 : It is obtained by heating caprolactum with water at a high temperature



Caprolactam Nylon 6

Polyesters

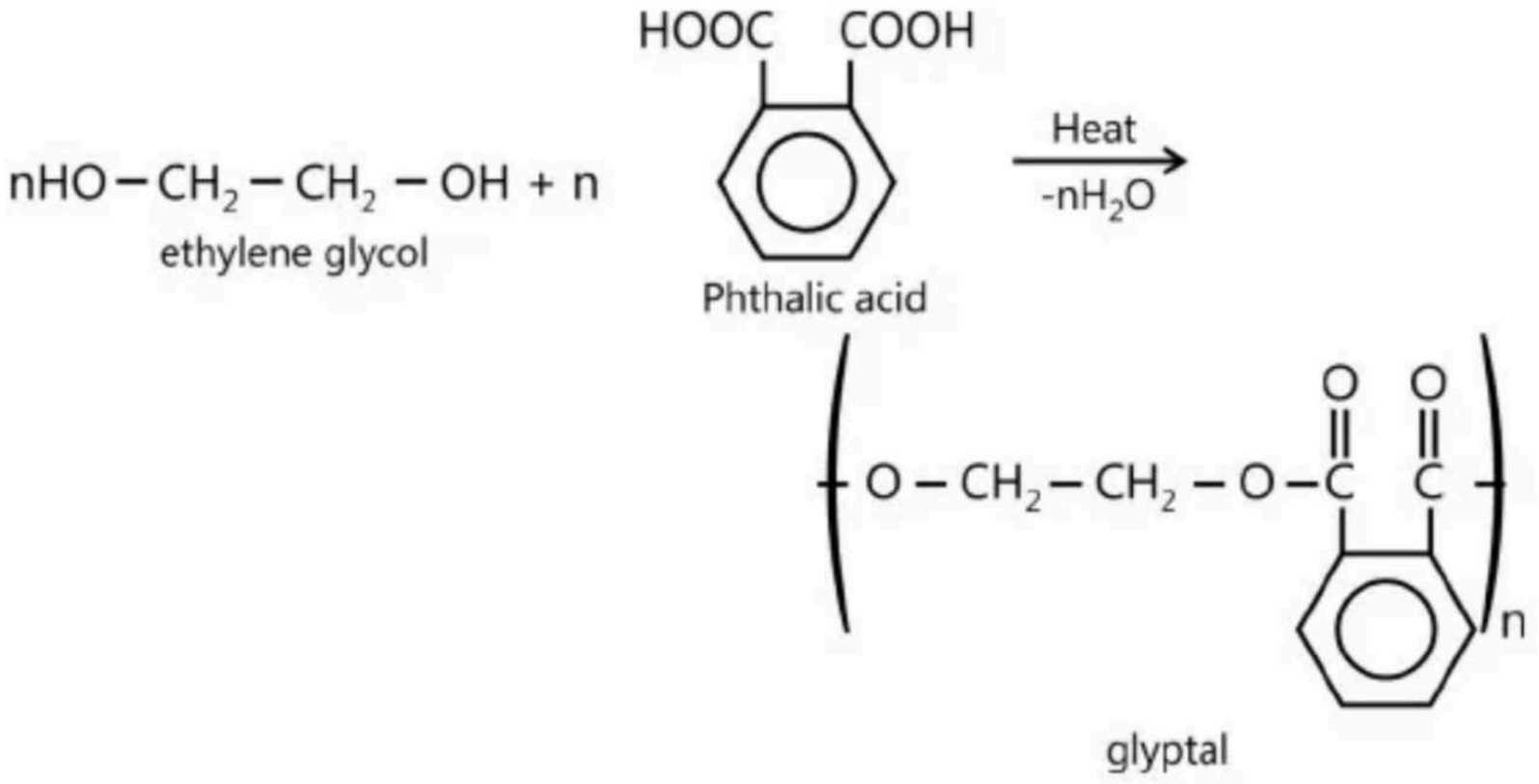
(a) Terylene or Dacron : It is manufactured by heating a mixture of ethylene glycol and terephtalic acid at 420 to 460 K in the presence of zinc acetate-antimony trioxide catalyst.



Ethylene glycol (Ethane-1, 2 - diol) Terephthalic acid (Benzene-1,4 - di Terylene or dacron

Monksays

2. Glyptal



ethylene glycol

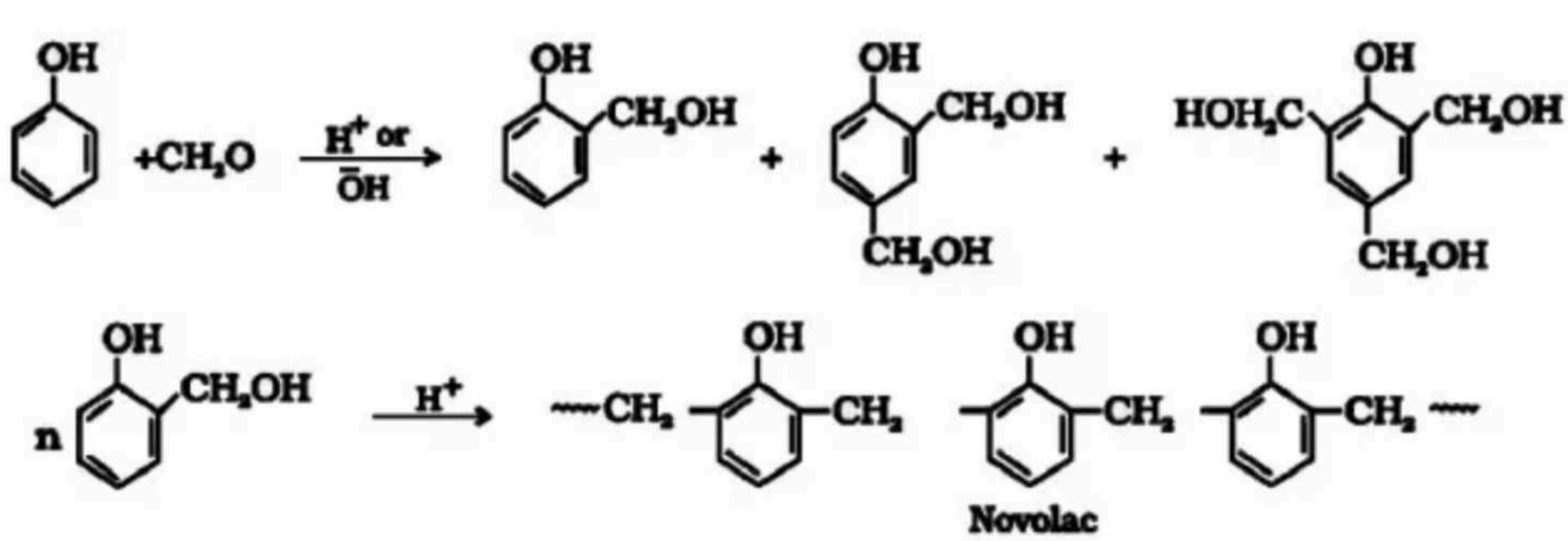
Polymethylmethacrylate (PMMA)

The polymer is known by several commercial names such as lucite, acrylite, plexiglass and perspex.

Resins

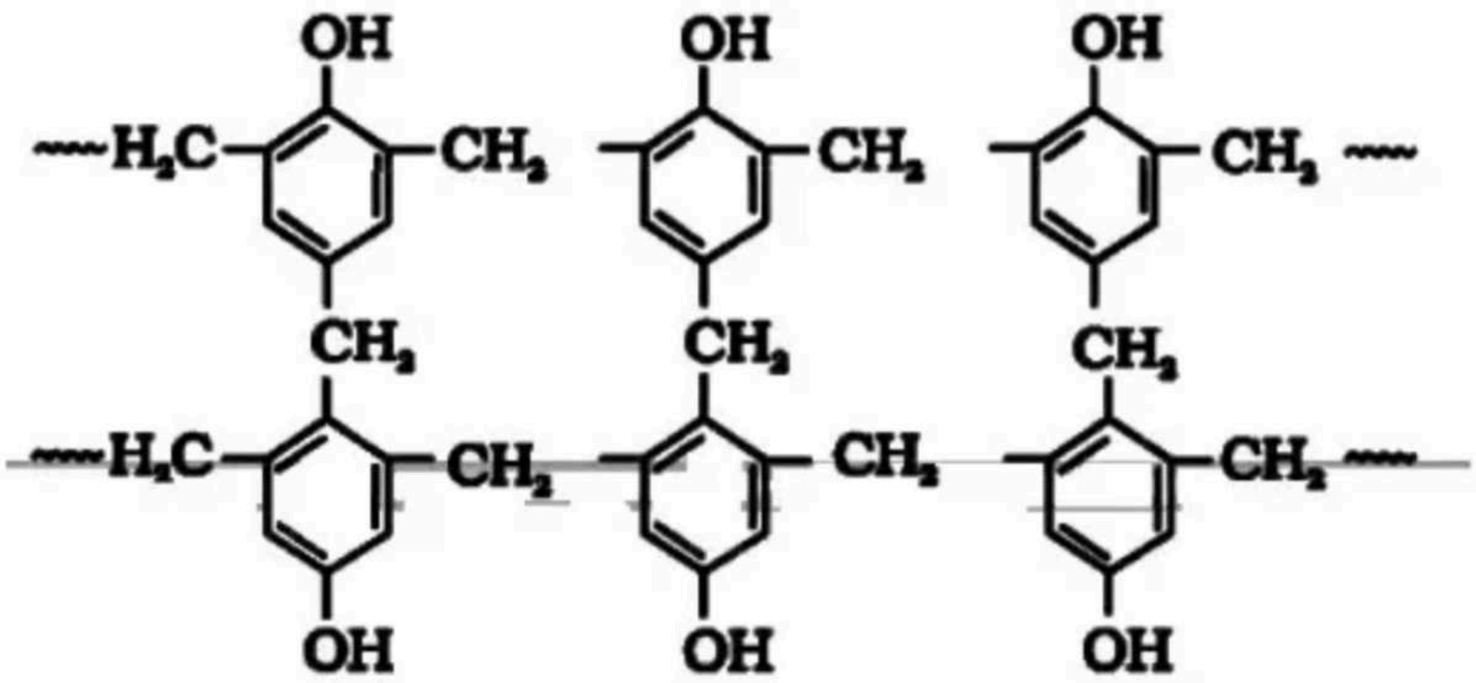
1. Phenol-Formaldehyde Polymer

These polymers are obtained by the condensation reaction of phenol with formaldehyde in the presence of either acid or a base catalyst. The reaction involves the formation of methylene bridge at ortho, para or both ortho and para positions. A linear or cross linked material is obtained depending upon the condition of reaction.

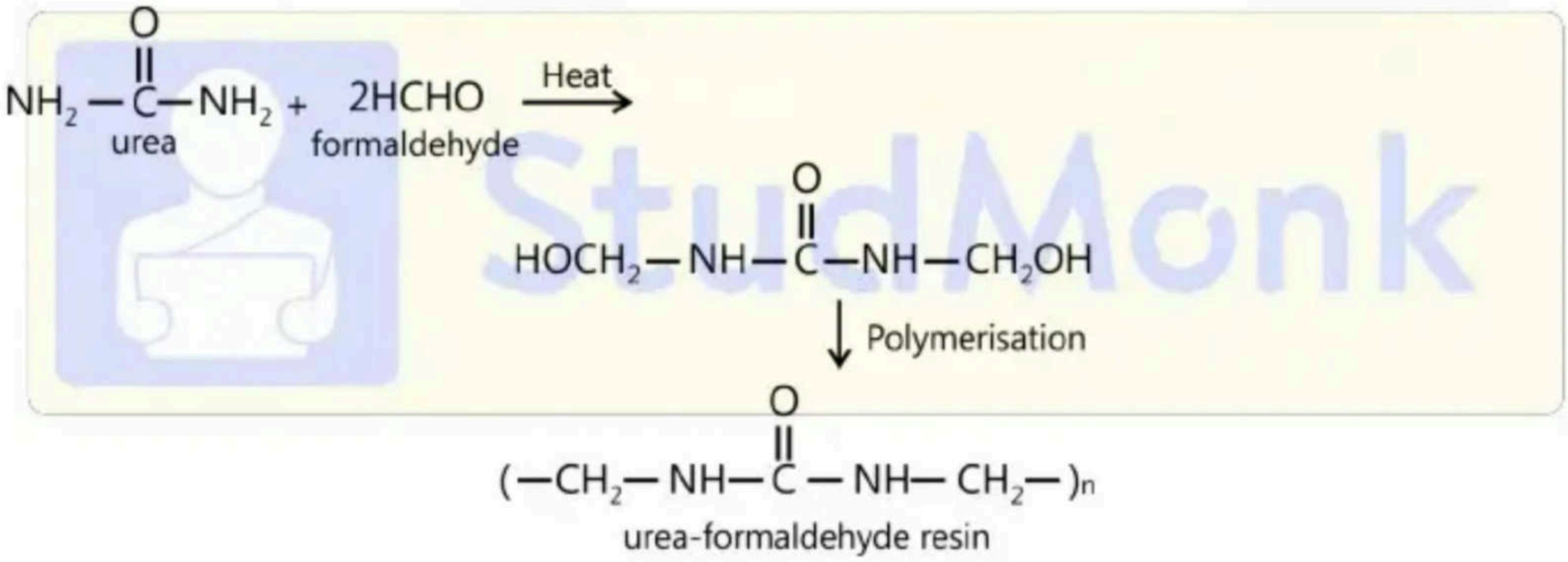


Monksays

(b) Novolac on heating with formaldehyde forms Bakelite

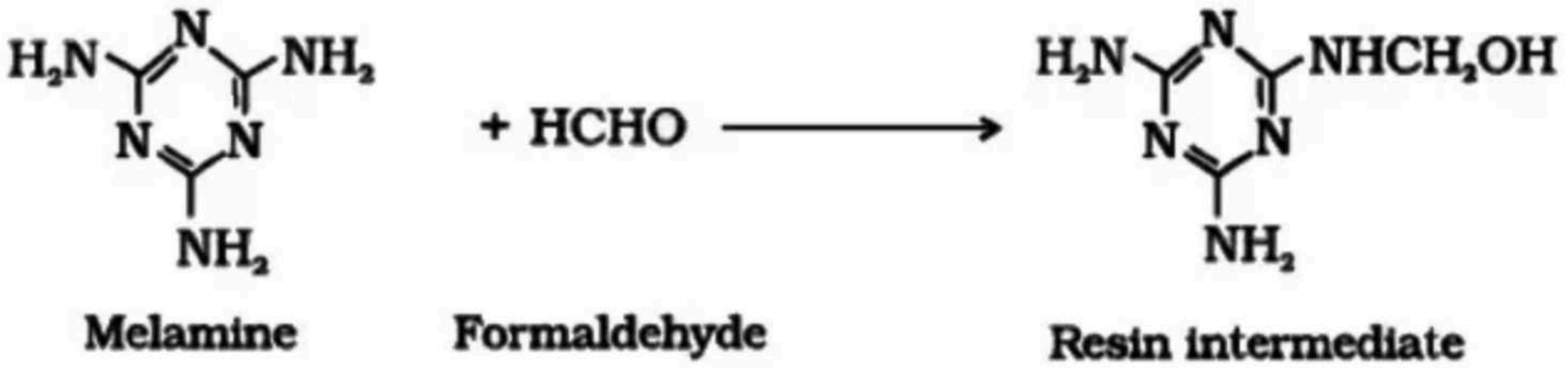


2. Urea-Formaldehyde Resin



4. Melamine – formaldehyde polymer : Melamine formaldehyde polymer is formed by the condensation polymerization of melamine and formaldehyde

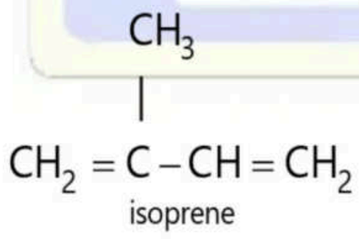
Monksays



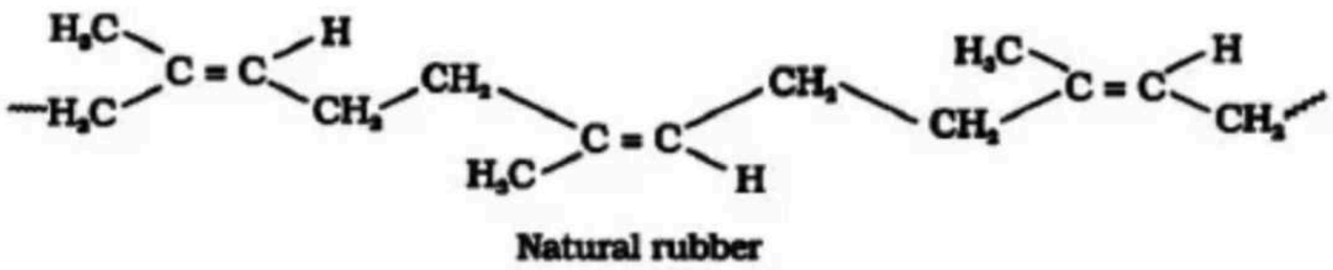
StudMonk

4. Natural Rubber

Natural rubber is a coiled linear 1, 4-polymer of isoprene.



(a) Natural rubber : Natural rubber is a linear polymer of isoprene (2-methyl-1, 3-butadiene) and is also called as cis-1, 4-polyisoprene.

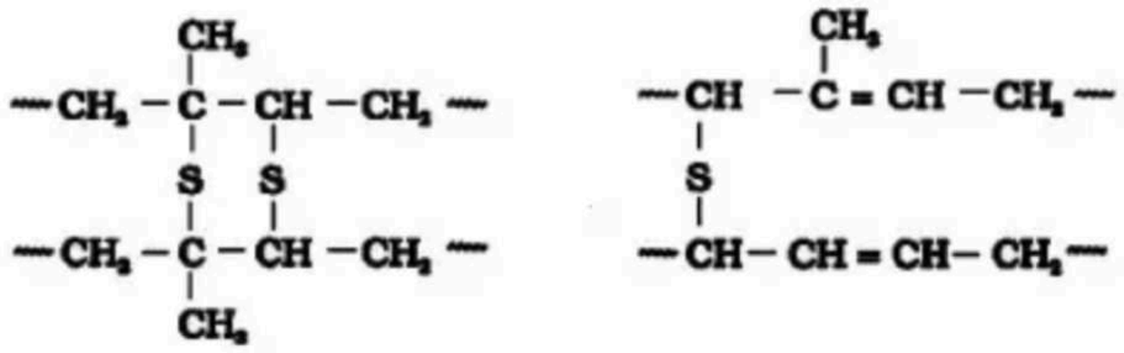


Vulcanisation of Rubber

The properties of natural rubber can be modified by introducing -S-S- polysulphide crosslinks in its structure.

Monksays

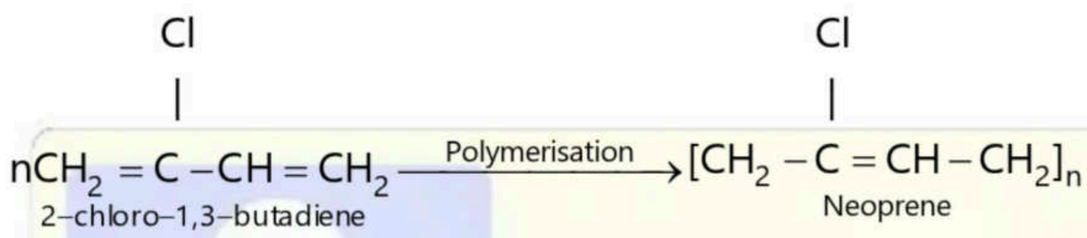
The process of heating a mixture of raw rubber with sulphur and an appropriate addictive in a temperature range between 373 K to 415 K to improve upon physical properties like elasticity, strength etc.



Sulphur cross links in vulcanised rubber

Synthetic rubber

(A) Neoprene or polychloroprene



2–chloro–1,3–butadiene

Neoprene

(B) Buna – N

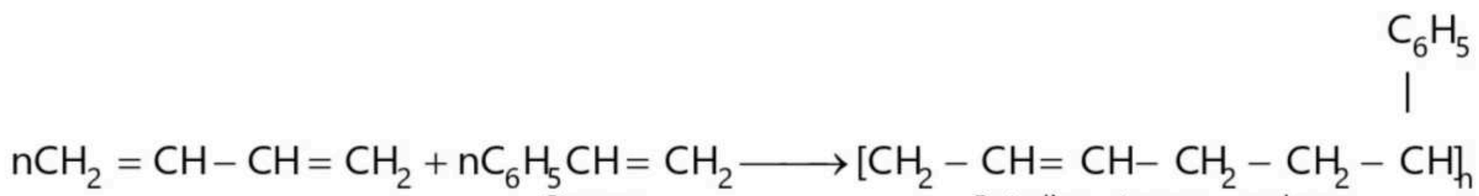


1,3–butaline

Acrylanitrile

Buna–N

(C) Buna – S



1,3–butadine

Styreme

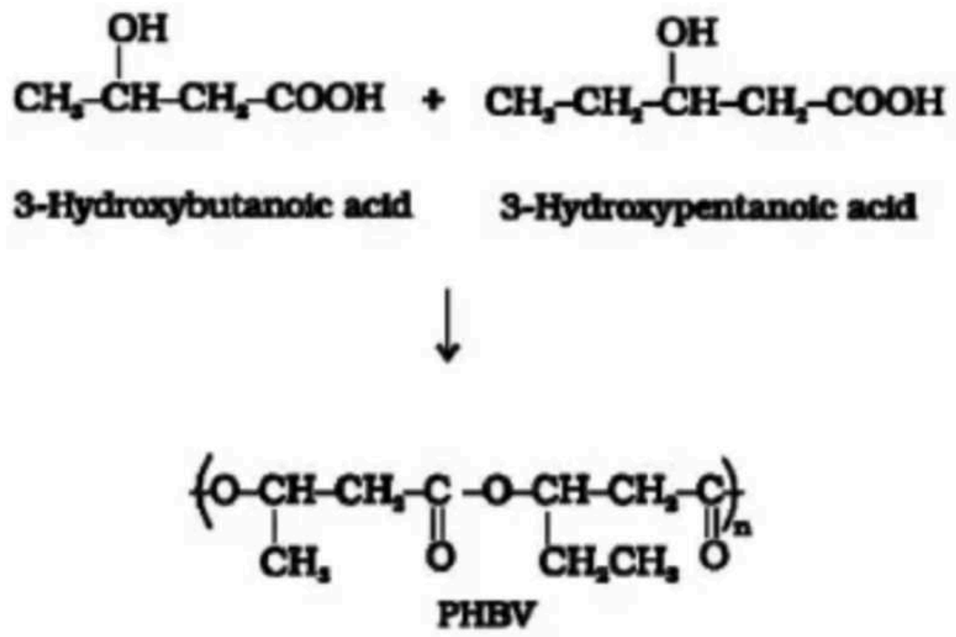
Butadine–styreme copolymer (Buna–S)

Biopolymers and Biodegradable Polymers

Aliphatic polyesters are the common examples of biodegradable Polymers.

(a) Poly-β-hydroxybutyrate-co-β-hydroxy valerate (PHBV):

It is obtained by the copolymerization of 3-hydroxybutanoic acid and 3-hydroxypentanoic acid



(b) Nylon 2-nylon 6 : It is an alternating polyamide copolymer of glycine (H_2N-CH_2-COOH) and amino caproic acid $(H_2N(CH_2)_5COOH)$

$$nH_2N-CH_2-COOH + nH_2N-(CH_2)_5-COOH \rightarrow (HN-CH_2-\overset{O}{\overset{\|}{C}}-HN-(CH_2)_5-\overset{O}{\overset{\|}{C}})_n$$

Glycine — Amino expronic acid

Some polymers:

S.N	Name of polymer	Class of polymer	Name/s of monomer/s	Uses
1.	Dynel	Copolymer	Vinyl chloride and acrylonitrile	Human hair wigs
2.	Glyptal	Copolymer	Ethylene glycol and phthalic acid	In paints
3.	Thiokol	Condensation	Ethylene chloride and sodium (Polysulphide rubber)	Rubber tetra sulphide
4.	Superglue	Homopolymer	Methyl α-cyanoacrylate	Glue
5.	Kevlar	Polyamide condensation	Terephthalic acid chloride and p-phenylene diamine	Bullet proof vests and helmets
6.	Nomex	Polyamide condensation	m-phthalic acid and m-dinitrobenzene	Protective clothes for race car drives and fire fighters.
7.	Lexan	Polycarbonate	Diethylcarbonate and bisphenol Ester condensation	Bullet proof windows and helmets
viii)	Polyurethane or Thermocole	Copolymer glycol	Toluene diisocyanate and ethylene	For padding and building insulation
ix)	Saran	Copolymer chloride	Vinyl chloride and vinylidene	Bumpers

Monksays